Leveraged Learning

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In 1948 Claude Shannon introduced the world to information entropy. He quantified the amount of information in a signal. What if we wanted to quantify learning in the same spirit, as the increased understanding of a problem? This work is an attempt to follow that thread. It models the problem as a boolean map, and the understanding as the aggregate ability to predict that map. And learning, as the speed at which that understanding increases, specifically as information is supplied to the learner. Some interesting relationships can be formed between the two.

Allow us to enumerate the seminal tenets of this treatise.

1. Learning is the incremental ability to answer questions correctly that follow a pattern.

2. The number of questions reviewed is not the right measure of a learner's investment. What counts is the information received from asking them. Not all questions are created equal: about some, the learner already knows something priori, and their answers count for *less*.

3. Conversely, with instincts about the world, one can infer *more* than a single bit from a yes–no question. One answer settles expectations on any number of yet unseen questions.

4. This ability to extrapolate from prior information is codified here as *leverage*

$$\text{Leverage of Yes/No} = \frac{\text{Learned} > 1 \text{ bit}}{\text{Received} < 1 \text{ bit}} > 1. \tag{1}$$

Contents

1 A thought experiment

Let us start with a small thought experiment, by introducing *Werner the learner* (Figure 1).

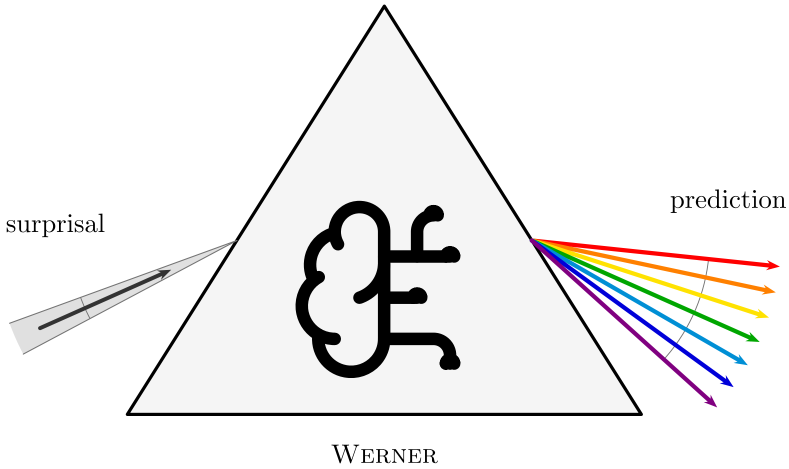


Figure 1: Werner the learner.

In this work, we model learning as the increased ability to give the correct answer to questions, as more of the ground truth is revealed. It is Werner's job to learn a pattern. The most basic shape of a pattern is a *binary map*, and the most basic binary map is a map from one bit to one bit.

The setup of the experiment:

- a **question** $q \in \mathcal{Q} = \{0, 1\}$;
- an **answer** $a \in \mathcal{A} = \{0, 1\}$;
- the **truth** $\psi : \mathcal{Q} \to \mathcal{A}$, the one fixed map nature uses to answer questions. ψ is unknown to Werner. By definition $\psi(q) = a$.

There are exactly four candidate maps ϕ_j that ψ could be (Table 1): each of the two inputs can be sent to either of the two outputs, independently. We index them by reading the truth table $\phi_j(1)\,\phi_j(0)$ as a binary numeral: that number *is* j.

j	$\phi_j(1)$	$\phi_j(0)$	name
0	0	0	constant 0
1	0	1	NOT ($\neg q$)
2	1	0	identity (q)
3	1	1	constant 1

Table 1: The four binary maps from one bit to one bit.

1.1 An unintelligent learner

Now, if Werner is unintelligent, he has no preconceived notion of which maps are more useful, more likely, or somehow more logical than others. In practice, we would have to tell Werner the answer a to every single question q, before he has learned the whole pattern.

We model this with a **uniform prior**: writing $p_j = \Pr(\psi = \phi_j)$ for Werner's belief, every candidate map carries the same weight.

EXAMPLE 1

For the four maps at hand:

$$p_j = \tfrac{1}{4}, \qquad j = 0, 1, 2, 3. \tag{2}$$

What does this belief predict? For each question q separately, we can tabulate the probability of each possible answer a: it is the total belief sum of the maps that would answer a to q,

$$M_{a,q} \;=\; \sum_{j\,:\,\phi_j(q)=a} p_j. \tag{3}$$

Arranged with one row per answer ($a = 0, 1$ from the top) and one column per question ($q = 1, 0$ from the left, descending like the digits of the numeral convention), these numbers form the matrix M. Each column is a probability distribution over answers, so M is column-stochastic. Werner reads it as follows: find the column of the question at hand. It's the marginal distribution over answers, according to his belief.

EXAMPLE 2

Under the uniform prior, two of the four maps answer 0 and two answer 1, at either question:

$$M \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & M_{0,1} & M_{0,0} \\ \mathbf{1} & M_{1,1} & M_{1,0} \end{array} \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p_0+p_1 & p_0+p_2 \\ \mathbf{1} & p_2+p_3 & p_1+p_3 \end{array} \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}. \tag{4}$$

Every column is a fair coin: the unintelligent Werner predicts nothing about any answer before it is revealed.

Werner gets to predict the answer to the question $q = 0$. By his own belief it is a coin toss: $M_{0,0} = M_{1,0} = \frac{1}{2}$. Let us say the answer comes back $a = 1$. The **surprisal** of this answer, to Werner, is 1 bit, because the probability he assigned to it was one half.

In general, when an event a belief holds with probability P actually occurs, its surprisal is

$$s \;=\; -\log_2 P \quad \text{bits}, \tag{5}$$

here $s = -\log_2 M_{a,q}$ for answer a arriving at question q. Surprisal measures how much the received information deviates from what was already expected: an answer held certain ($P = 1$) carries 0 bits: nothing new. A coin toss carries exactly 1 bit; and the surprisal grows without bound as the answer gets more unexpected, $P \to 0$. It is additive: the surprisal of two independent events occurring is the sum of their surprisals, just as their probabilities multiply.

Moreover, Werner can use a process of elimination (an extreme form of Bayes' rule) to update his prior into a posterior. Semantically: every candidate map that would have answered the question wrongly is now *ruled out*, not merely disfavored. They are inconsistent with what's been learned.

EXAMPLE 3

The truth ψ answered 1 at $q = 0$, and a map with $\phi_j(0) = 0$ cannot be ψ. This kills the constant-0 map ($j = 0$) and the identity ($j = 2$). The two survivors, NOT ($j = 1$) and constant 1 ($j = 3$), both predicted the observed answer, so this round of evidence does not distinguish between them at all (Table 2).

j	$\phi_j(1)$	$\phi_j(0)$	name
~~0~~	~~0~~	~~**0**~~	~~constant 0~~
1	0	1	NOT ($\neg q$)
~~2~~	~~1~~	~~**0**~~	~~identity (q)~~
3	1	1	constant 1

Table 2: The surviving candidate maps after the answer $\psi(0) = 1$.

The surviving ratio of believabilities must be preserved, and the only update that does so is to renormalize the surviving weights to sum to one. (This is Bayes' rule with a 0/1 likelihood: $\Pr(\psi = \phi_j \mid a) \propto \Pr(a \mid \phi_j)\, p_j$, and $P(a \mid \phi_j)$ is 1 for a map that answers a and 0 for a map that does not.)

EXAMPLE 4

The posterior is

$$p' = \left(0,\ \tfrac{1}{2},\ 0,\ \tfrac{1}{2}\right). \tag{6}$$

Constructing the new answer matrix from p' by the same recipe as before:

$$M' = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p_0' + p_1' & p_0' + p_2' \\ \mathbf{1} & p_2' + p_3' & p_1' + p_3' \end{array} = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & 0 \\ \mathbf{1} & \frac{1}{2} & 1 \end{array}. \tag{7}$$

Indeed, Werner knows the answer to $q = 0$: that column has collapsed onto $a = 1$. He is still as agnostic as possible about the other answer: the $q = 1$ column remains a fair coin. He gained 1 bit of surprisal information, and cleaned up his table to the tune of 1 bit: the $q = 0$ column went from a full bit of answer-uncertainty to none, while the $q = 1$ column is unchanged.

After the next question, $q = 1$, he learns $a = 1$, again a coin toss beforehand ($M'_{1,1} = \frac{1}{2}$), so again worth 1 bit of surprisal. Elimination now rules out NOT ($\phi_1(1) = 0$), leaving a single survivor:

$$p'' = \left(0,\ 0,\ 0,\ 1\right), \qquad M'' = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 0 & 0 \\ \mathbf{1} & 1 & 1 \end{array}. \tag{8}$$

So this is the constant map: $\psi = \phi_3$. Werner is done. Every column is one-hot, and he will be able to answer all questions correctly from now on.

1.2 Intelligent priors

Now, let us assume that Werner knows *something* about the world before he starts.

EXAMPLE 5

Let's restart, with a different ψ. Moreover, let Werner know the map is not constant. His belief then spreads evenly over the two remaining candidates, NOT ($j = 1$) and the identity ($j = 2$):

$$p = \left(0,\ \tfrac{1}{2},\ \tfrac{1}{2},\ 0\right). \tag{9}$$

Is he any better at predicting $\psi(0)$? Constructing M by the usual recipe:

$$M = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p_0 + p_1 & p_0 + p_2 \\ \mathbf{1} & p_2 + p_3 & p_1 + p_3 \end{array} = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}. \tag{10}$$

He is still just as bad as before: every column a fair coin.

So what has all this veteran wisdom about the nature of patterns in the world bought him? His increased ability for *inference*.

EXAMPLE 6

Ask $q = 0$ as the first question, and let the answer again come back $a = 1$. Again a coin toss beforehand, so again worth exactly 1 bit of surprisal. Elimination rules out the identity ($\phi_2(0) = 0$), and the lone survivor is NOT:

$$p' = \left(0,\ 1,\ 0,\ 0\right), \qquad M' = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 1 & 0 \\ \mathbf{1} & 0 & 1 \end{array}. \tag{11}$$

This time *both* columns collapse at once. Werner received the same single bit of surprisal as before, but cleaned up 2 bits of table: the answer at $q = 0$ he observed, and the answer at $q = 1$ he *deduced*, without ever asking. Pre-emting the constants means $\phi(1) \neq \phi(0)$, so his prior held the two answers perfectly anticorrelated, and settling one settles the other. He is done after one question instead of two.

We say the information from the first answer was **leveraged**. We will make this more precise in the upcoming sections.

1.2.1 A one-parameter world

For more structure, let us relax the claim that there are *no* constant maps. Instead, we create a one-parameter world where the prior probability that the map is constant is p, spread evenly within each class (Table 3):

j	$\phi_j(1)$	$\phi_j(0)$	name	p_j
0	0	0	constant 0	$p/2$
1	0	1	NOT ($\neg q$)	$(1-p)/2$
2	1	0	identity (q)	$(1-p)/2$
3	1	1	constant 1	$p/2$

Table 3: The one-parameter prior: constant with probability p.

In vector form, and with the answer matrix constructed by the usual recipe:

$$\big(p_0,\ p_1,\ p_2,\ p_3\big) \ = \ \tfrac{1}{2}\big(p,\ 1-p,\ 1-p,\ p\big), \qquad M \ = \ \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}, \tag{12}$$

fair coins in every column, for every value of p.

How much is there to learn, in total? We account for the table as a whole by declaring entropy *extensive* over the columns. That is, the answer-uncertainties of the columns simply add:

$$H(M) \ := \ \sum_q H\big(M_{\cdot,q}\big), \qquad H\big(M_{\cdot,q}\big) \ := \ -\sum_a M_{a,q}\,\log_2 M_{a,q}. \tag{13}$$

Here both columns are fair coins carrying 1 bit each, so the table starts at its maximum: $H(M) = 2$ bits.

Now let us ask a question. Again we could use $q = 0$; by the symmetry of the prior it actually does not matter for this argument. And we are now going to talk about *expected* values, instead of realized ones: averages over the possible maps, denoted $\langle\cdot\rangle$. We must assume our prior is *a priori* correct, if we had better assumptions, those would be reflected in the prior. In other words, we model nature to draw the truth from the very distribution Werner believes. Therefore, the average over ψ simply zips with the weights: $\langle X\rangle = \sum_j p_j\,X(\psi{=}\phi_j)$.

How much surprisal should Werner expect from asking a question q? The answer comes back a exactly when the truth answers a, which under the zipped average happens with probability $\sum_{j:\phi_j(q)=a} p_j = M_{a,q}$ — the very probability Werner assigned to it — and when it does, it costs him $-\log_2 M_{a,q}$ of surprisal. Grouping the average over maps by the answer each map gives:

$$\langle s_q\rangle \ = \ \sum_j p_j\,\big(-\log_2 M_{\phi_j(q),\,q}\big) \ = \ \sum_a \underbrace{\sum_{j\,:\,\phi_j(q)=a} p_j}_{M_{a,q}}\,\big(-\log_2 M_{a,q}\big) \ = \ -\sum_a M_{a,q}\,\big(\log_2 M_{a,q}\big). \tag{14}$$

The right-hand side is exactly the column entropy $H(M_{\cdot,q})$ of equation (13): the expected surprisal of a question is the entropy of its column.

Each answer occurs with exactly the probability Werner assigned to it, so his average surprisal is his own uncertainty about that answer. This identity will do a great deal of work in what follows.

For a two-answer column, this entropy is a function of a single number, the probability x of one of the two answers: the binary entropy function:

$$h_2(x) \ := \ -x\log_2 x \ - \ (1-x)\log_2(1-x), \tag{15}$$

zero at $x \in \{0,1\}$ (a settled answer) and maximal, 1 bit, at the fair coin $x = \frac{1}{2}$. For the first question here the column is a fair coin, so

$$\langle s_1\rangle \ = \ H\big(M_{\cdot,0}\big) \ = \ h\big(\tfrac{1}{2}\big) \ = \ 1 \text{ bit}, \tag{16}$$

whatever the value of p.

And the other column is now cleaned up. Breaking the update down over the possible maps: with probability $\frac{1}{2}$ the answer is 1 (truth is NOT or constant 1), with probability $\frac{1}{2}$ it is 0 (truth is FALSE or the identity), and

$$M' \ = \ \underbrace{\begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 1-p & 0 \\ \mathbf{1} & p & 1 \end{array}}_{a_1=1,\ \text{prob } 1/2} \quad \text{or} \quad \underbrace{\begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p & 1 \\ \mathbf{1} & 1-p & 0 \end{array}}_{a_1=0,\ \text{prob } 1/2}. \tag{17}$$

Either way, the asked column is one-hot and the unasked column is a coin of bias p: what survives on $q = 1$ is precisely the class uncertainty "constant or not". The total entropy of the table is now

$$H(M') \;=\; 0 + h_2(p) \;=\; h_2(p), \tag{18}$$

a reduction of $2 - h_2(p)$ bits, purchased with a single bit of surprisal. Define the **leverage** after one question as the ratio of table entropy destroyed to surprisal received:

$$L_1 \;:=\; \frac{2 - h_2(p)}{1} \;=\; 2 - h_2(p) \;\in\; [1, 2]. \tag{19}$$

The expected surprisal on the second question is, once more, the entropy of its column,

$$\langle s_2 \rangle \;=\; h_2(p) \;\leq\; 1 \text{ bit}, \tag{20}$$

after which the table is empty. Cumulatively over the two questions, $1+h_2(p)$ bits of received information reduced the table by the full 2 bits it started with, so the **cumulative leverage** after two questions is

$$L_2 \;:=\; \frac{2}{1 + h_2(p)} \;\in\; [1, 2]. \tag{21}$$

Our knowledge about the world has allowed us to be more certain after our inference. Set $p = \frac{1}{2}$, and the prior is the uniform one of Section 1.1: $h = 1$, $L_1 = L_2 = 1$, and nothing is leveraged at all. As p skews further from $\frac{1}{2}$, the first question unlocks more. The second question carries less and less expected information, and Werner's potential for learning from it goes down with it. The column it could clean up is nearly settled already. In the limit $p = 0$, we recover the non-constant world from the start of Section 1.2. There is no surprisal and no improvement to M left in the second question at all: it contributes nothing to denominator or enumerator, and $L_1 = L_2$. Figure 2 traces all three curves between these extremes.

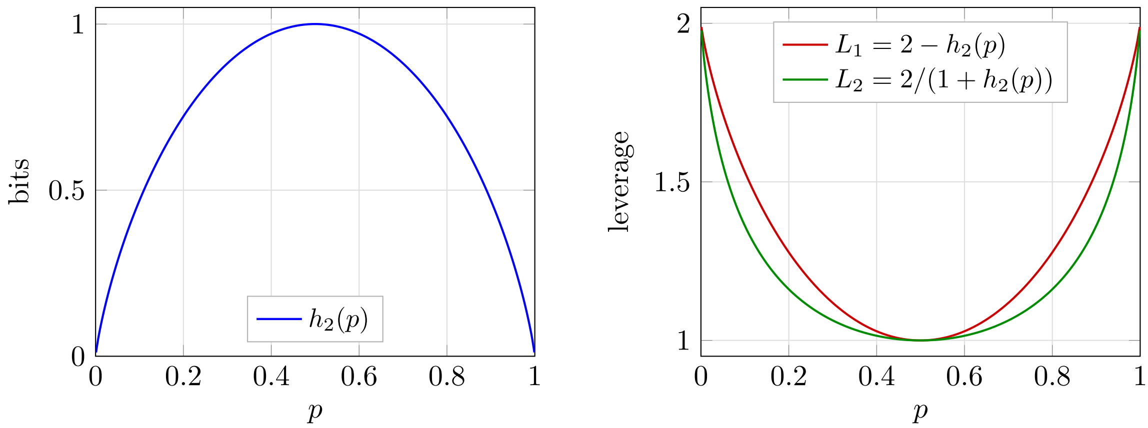


Figure 2: The one-parameter world as a function of the constant-map probability p. Left: the table entropy $h_2(p)$ remaining after one question, in bits. Right: the one-question leverage L_1 and the cumulative leverage L_2 after both questions. At $p = \frac{1}{2}$ (the uniform prior) nothing is leveraged; toward the deterministic-class extremes the first answer doubles its value.

In general, in a larger world, if there are d questions to disagree on between the two maps, the reader can convince themselves that the limit as $p \to 1$ is $L_1 = d$.

2 The general case

Now let us give the definitions in full generality. Let questions be bit strings of length n, and answers are bit strings of length m:

$$q \in \mathcal{Q} = \{0,1\}^n \equiv \mathbb{F}_2^{\otimes n}, \qquad a \in \mathcal{A} = \{0,1\}^m \equiv \mathbb{F}_2^{\otimes m}, \tag{22}$$

reminding the binary field $\mathbb{F}$. We write the sizes of those two sets without bars,

$$Q \;:=\; |\mathcal{Q}| \;=\; 2^n, \qquad A \;:=\; |\mathcal{A}| \;=\; 2^m, \tag{23}$$

so there are Q distinct questions and A possible answers to each. As before, we read bit strings as the numbers they spell, using the two interchangeably: $q \in \{0, \dots, Q{-}1\}$ and $a \in \{0, \dots, A{-}1\}$. The truth $\psi : \mathcal{Q} \to \mathcal{A}$ is one of the candidate maps ϕ_j; since each of the Q inputs can be sent to any of the A outputs independently, there are

$$N \;=\; A^Q \;=\; 2^{mQ} \tag{24}$$

of them, and the index j is again the truth table read as a numeral, now with Q digits in base A: $j = \sum_q \phi_j(q)\, A^q$, running from the constant-0 map ($j = 0$) to the constant-($A{-}1$) map ($j = N{-}1$). Werner's belief is a distribution over all of them, $\sum_{j=0}^{N-1} p_j = 1$.

EXAMPLE 7

The example we will use to guide the general case is $n = 2$, $m = 1$, which has $N = 2^{1\cdot 4} = 16$ maps: every Boolean function of the two input bits b_1, b_0, with q represented as $b_1 b_0$, or read as an ordinary binary number, $q = 2b_1 + b_0$. Each ϕ_j in Table 4 is shown as its truth table, printed in descending question order $(11, 10, 01, 00)$ so that the answer to question q occupies the digit of weight 2^q. The printed string is exactly j in binary (e.g. AND, which answers 1 only to $q = 11$, reads 1000, so it is $j = 8$). An arbitrary prior p_j alongside is conjured up for illustration.

	$\phi_j(q)$					
j	11	10	01	00	name	p_j
0	0	0	0	0	FALSE (constant 0)	0.394
1	0	0	0	1	$\neg(b_1 \vee b_0)$ (NOR)	0.010
2	0	0	1	0	$\neg b_1 \wedge b_0$	0.010
3	0	0	1	1	$\neg b_1$	0.010
4	0	1	0	0	$b_1 \wedge \neg b_0$	0.131
5	0	1	0	1	$\neg b_0$	0.010
6	0	1	1	0	$b_1 \oplus b_0$ (XOR)	0.010
7	0	1	1	1	$\neg(b_1 \wedge b_0)$ (NAND)	0.010
8	1	0	0	0	$b_1 \wedge b_0$ (AND)	0.213
9	1	0	0	1	$b_1 = b_0$ (XNOR)	0.092
10	1	0	1	0	b_0 (projection)	0.040
11	1	0	1	1	$b_1 \to b_0$ (IF THEN)	0.010
12	1	1	0	0	b_1 (projection)	0.010
13	1	1	0	1	$b_0 \to b_1$ (IF THEN)	0.030
14	1	1	1	0	$b_1 \vee b_0$ (OR)	0.010
15	1	1	1	1	TRUE (constant 1)	0.010

Table 4: The guiding example: all 16 maps of the $n = 2$, $m = 1$ hypothesis space, with a contrived prior ($\sum_j p_j = 1.000$).

2.1 The answer matrix and the update rule

The answer matrix is defined by the same rule as before: the total belief in the maps that would answer a to q:

$$M_{a,q} \;=\; \sum_{j\,:\,\phi_j(q)=a} p_j, \tag{25}$$

now an $A \times Q$ matrix, still with each column a probability distribution over the possible answers: column-stochastic.

It is worth taking a moment to describe the constitution of this matrix. Because of the enumeration of j (the index *is* the truth table) the answer matrix has a natural structure: it is the contraction of the belief vector p with a fixed tensor with $\{0,1\}$ entries E, summed along the map index,

$$M_{a,q} \;=\; \sum_{j=0}^{N-1} E_{a,q,j}\, p_j, \qquad E_{a,q,j} \;:=\; \delta\big(\phi_j(q),\, a\big) \;=\; \delta\Big(\big\lfloor j \cdot 2^{-qm} \big\rfloor \bmod A,\; a\Big). \tag{26}$$

The second form is pure arithmetic on the index: by the numeral convention, $\phi_j(q)$ is nothing but the q-th digit of j in base A. For $m = 1$ this reduces to: bit q of j equals a. This means E never needs to be stored, membership of j in the entry $M_{a,q}$ is read directly off the digits of j. This is a good thing, E is astronomically large for even moderate m and n. The tensor E is moreover sparse and perfectly regular: each question-slice (q fixed) holds exactly N nonzero entries: every map answers exactly one a divided evenly over the A answer rows, N/A apiece. Within a slice the nonzero row follows the odometer pattern of a numeral system: runs of A^q consecutive j share a digit, cycling through all A rows with period A^{q+1}.

The odometer pattern gives every entry of M the same compact closed form: for $m = 1$, bit q of j equals 0 exactly when $j \bmod 2^{q+1} < 2^q$, and equals 1 otherwise.

EXAMPLE 8

Each entry of M collects the eight maps from two to one bit of Table 4 whose truth tables carry the right digit. Filling in the entries - note the period halves column by column. The outer columns are written in their simplified forms, $j \bmod 16 < 8$ being simply $j < 8$ and $j \bmod 2 < 1$ simply "j even".

$$M \;=\; \begin{array}{c|cccc} \mathbf{a\backslash q} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} \\ \hline \mathbf{0} & \sum\limits_{j<8} p_j & \sum\limits_{j \bmod 8<4} p_j & \sum\limits_{j \bmod 4<2} p_j & \sum\limits_{j \text{ even}} p_j \\ \mathbf{1} & \sum\limits_{j\geq 8} p_j & \sum\limits_{j \bmod 8\geq 4} p_j & \sum\limits_{j \bmod 4\geq 2} p_j & \sum\limits_{j \text{ odd}} p_j \end{array}$$
$$=\; \begin{array}{c|cccc} \mathbf{a\backslash q} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} \\ \hline \mathbf{0} & 0.585 & 0.779 & 0.890 & 0.818 \\ \mathbf{1} & 0.415 & 0.221 & 0.110 & 0.182 \end{array}\,. \tag{27}$$

The update rule also carries over verbatim. When question q is asked and the answer $a = \psi(q)$ comes back, the process of elimination reads, in one formula,

$$p'_j \;=\; \frac{\delta\big(\phi_j(q),\, a\big)\, p_j}{M_{a,q}} \;=\; \frac{E_{a,q,j}\, p_j}{M_{a,q}}. \tag{28}$$

The Kronecker delta performs the elimination. Only maps that agree with $\psi(q) = a$ survive and the division renormalizes the survivors, whose total prior mass is exactly $M_{a,q}$ by equation (25). Note that one and the same number prices the answer and funds the update:

the event arrives with the probability $M_{a,q}$, costs Werner $-\log_2 M_{a,q}$ bits of surprisal, and its belief mass becomes the normalization. Every update shrinks the support by A^{-1}, radically thinning, especially for large m. The survivors live on a block-wave in j-space. The map determines the offset and frequency. The total set of survivors depends on the set of asked questions, those that fell through the slits, not the order: updating commutes. This is illustrated in the example below.

EXAMPLE 9

Figure 3 makes the process visual for $n = m = 2$: a schematic prior $\{p_j\}$ over $N = 4^4 = 256$ maps, drawn as vertical lines with thickness representing weight. Each answer stops 3 out of every $A = 4$ surviving lines: maps whose truth table carry a wrong digit at the asked question. Renormalization inflates the weight of the rest, so the total probability is always unit. Four questions cut $256 \to 64 \to 16 \to 4 \to 1$, and the last line standing is the truth. Note the regularity of the cull: each answer keeps a single residue class of j, so the cadence of the survivors is regular.

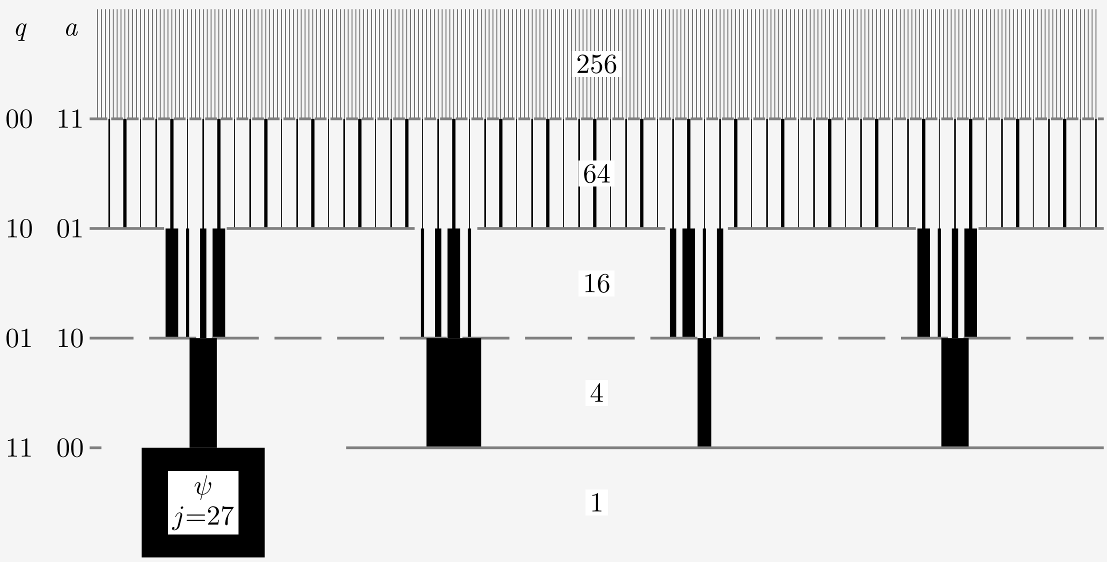


Figure 3: The update rule as a cull, for $n = m = 2$. All 256 candidate maps start as vertical lines under a schematic prior. The truth is the bitwise negation $\psi(q) = \neg q$ ($j = 27$, '0123' base 4.), and the four questions are asked in the order $00, 10, 01, 11$ ('0, 2, 1, 3' base 4). Each answer terminates the 3/4 of surviving maps whose truth-table digit at the asked question disagrees. Every survivor's thickness, proportional to its weight, is rescaled by the renormalization factor of that update, preserving their ratios. After four questions a single map remains. Interestingly, the first cull's survivors are evenly spaced because it's the 'smallest' question. After the three smallest q, even spacing returns.

2.2 Entropy bookkeeping

The total entropy of the table is extensive over the columns, exactly as in equation (13):

$$H(M) = \sum_{q\in\mathcal{Q}} H\big(M_{\cdot,q}\big) = -\sum_{q\in\mathcal{Q}}\sum_{a\in\mathcal{A}} M_{a,q}\log_2 M_{a,q} \leq mQ, \tag{29}$$

each column carrying at most m bits (a uniform spread over its A answers). The ceiling mQ is the size of the entire truth table in bits, and it is attained precisely (but not exclusively) by the uniform prior: maximal ignorance prices every answer at full cost.

The belief itself carries an entropy as well, and we give its gap to the table a name:

$$H(p) \;=\; -\sum_{j=0}^{N-1} p_j \log_2 p_j \le mQ, \tag{30}$$

$$C \;:=\; H(M) - H(p) \;\geq\; 0. \tag{31}$$

$H(p)$, Werner's uncertainty about the *identity* of the truth, obeys the same ceiling: a distribution over N maps has entropy at most $\log_2 N = mQ$, attained again by the uniform prior. The information in the prior is not directly useful for prediction, but it is good to track.

Why is $C > 0$? The answers jointly determine the map, so $H(p)$ is the *joint* entropy of all Q answers, while $H(M)$ sums their *marginal* entropies and a joint entropy never exceeds the sum of its marginals. Hence $H(p) \le H(M)$, and the gap C, the **total correlation** of the answers, is nonnegative: knowledge stored in correlations *between* answers, invisible to every single column of M.

We choose this convention for $H(M)$ because *in the ignorant case*

1. $H(M)$ is the total amount of information we must collect from questions to settle the map
2. $H(M) = H(p)$, so the ignorance agrees in both bases, without correlations.

Therefore, the scaling is sensible. How these quantities respond to an *intelligent* prior, is the main question of this work.

2.3 Playing the guiding example

Let us watch these ledgers in motion by playing the guiding example through all four questions. Werner can compute two forecasts for every unasked question, in advance, from M alone: the expected surprisal $\langle s\rangle$, which by equation (14) is the entropy of that question's column, and the expected drop $\langle \Delta H(M)\rangle$ of the *whole* table if that question is asked next. This is the average over possible a, weighted by their probability (according to M):

$$\langle s\rangle(q) \;=\; \sum_a M_{a,q}\left(-\log_2 M_{a,q}\right) = H(M_{\cdot,q}), \tag{32}$$

$$\langle \Delta H(M)\rangle(q) \;=\; \sum_a M_{a,q}\left[H(M) - H\big(M^{(q,a)}\big)\right], \tag{33}$$

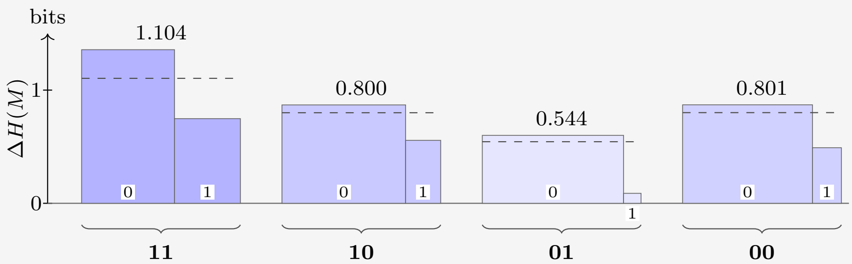
where $M^{(q,a)}$ is the table after the update rule (28) processes answer a to question q. We know the entropy of column q will drop to zero, ditching $\langle s\rangle(q)$ entropy from M. But other columns will also respond. The excess of the second over the first is the expected transfer to unasked columns, and it can only be paid out of correlations. We append both forecasts as extra rows under M.

EXAMPLE 10

Werner's belief is the prior of table 4 as before. He plays greedily: always asking the question with the largest forecast reduction in matrix entropy. The truth is AND ($\psi = \phi_8$).

Question 1. The starting state:

a\q	**11**	**10**	**01**	**00**
0	0.585	0.779	0.890	0.818
1	0.415	0.221	0.110	0.182
$\langle s \rangle$	0.979	0.762	0.500	0.684
$\langle \Delta H(M) \rangle$	1.104	0.800	0.544	0.801



To see where the forecast entries come from, take $q = 11$: the answer $a = 0$ arrives with probability $M_{0,11} = 0.585$ and would drop $H(M)$ from 2.925 to 1.569; the answer $a = 1$ arrives with probability 0.415 and drops it only to 2.178, so

$$\langle s \rangle(11) = 0.585\,(0.773) + 0.415\,(1.269) = 0.979,$$
$$\langle \Delta H(M) \rangle(11) = 0.585\,(1.357) + 0.415\,(0.748) = 1.104.$$

EXAMPLE 11

Both forecast rows point to $q = 11$: it is the closest thing to a coin on offer ($\langle s \rangle = 0.979$) and moves the whole table most, by a wide margin. Part of the $\Delta H(M)$ is the $\langle s \rangle$. Werner asks it, and nature answers $a = 1$ (probability 0.415, surprisal $s_1 = 1.269$). Eight maps die:

j	$\phi_j(11)$	name	p_j	p'_j
~~0~~	~~0~~	~~FALSE (constant 0)~~	~~0.394~~	~~—~~
~~1~~	~~0~~	~~$\neg(b_1 \vee b_0)$ (NOR)~~	~~0.010~~	~~—~~
~~2~~	~~0~~	~~$\neg b_1 \wedge b_0$~~	~~0.010~~	~~—~~
~~3~~	~~0~~	~~$\neg b_1$~~	~~0.010~~	~~—~~
~~4~~	~~0~~	~~$b_1 \wedge \neg b_0$~~	~~0.131~~	~~—~~
~~5~~	~~0~~	~~$\neg b_0$~~	~~0.010~~	~~—~~
~~6~~	~~0~~	~~$b_1 \oplus b_0$ (XOR)~~	~~0.010~~	~~—~~
~~7~~	~~0~~	~~$\neg(b_1 \wedge b_0)$ (NAND)~~	~~0.010~~	~~—~~
8	1	$b_1 \wedge b_0$ (AND)	0.213	0.513
9	1	$b_1 = b_0$ (XNOR)	0.092	0.222
10	1	b_0 (projection)	0.040	0.096
11	1	$b_1 \to b_0$ (IF THEN)	0.010	0.024
12	1	b_1 (projection)	0.010	0.024
13	1	$b_0 \to b_1$ (IF THEN)	0.030	0.072
14	1	$b_1 \vee b_0$ (OR)	0.010	0.024
15	1	TRUE (constant 1)	0.010	0.024

The ledger:

$$H(M)\colon 2.925 \to 2.178, \qquad H(p)\colon 2.707 \to 2.093, \qquad C\colon 0.218 \to 0.085.$$

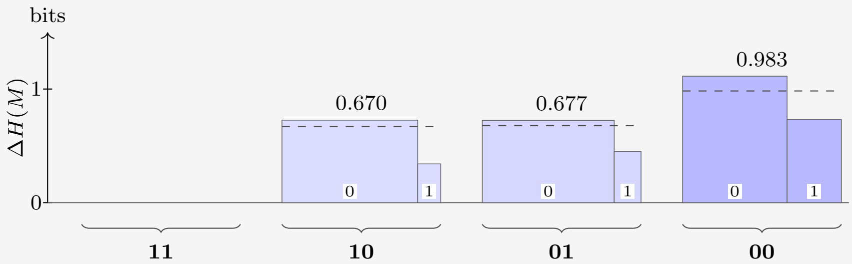
Werner received 1.269 bits but destroyed only 0.748 bits of table. Correlations point the right way only on average, not on every draw. In this case, Werner got the much less informative branch, and paid the premium price for it. The destroyed-per-received ratio of this round is 0.589: well below 1, anti-leverage.

Question 1 is a reminder that the forecast rows are *expectations*: they hold on average over many repetitions of the experiment, and only if the world really does draw its truth ψ from Werner's prior. On any single run the realized numbers are free to deviate.

EXAMPLE 12

Question 2. The state after the collapse, with fresh forecasts:

a\q	**11**	**10**	**01**	**00**
0	0	0.855	0.831	0.658
1	1	0.145	0.169	0.342
$\langle s \rangle$	—	0.596	0.655	0.927
$\langle \Delta H(M) \rangle$	—	0.670	0.677	0.983



Greedy Werner picks $q = 00$. Nature answers $a = 0$, which is the answer Werner favored (probability 0.658), so the surprisal is only $s_2 = 0.604$ bits. Four of the eight die:

j	$\phi_j(00)$	name	p_j	p'_j
8	0	$b_1 \wedge b_0$ (AND)	0.513	0.780
~~9~~	~~1~~	~~$b_1 = b_0$ (XNOR)~~	~~0.222~~	~~—~~
10	0	b_0 (projection)	0.096	0.147
~~11~~	~~1~~	~~$b_1 \to b_0$ (IF THEN)~~	~~0.024~~	~~—~~
12	0	b_1 (projection)	0.024	0.037
~~13~~	~~1~~	~~$b_0 \to b_1$ (IF THEN)~~	~~0.072~~	~~—~~
14	0	$b_1 \vee b_0$ (OR)	0.024	0.037
~~15~~	~~1~~	~~TRUE (constant 1)~~	~~0.024~~	~~—~~

The ledger:

$$H(M)\colon 2.178 \to 1.065, \qquad H(p)\colon 2.093 \to 1.035, \qquad C\colon 0.085 \to 0.030.$$

This time 1.113 bits are destroyed for 0.604 received. Where did the extra 0.509 come from? Partly from the favored answer arriving - the belief dropped 1.058 against only 0.604 paid - and partly from the store: C fell by 0.055. Step ratio 1.842. Cumulatively, however, 1.860 destroyed per 1.873 received gives a leverage of 0.993 after 2 steps: still a hair below 1. Even a strong second round has not quite repaid the expensive shock of question 1.

Why track C, when its step-by-step changes visibly do not match the amounts deduced? Because on a single realized run the deduced part has *two* sources, and C is the only quantity that separates them. Writing Δ for the realized drops in one step,

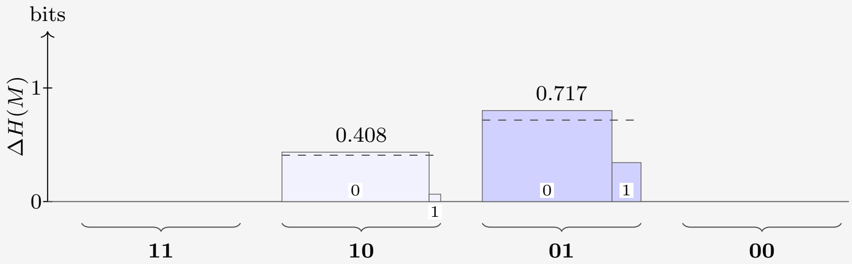
$$\underbrace{\Delta H(M) - s}_{\text{deduced}} = \underbrace{\big[\Delta H(p) - s\big]}_{\text{windfall}} + \underbrace{\big[-\Delta C\big]}_{\text{tower}}, \tag{34}$$

an identity, since $H(M) = H(p) + C$ at every stage. The windfall is luck: the realized surprisal undershooting (or overshooting) the realized drop in belief entropy. Windfall is zero in expectation. The tower term is structure: a genuine withdrawal from the correlation store, and the channel with a global guarantee. Over a complete run its withdrawals sum to exactly the initial C, while the windfalls average away.

EXAMPLE 13

Question 3. Two questions remain:

a\q	**11**	**10**	**01**	**00**
0	0	0.927	0.817	1
1	1	0.073	0.183	0
$\langle s \rangle$	—	0.378	0.687	—
$\langle \Delta H(M) \rangle$	—	0.408	0.717	—



Greedy Werner picks $q = 01$; nature answers $a = 0$ (probability 0.817, surprisal $s_3 = 0.292$). Two die:

j	$\phi_j(01)$	name	p_j	p'_j	
8	0	$b_1 \wedge b_0$ (AND)	0.780	0.955	
~~10~~	~~1~~	~~b_0 (projection)~~	~~0.147~~	—	
12	0	b_1 (projection)	0.037	0.045	
~~14~~	~~1~~	~~$b_1 \vee b_0$ (OR)~~	~~0.037~~	—	

The ledger:

$$H(M)\colon 1.065 \to 0.264, \qquad H(p)\colon 1.035 \to 0.264, \qquad C\colon 0.030 \to 0.$$

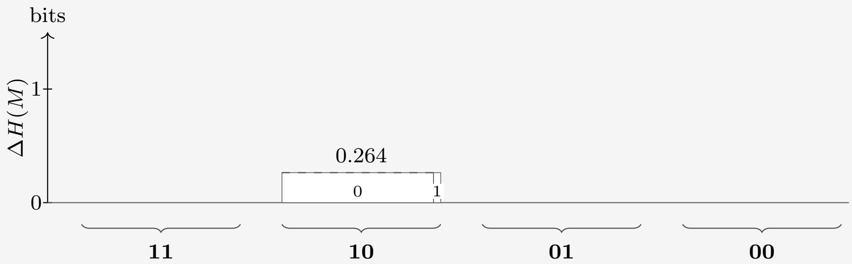
Again a bargain: 0.801 destroyed for a mere 0.292 received, and the cumulative ratio finally clears unity, at 1.229. And note that C has dropped to zero.

C measures correlations that are not visible from M. If there's only one unknown column, there's nothing to infer. Put another way, usually, the binary maps are an overcomplete basis for M, because their number $N \gg \dim(M) = 2^{m+n}$ Expressing M as a linear combination of maps drops knowledge. That's not the case when the number of free parameters has been reduced by learning to 1 each.

EXAMPLE 14

Question 4. One question, one heavily loaded coin:

a\q	**11**	**10**	**01**	**00**
0	0	0.955	1	1
1	1	0.045	0	0
$\langle s \rangle$	—	0.264	—	—
$\langle \Delta H(M) \rangle$	—	0.264	—	—



The two forecasts now coincide: with $C = 0$ there is no transfer left to hope for, and the last question can only be observed, not leveraged. Nature answers $a = 0$ (probability 0.955, surprisal $s_4 = 0.066$):

j	$\phi_j(10)$	name	p_j	p'_j
8	0	$b_1 \wedge b_0$ (AND)	0.955	1.000
~~12~~	~~1~~	~~b_1 (projection)~~	~~0.045~~	~~—~~

The ledger closes:

$$H(M)\colon 0.264 \to 0, \qquad H(p)\colon 0.264 \to 0, \qquad C\colon 0 \to 0.$$

The truth is AND. The step destroyed 0.264 for 0.066 received (ratio 3.990) — not deduction this time, but luck: the destroyed amount was fixed in advance (the column's full entropy), while the realized surprisal undershot it because the likelier answer arrived. In the split of equation (34): pure windfall, tower 0.

EXAMPLE 15

Cumulatively: 2.925 bits of table destroyed, all of it, for 2.231 bits of surprisal received: a run ratio of 1.311. Most of the realized 1.311 was luck, not intelligence. Figure 4 stacks the bits after each question: what uncertainty remains, plus what was received. Once it dips below the starting entropy, our prior has done net work.

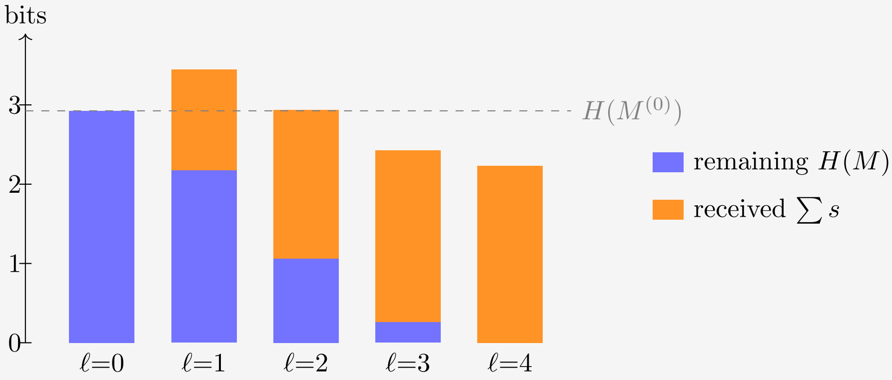


Figure 4: The realized run through the guiding example: remaining table entropy plus cumulative surprisal received, after each question, against the initial total (dashed). The gap between each stack and the line is the cumulative deduced part. At $\ell = 1$ the stack tops out 0.521 bits *above* the line — the negative deduction of question 1, plainly visible. At $\ell = 2$ the stack still grazes the line from above (0.013 bits); only from $\ell = 3$ on do the stacks fall short of it, as genuine deduction overtakes the early loss.

2.4 Leverage

If the prior is good, meaning it actually models reality (the prevalence of maps), we expect graphs such as figure 4 to drop as questions come in. The ratio of table entropy destroyed per bit of surprisal received deserves a name.

The **leverage**. Run the protocol for ℓ rounds: questions $q_1, \ldots, q_\ell$ are asked, the answers $a_k = \psi(q_k)$ are each processed by the update rule (28), producing the tables $M^{(k)}$, and round k costs the surprisal $s_k = -\log_2 M^{(k-1)}_{a_k, q_k}$. The cumulative leverage after ℓ rounds is the table entropy destroyed per bit of surprisal received:

$$L_\ell \;:=\; \frac{H\big(M^{(0)}\big) - H\big(M^{(\ell)}\big)}{\sum_{k=1}^{\ell} s_k}\,. \tag{35}$$

A leverage of 1 is the uniform-prior baseline: every bit of table entropy destroyed was paid for at face value, one bit of surprisal each. Anything above 1 is deduction: table entropy removed without ever being observed.

2.5 Correlators

This section can be skipped on a first reading: the later chapters do not depend on it.

What kind of operator is the update rule (28)? In map space it is as simple as an operator gets: multiplication by a diagonal 0/1 mask, then renormalization. But there is a change of basis in which the same operation becomes a *convolution* with a two-term kernel, and in which the correlation store C stops being a single number and becomes visible coordinate by coordinate. The main advantage is that the effect of each update is well tabulated beforehand into a small number of targeted parameters.

This section builds that basis for $m = 1$; the final paragraph deals with general m.

First, re-encode each answer as a spin,

$$\sigma_q(\phi_j) \;:=\; (-1)^{\phi_j(q)} \;\in\; \{+1, -1\}, \tag{36}$$

so answer $0 \mapsto +1$ and $1 \mapsto -1$. A pure relabeling, but it buys the one algebraic fact everything below turns on: $\sigma_q^2 = 1$.

The **Walsh–Hadamard transform** is built from a single matrix. For one bit,

$$W \;=\; \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad W_{c,b} \;=\; (-1)^{c\,b}, \quad c, b \in \{0, 1\}. \tag{37}$$

EXAMPLE 16

Acting on a one-bit distribution,

$$W \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} \;=\; \begin{pmatrix} p_0 + p_1 \\ p_0 - p_1 \end{pmatrix} \;=\; \begin{pmatrix} 1 \\ \langle \sigma \rangle \end{pmatrix} :$$

row $c = 0$ computes the normalization, row $c = 1$ the spin bias. Distribution in, correlators out.

The normalization riding along is a redundancy. The original distribution also uses one variable more than its degrees of freedom.

EXAMPLE 17

Tensoring two copies (the $n = 1$ case) does the same for two bits at once:

$$W \otimes W \;=\; \begin{pmatrix} W & W \\ W & -W \end{pmatrix} \;=\; \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \quad \begin{array}{l} \leftarrow c = 00\text{: sums to } 1 \\ \leftarrow c = 01\text{: } \langle \sigma_0 \rangle \\ \leftarrow c = 10\text{: } \langle \sigma_1 \rangle \\ \leftarrow c = 11\text{: } \langle \sigma_1 \sigma_0 \rangle \end{array} \tag{38}$$

The Kronecker product multiplies signs, so for any number l of tensor factors the entry at row c, column j is

$$\left(W^{\otimes l}\right)_{c,j} \;=\; \prod_{i=0}^{l-1} (-1)^{c_i\, j_i} \;=\; (-1)^{\sum_i c_i\, j_i}, \tag{39}$$

with c_i, j_i the i-th bits of the row and column indices; every further tensor factor appends one more bit.

For $m = 1$ a map *is* its truth table of Q bits (the numeral j) so Werner's belief is a vector of length $N = 2^Q$, and one transform computes every correlator at once:

$$\chi \;=\; W^{\otimes Q} p, \qquad \chi_{\mathcal{S}} \;=\; \Big\langle \prod_{q \in \mathcal{S}} \sigma_q \Big\rangle \;=\; \sum_{j=0}^{N-1} (-1)^{\sum_{q \in \mathcal{S}} \phi_j(q)}\, p_j. \tag{40}$$

Each row computes the **correlator** of one subset $\mathcal{S} \subseteq \mathcal{Q}$ of questions. In the subscript we may encode the set as its *mask*: written in the same descending question order as everywhere else, bit q of the subscript records whether $q \in \mathcal{S}$. The row index of the matrix, read as a numeral, is the mask.

EXAMPLE 18

For $n = 2$, the question order is $(11, 10, 01, 00)$ so χ_{0110} is the correlator of $\mathcal{S} = \{10, 01\}$.

the sixteen rows of $W^{\otimes 4}$ probe the sixteen subsets of $Q = \{11, 10, 01, 00\}$. A few entries of the dictionary:

mask	set $\mathcal{S}$	correlator
0000	$\emptyset$	$\chi_{0000} = 1$ (normalization)
0001	$\{00\}$	$\chi_{0001} = \langle \sigma_{00} \rangle$
1000	$\{11\}$	$\chi_{1000} = \langle \sigma_{11} \rangle$
0110	$\{10, 01\}$	$\chi_{0110} = \langle \sigma_{10}\, \sigma_{01} \rangle$
1111	$\{11, 10, 01, 00\}$	$\chi_{1111} = \langle \sigma_{11}\, \sigma_{10}\, \sigma_{01}\, \sigma_{00} \rangle$, the full parity

Sort the N correlators by $|\mathcal{S}|$, the number of questions probed, into *shells*. Shell 0 is the normalization, $\chi_\emptyset = 1$, riding along as a correlator. Shell 1 is M in different clothes:

$$\chi_{\{q\}} \;=\; M_{0,q} - M_{1,q}, \qquad M_{a,q} \;=\; \tfrac{1}{2}\big(1 + (-1)^a\, \chi_{\{q\}}\big), \tag{41}$$

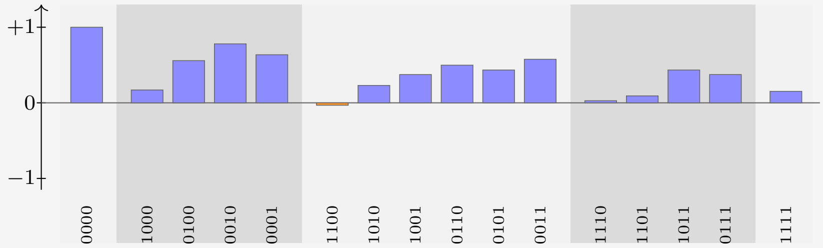
the bias of column q. For $m = 1$, we elegantly see the first shell correlators are the distance from fair coins.

Shell 2 holds the agreement biases of pairs of questions, knowledge M cannot see: two priors can share every column of M and differ here. Higher shells store subtler and subtler parity knowledge. Nothing is lost in the sorting: $W^2 = 2I$, so $(W^{\otimes Q})^2 = NI$. The transform is its own inverse up to a factor N, and knowing every correlator is knowing the belief. The ledger also gives C a signature: a product prior ($C = 0$) factorizes every correlator, $\chi_{\mathcal{S}} = \prod_{q \in \mathcal{S}} \chi_{\{q\}}$, so its upper shells repeat what shell 1 already said; correlated knowledge is precisely *non-factorizing* correlators.

EXAMPLE 19

The prior of Table 4 transforms to the ledger

shell	correlators
0	$\chi_{0000} = 1$
1	$\chi_{1000} = +0.170$, $\chi_{0100} = +0.558$, $\chi_{0010} = +0.780$, $\chi_{0001} = +0.636$
2	$\chi_{1100} = -0.032$, $\chi_{1010} = +0.230$, $\chi_{1001} = +0.374$, $\chi_{0110} = +0.498$, $\chi_{0101} = +0.434$, $\chi_{0011} = +0.576$
3	$\chi_{1110} = +0.028$, $\chi_{1101} = +0.092$, $\chi_{1011} = +0.434$, $\chi_{0111} = +0.374$
4	$\chi_{1111} = +0.152$



Shell 1 reproduces the columns of equation (27): $(1 + 0.170)/2 = 0.585 = M_{0,11}$, and likewise 0.779, 0.890, 0.818. The negative entry χ_{1100} says questions 11 and 10 disagree slightly more often than they agree; every other pair leans toward agreement. This is a reflection of the strong coherent 0-preference encoded in the prior. Answering 1 would give negative 1-shells.

Now look at the *rows* of $W^{\otimes Q}$ as functions of j. They are **block waves**. The observation mask is built of the same material:

$$\delta\big(\phi_j(q), a\big) \;=\; \frac{1}{2}\Big(1 + (-1)^{a+\phi_j(q)}\Big)\,, \tag{42}$$

a 0/1 block wave: one constant plus one Hadamard row. And multiplication and convolution swap under this transform exactly as under Fourier: multiplying two functions of j pointwise convolves their ledgers. And the observation mask's ledger has only two nonzero entries - at $\emptyset$ and at $\{q\}$ - so this convolution is no great sum: it combines each correlator with exactly one other.

The update rule, carried out entirely in the ledger, is therefore that two-term combination, followed by renormalization by the new shell 0. In index notation, learning $a = \psi(q)$ sends every set $\mathcal{S}$ to

$$\chi'_{\mathcal{S}} \;=\; \frac{\chi_{\mathcal{S}} + (-1)^a\,\chi_{\mathcal{S}\triangle\{q\}}}{1 + (-1)^a\,\chi_{\{q\}}}, \tag{43}$$

where $\mathcal{S}\triangle\{q\}$ is $\mathcal{S}$ with the membership of q toggled: q is added if absent, removed if present (on masks, bit q is flipped). The numerator is the monomial algebra of $\sigma_q^2 = 1$: multiplying $\prod_{r\in\mathcal{S}} \sigma_r$ by σ_q toggles q's membership of the product, so every correlator is averaged with its **partner**: the correlator differing by q, signed by the answer. The denominator is the $\mathcal{S} = \emptyset$ instance of the numerator, and by equation (41) it equals $2M_{a,q}$: the same number that priced the answer in equation (28) renormalizes the ledger.

EXAMPLE 20

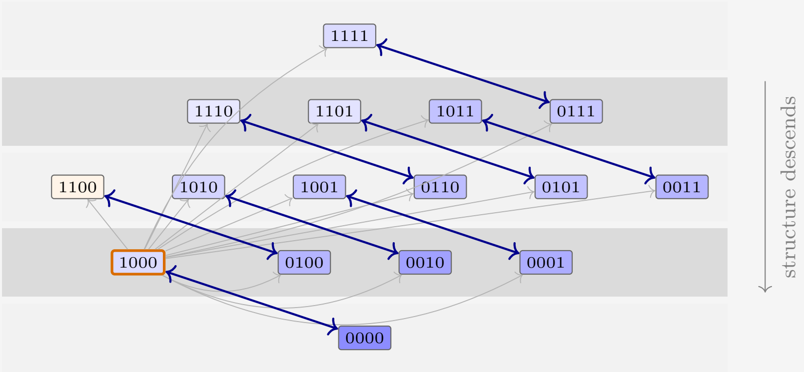


Figure 5: The update rule in the ledger, drawn for the guiding example's first question ($q = 11$). Every set is paired with the partner whose membership of 11 is toggled (arrows); each pair is averaged with the answer's sign and renormalized, per equation (43). The pair $\emptyset \leftrightarrow \{11\}$ at the orange-bordered asked set fixes the normalization and saturates the asked bias; each blue double arrow averages a pair into both of its members, and the net flow of structure is one shell down, toward M. The thin gray arrows from the asked q to every other pill remind that this sets the renormalization. The fill encodes each prior value to scale: darker is larger, orange is negative.

Read formula (43) by cases:

- $\mathcal{S} = \emptyset$: partner $\{q\}$, giving $(1 + (-1)^a \chi_{\{q\}})/(1 + (-1)^a \chi_{\{q\}}) = 1$. Normalization is preserved.
- $\mathcal{S} = \{q\}$, the asked question: partner $\emptyset$, giving $(\chi_{\{q\}} + (-1)^a)/(1 + (-1)^a \chi_{\{q\}}) = (-1)^a$. The bias saturates to certainty — the one-hot collapse of column q, derived rather than decreed.
- $\mathcal{S} = \{q'\}$, an unasked question: the partner is the pair $\{q', q\}$, and one line of algebra puts the shift in terms of the *connected* correlator:

$$\chi'_{\{q'\}} - \chi_{\{q'\}} = \frac{(-1)^a \big(\chi_{\{q',q\}} - \chi_{\{q'\}}\, \chi_{\{q\}}\big)}{2M_{a,q}} = \frac{(-1)^a \operatorname{Cov}\big(\sigma_{q'}, \sigma_q\big)}{2M_{a,q}}. \tag{44}$$

  Unasked columns move in proportion to their covariance with the asked question: knowledge stored one shell up descends into the predictions. A product prior has zero covariance everywhere, so its unasked columns are frozen — the "$C = 0$ means no transfer" fact, now a one-line consequence.
- Higher shells: the same cascade. Each update marries shell-k sets containing q to shell-$(k-1)$ sets without it; whatever structure was stored jointly with q moves one shell down, toward M.

EXAMPLE 21

Question 1 of the walkthrough asked $q = 11$ and received $a = 1$: sign $(-1)^a = -1$, toggled set $\{11\}$, mask 1000. Equation (43) maps the ledger of the previous box to

shell	updated correlators
0	$\chi'_{0000} = 1$
1	$\chi'_{1000} = -\mathbf{1}$, $\chi'_{0100} = +0.711$, $\chi'_{0010} = +0.663$, $\chi'_{0001} = +0.316$
2	$\chi'_{1100} = -0.711$, $\chi'_{1010} = -0.663$, $\chi'_{1001} = -0.316$, $\chi'_{0110} = +0.566$, $\chi'_{0101} = +0.412$, $\chi'_{0011} = +0.171$
3	$\chi'_{1110} = -0.566$, $\chi'_{1101} = -0.412$, $\chi'_{1011} = -0.171$, $\chi'_{0111} = +0.267$
4	$\chi'_{1111} = -0.267$

The asked bias saturates: $\chi'_{1000} = (0.170 - 1)/(1 - 0.170) = -1$. The other singletons move by equation (44): for $q' = 10$ the covariance is $-0.032 - (0.558)(0.170) = -0.127$, so the shift is $(-1)(-0.127)/(2 \times 0.415) = +0.153$, landing on $+0.711$ — and indeed $(1 + 0.711)/2 = 0.855$, the column $M_{0,10}$ of the Question 2 box. Note also that every mask containing the asked bit is now just $(-1)^a$ times its partner ($\chi'_{1010} = -\chi'_{0010}$, etc.): with σ_{11} pinned, no correlator involving question 11 carries independent knowledge anymore.

EXAMPLE 22

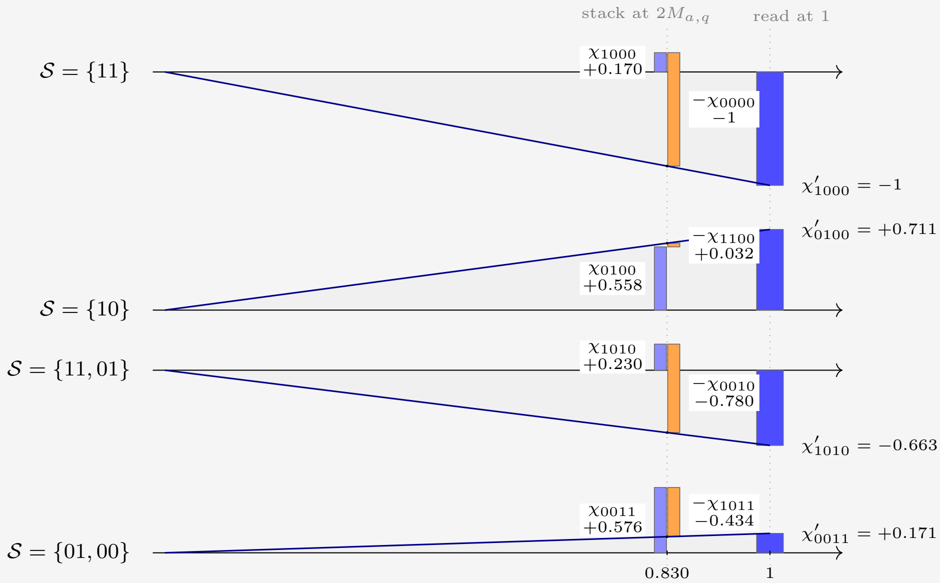


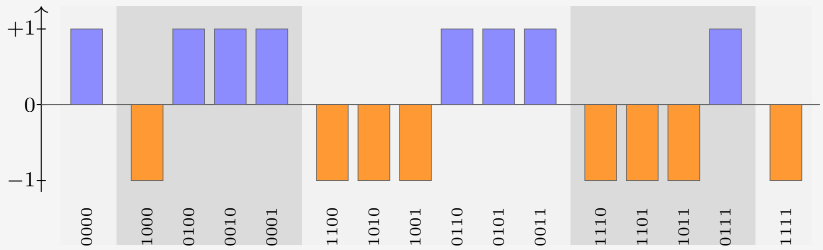
Figure 6: The update rule of equation (43), drawn for the guiding example's first update ($q = 11$, $a = 1$, so $(-1)^a = -1$ and toggled bit 1000). One row per set $\mathcal{S}$. At $x = 2M_{a,q} = 1 + (-1)^a\chi_{1000} = 0.830$ stands the numerator as a waterfall: the bar of $\chi_{\mathcal{S}}$ (blue), and beside it the partner's signed contribution $(-1)^a\chi_{\mathcal{S}\triangle\{q\}}$ (orange), hanging from the level where the first bar ends. The two subscripts differ by the flipped bit. The ray from the origin through the waterfall's endpoint rescales it, by similar triangles, to $x = 1$, where its height is the updated correlator $\chi'_{\mathcal{S}}$.

As Werner moves through the questions, more correlators settle to ± 1. The 1-shell models the belief of the bias of each question. The 2-shell answers questions such as 'If the answer to $\psi(10) = 0$, then $\psi(01) = 1$'. And the higher orders are even more conditional, measuring belief about such statements, in the case of certain answers, etc. As we learn more, the conditions become satisfied or not, and the information becomes less nested.

EXAMPLE 23

The remaining questions of the run (00, 01, 10, all answered 0, so with sign +1) each saturate their own bit the same way, and since XOR toggles commute, the order does not matter — the commutativity of updating, seen again. After all four, the ledger is pure signs, $\chi_{\mathcal{S}} = (-1)^{|\mathcal{S}\cap\{11\}|}$, which is -1 on every set containing 11, $+1$ on the rest: the transform of the point mass on $\psi = \phi_8$:

shell	final correlators
0	$\chi_{0000} = 1$
1	$\chi_{1000} = -1,\ \ \chi_{0100} = +1,\ \ \chi_{0010} = +1,\ \ \chi_{0001} = +1$
2	$\chi_{1100} = -1,\ \ \chi_{1010} = -1,\ \ \chi_{1001} = -1,$ $\chi_{0110} = +1,\ \ \chi_{0101} = +1,\ \ \chi_{0011} = +1$
3	$\chi_{1110} = -1,\ \ \chi_{1101} = -1,\ \ \chi_{1011} = -1,\ \ \chi_{0111} = +1$
4	$\chi_{1111} = -1$



2.5.1 General m

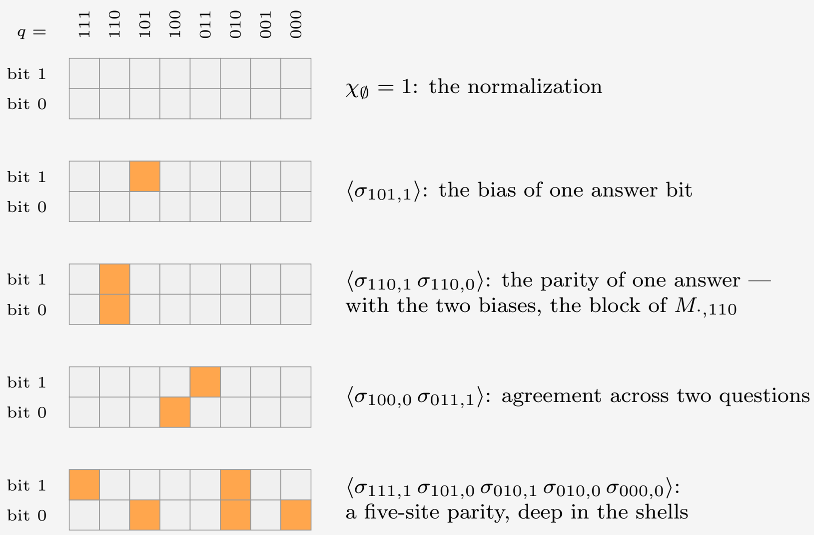
It turns out that taking $m > 1$ is more general but not much more instructive. With multiple output bits it is as if we have m a priori independent maps per question; they just happen always to be asked and answered simultaneously, in unison. Therefore we do not have to think about what one of them teaches us about another. It would mean we have more than 2 options every question, and therefore the factor by which every update thins the hypotheses is not 2 but A. The guiding example helps because the maps are represented as binary numbers, which are more familiar.

For completeness, the main formulas. The truth table is now mQ bits: one *site* (q, i) per bit i of each answer, each carrying a spin $\sigma_{q,i}(\phi_j) := (-1)^{\phi_j(q)_i}$, where $\phi_j(q)_i$ is the i-th bit of the answer. Correlators are indexed by sets $\mathcal{S}$ of sites: masks of mQ bits under the same numeral convention, m of them per question:

$$\chi_{\mathcal{S}} = \Big\langle \prod_{(q,i)\in\mathcal{S}} \sigma_{q,i} \Big\rangle = \sum_{j=0}^{N-1} (-1)^{\sum_{(q,i)\in\mathcal{S}} \phi_j(q)_i}\, p_j. \tag{45}$$

Example 24

For $n = 3$, $m = 2$ the sites form a 2×8 grid: one column per question, one row per bit of the answer. A set $\mathcal{S}$ is a pattern of hot cells in this grid, and $\chi_{\mathcal{S}}$ multiplies the corresponding spins:



Any of the $2^{16} = 65536$ hot/cold patterns is a correlator, and sorting by the number of hot cells gives the shells. The mask of $\mathcal{S}$ is the grid read as a 16-bit numeral.

Write $\mathcal{B}_q$ for the block of q's m sites, and $a \cdot \mathcal{T} := \sum_{i \in \mathcal{T}} a_i$ for the parity of the answer a on a subset $\mathcal{T} \subseteq \mathcal{B}_q$. The answer matrix is read off the correlators supported on one block. There are A of them per column:

$$M_{a,q} = \frac{1}{A} \sum_{\mathcal{T} \subseteq \mathcal{B}_q} (-1)^{a \cdot \mathcal{T}}\, \chi_{\mathcal{T}}. \tag{46}$$

The observation mask at (q, a) is a function of the m bits of one block, so its ledger has A nonzero entries, and hearing $a = \psi(q)$ combines every correlator with its A partners — one per toggle $\mathcal{T}$ within the asked block:

$$\chi'_{\mathcal{S}} = \frac{\sum_{\mathcal{T} \subseteq \mathcal{B}_q} (-1)^{a \cdot \mathcal{T}}\, \chi_{\mathcal{S} \triangle \mathcal{T}}}{A\, M_{a,q}}, \tag{47}$$

the denominator once more the $\mathcal{S} = \emptyset$ instance of the numerator. For $m = 1$ the three formulas collapse to equations (40), (41) and (43).

EXAMPLE 25

To draw the update, take a sparse prior at $n=3$, $m=2$: four maps, $\phi(q)=q \bmod 4$ (weight 0.45), constant 00 (0.35), "11 iff $q \geq 4$" (0.12), constant 11 (0.08). Werner asks $q=110$ and hears $a=11$: toggle signs $(+,-,-,+)$ on $(\emptyset, 110_0, 110_1, \text{both})$, where q_i denotes site (q,i), and denominator $AM_{11,110}=0.8$. The pairs of the $m=1$ figure become *quartets*:

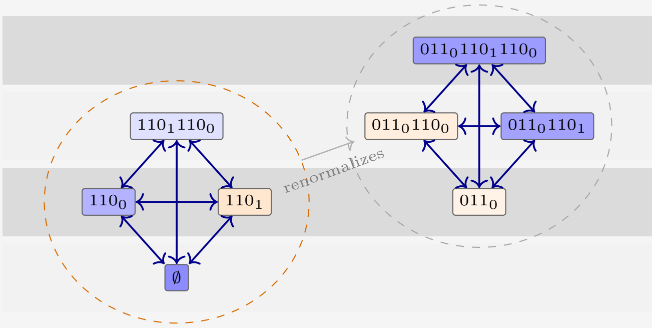


Figure 7: For $m=2$, toggling any subset of the asked block's two sites ties four correlators into a clique that averages jointly, per equation (47). The on-block quartet (orange hull) contains the normalization and the entire block of $M_{\cdot,110}$; its signed sum is the denominator that renormalizes every other quartet. Fills to scale for the sparse prior, as before.

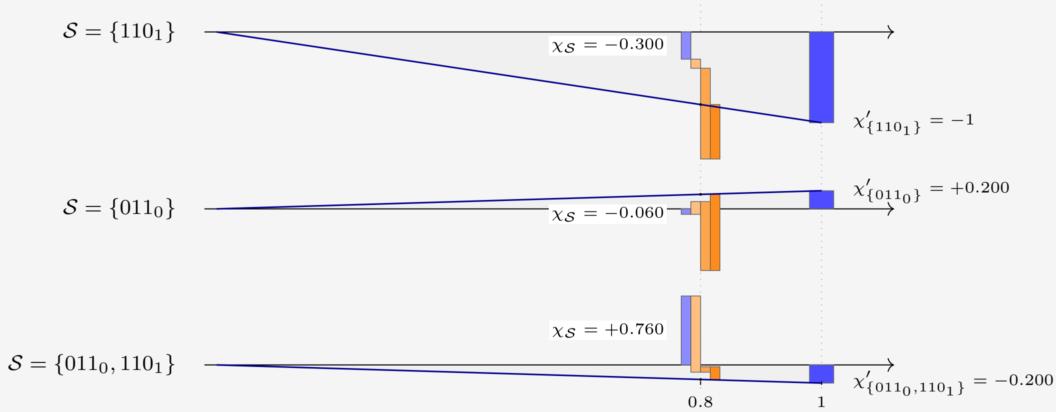


Figure 8: The $m=2$ update, three sets from the sparse prior. The waterfall now has four segments: $\chi_{\mathcal{S}}$ (blue) and its three signed quartet partners, in toggle order 110_0, 110_1, both (light to dark orange). The ray from the origin rescales the endpoint from $x=AM_{a,q}=0.8$ to $x=1$. The asked site saturates to -1; the foreign site 011_0 flips sign, $-0.060 \to +0.200$; and $\{011_0, 110_1\}$ lands on -0.200, the renormalized value $-\chi'_{\{011_0\}}$.

3 Expectations

While the games of the previous sections may be entertaining, they don't readily divulge any universal laws. When we average (integrate) out the question order, and the specific map drawn, many more robust quantities appear. This is a stepping stone to our terminus: the thermodynamic limit, where we expect questions to flow in and learning to happen on a continuum. Where such ensemble average statements are physically instructive and often quite descriptive of nature.

Let $\langle \ldots \rangle$ denote expectation over both

- the truth $\psi \sim \{p_j\}$,
- uniformly over the question order $\{q_k\}$.

We take the ratio of the expected entropy cleared from the table M to the surprisal received:

$$\langle L_\ell \rangle \;:=\; \frac{H\big(M^{(0)}\big) - \big\langle H\big(M^{(\ell)}\big)\big\rangle}{\big\langle \sum_{i=1}^{\ell} s_i \big\rangle}\,. \tag{48}$$

We did this implicitly in Section 1.2.1. There, the problem was actually symmetric in $q_0 \leftrightarrow q_1$. That is not the case in general.

As stated above, in order to progress, we will be interested in such averages. Both are natural. The question order is uniform, without replacement: Without control of the questions, Werner has no reason to expect one to arrive before another. And the truth is drawn from the prior itself: Werner scores his future against his own beliefs, which is the best he can do.

The bookkeeping, especially for questions asked, simplifies at once. Updating commutes, so after ℓ rounds the state depends only on the asked *set* $\mathcal{S}$, a uniform random subset of size ℓ. Let us say the answers received form a vector $y = \{y_i = \psi(q_i)\}$, for $q_i \in \mathcal{S}$. Each surprisal is paid at the posterior odds of its own round, $s_k = -\log_2 M^{(k-1)}_{y_k,q_k}$, and the posterior column is, by construction of the update rule, exactly the conditional probability of the next answer given everything seen so far: $M^{(i-1)}_{y_i,q_i} = \Pr(\psi(q_i) = y_i \mid \psi(q_1) = y_1, \ldots, \psi(q_{i-1}) = y_{i-1})$. The chain rule of probability multiplies these conditionals into the joint probability of the whole pattern, so the logs add and the history drops out:

$$\sum_{i=1}^{\ell} s_i \;=\; -\log_2 P_{\mathcal{S},y}, \qquad P_{\mathcal{S},y} \;:=\; \Pr\big(\psi(\mathcal{S}) = y\big) \;=\; \sum_{j\,:\,\phi_j(q_i)=y_i\ \forall q_i\in\mathcal{S}} p_j. \tag{49}$$

Conditioning on what came before is not an obstacle to the sum but its mechanism: it is precisely what makes the product of the round-by-round probabilities close into the block probability. The maps that answer the pattern y on $\mathcal{S}$ form a residue class, and $P_{\mathcal{S},y}$ is its mass. Averaging (49) over ψ gives the entropy of the block distribution $P_{\mathcal{S},\cdot}$: the distibution over such sets. Everything is generated by one sequence, the **mean block entropy** at size k:

$$G_k \;:=\; \binom{Q}{k}^{-1} \sum_{|\mathcal{S}|=k} H(\psi(\mathcal{S})) \;=\; -\binom{Q}{k}^{-1} \sum_{|\mathcal{S}|=k} \sum_{y} P_{\mathcal{S},y} \log_2 P_{\mathcal{S},y}, \tag{50}$$

the inner sum running over the A^k answer patterns on the block; nothing here is specific to $m = 1$. The endpoints are

$$G_0 \;=\; 0, \qquad G_Q \;=\; H(p). \tag{51}$$

G_1 is by definition the mean column entropy of the first question, so the prior table entropy is

$$H\big(M^{(0)}\big) \;=\; Q\, G_1. \tag{52}$$

EXAMPLE 26

What a block entropy is, on the guiding space ($n = 2$, $m = 1$). Take the block $\mathcal{S} = \{11, 10\}$, two questions. An answer pattern is $y = (y_{11}, y_{10})$, with four options, and by the numeral convention the maps answering each pattern are four consecutive rows of Table 4: the residue classes are read off the two leading truth-table digits.

y	maps j	names	$P_{\mathcal{S},y}$
$(0,0)$	0–3	FALSE, NOR, $\neg b_1 \wedge b_0$, $\neg b_1$	0.424
$(0,1)$	4–7	$b_1 \wedge \neg b_0$, $\neg b_0$, XOR, NAND	0.161
$(1,0)$	8–11	AND, XNOR, b_0, $b_1 \to b_0$	0.355
$(1,1)$	12–15	b_1, $b_0 \to b_1$, OR, TRUE	0.060

The four masses sum to one, and the block entropy is the Shannon entropy of this four-outcome distribution, $H(\psi(11,10)) = -\sum_y P_{\mathcal{S},y} \log_2 P_{\mathcal{S},y} = 1.723$ bits: Werner's joint uncertainty about the two answers before either is asked. It is one of the six size-two terms averaged into G_2 in the next example.

Two exact laws follow. The first is equation (49) averaged:

$$\left\langle \sum_{i=1}^{\ell} s_i \right\rangle = G_\ell. \tag{53}$$

The second law describes the total column entropy of $M^{(\ell)}$, which is the answer matrix after $\ell = |\mathcal{S}|$ questions have been settled. Fix $\mathcal{S}$ and average the posterior table over the answers. Each unasked q' column's entropy becomes a conditional entropy,

$$H(\psi(q') \mid \psi(\mathcal{S})) = H(\psi(\mathcal{S} \cup \{q'\})) - H(\psi(\mathcal{S})), \tag{54}$$

and the already answered columns contribute zero. Now average over $\mathcal{S}$: every set of size $\ell + 1$ arises as $\mathcal{S} \cup \{q'\}$ once per element, and substituting $(\ell{+}1)\binom{Q}{\ell+1} = (Q{-}\ell)\binom{Q}{\ell}$,

$$\left\langle H\big(M^{(\ell)}\big)\right\rangle = (Q - \ell)\,\big(G_{\ell+1} - G_\ell\big). \tag{55}$$

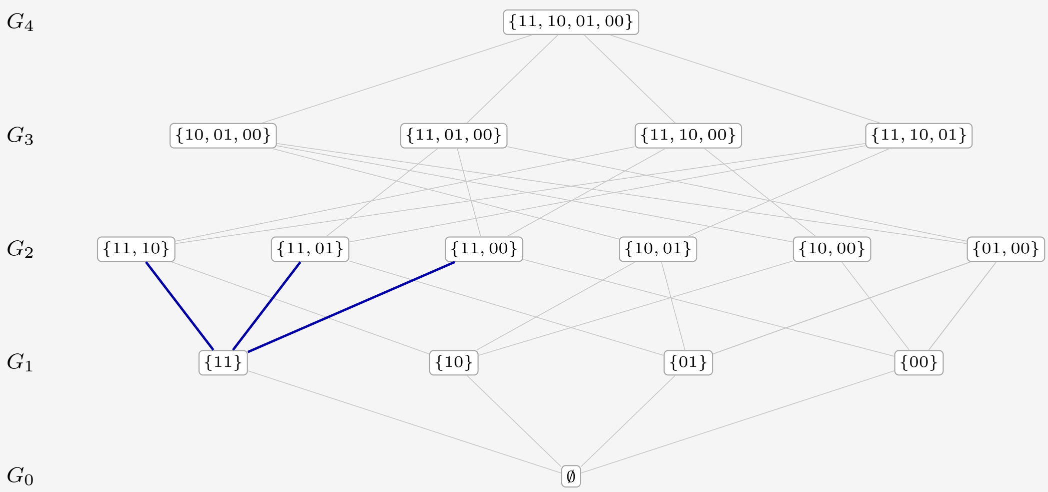
The whole expected learning curve is the discrete derivative of one monotone sequence.

EXAMPLE 27

This example illustrates equation (55): the total column entropy of the table M after ℓ questions, read off a lattice. Take the guiding space, $\mathcal{Q} = \{11, 10, 01, 00\}$ with $Q = 4$ and $m = 1$, and abbreviate $\psi(11, 10)$ for $\psi(\{11, 10\})$. Definition (50) averages $H(\psi(\mathcal{S}))$ over the sets of each size, so the five block entropies collect $1, 4, 6, 4, 1$ terms apiece:

$$\begin{aligned}
G_0 &= 0, \\
G_1 &= \tfrac{1}{4}\big[\, H(\psi(11)) + H(\psi(10)) + H(\psi(01)) + H(\psi(00)) \,\big], \\
G_2 &= \tfrac{1}{6}\big[\, H(\psi(11,10)) + H(\psi(11,01)) + H(\psi(11,00)) \\
&\qquad + H(\psi(10,01)) + H(\psi(10,00)) + H(\psi(01,00)) \,\big], \\
G_3 &= \tfrac{1}{4}\big[\, H(\psi(10,01,00)) + H(\psi(11,01,00)) \\
&\qquad + H(\psi(11,10,00)) + H(\psi(11,10,01)) \,\big], \\
G_4 &= H(\psi(11,10,01,00)) \ = \ H(p).
\end{aligned}$$

Those sixteen sets are the nodes of a lattice, one rank per value of ℓ, and each G_ℓ is the average of the entropies on rank ℓ:



Every edge up is the addition of one question, and it weighs the conditional entropy of the answer it adds: the edge from $\mathcal{S}$ up to $\mathcal{S} \cup \{q'\}$ represents $H(\psi(q') \mid \psi(\mathcal{S}))$. Fix a single node. Its upward edges run to the questions still unasked, and their weights are exactly the column entropies of the posterior table. The remaining table entropy of an asked set is the total marginal weight of the up-edges leaving its node. There are $Q - \ell$ of them, drawn in blue above for $\mathcal{S} = \{11\}$; summed and averaged over the rank, they are exactly equation (55)'s $H^{(\ell)}$, a component of the leverage.

Putting together (53) and (55), equation (48) becomes

$$\langle L_\ell \rangle \;=\; \frac{H(M^{(0)}) - (Q-\ell)\big(G_{\ell+1} - G_\ell\big)}{G_\ell}\,, \qquad 1 \le \ell \le Q, \tag{56}$$

the vanishing factor $Q - \ell$ retiring the undefined increment at $\ell = Q$; at $\ell = 0$ nothing has been asked or paid and the ratio is an empty $0/0$. Some observations.

- Expectation kills the windfall. Each step has $\langle \Delta H(p) - s \rangle = 0$, so the expected deduced part is a pure tower withdrawal: $H(M^{(0)}) - \langle H(M^{(\ell)}) \rangle - G_\ell = C_0 - \langle C_\ell \rangle$, with $\langle C_\ell \rangle = \langle H(M^{(\ell)}) \rangle - \big(H(p) - G_\ell\big)$. It is nonnegative: anti-leverage is only ever bad luck.

- Deduction saturates one round early: substituting $G_Q = H(p)$ at $\ell = Q-1$ already gives $\langle C_{Q-1}\rangle = 0$. The store is empty before the last question, which can only be observed, never leveraged.

- $\langle L_\ell\rangle \geq 1$ always, but it need not be monotone. The endpoint is forced: $\langle L_Q\rangle = H(M^{(0)})/H(p) = 1 + C_0/H(p)$, the run ratio quoted in Section 2. This is also why equation (48) is a ratio of expectations: the expectation of the ratio is dominated by lucky runs with vanishing denominators.

There is also a monotonicity property hiding in G. Entropy is *submodular*: a fresh answer is worth at most in a larger context $\mathcal{S}$ what it was worth in a smaller one $\mathcal{V}$,

$$H(\psi(q) \mid \psi(\mathcal{S})) \leq H(\psi(q) \mid \psi(\mathcal{V})); \quad \forall \mathcal{V} \subseteq \mathcal{S}, \tag{57}$$

because conditioning on more answers can only explain away part of its novelty, never add to it. Coupling a random ℓ-set to a random $(\ell+1)$-set and averaging, the increments $G_\ell - G_{\ell-1}$, which are the expected surprisals of the ℓ-th answer, decrease with ℓ: answers get cheaper as evidence accumulates. This only holds on uniformly random question order.

EXAMPLE 28

For the guiding prior of Table 4, recall the block-entropy sequence is $G_k = \left\langle \sum_i^k s_i \right\rangle$, and the laws unfold it into the expected run:

ℓ	received G_ℓ	remaining $\langle H(M^{(\ell)})\rangle$	store $\langle C_\ell\rangle$	leverage $\langle L_\ell\rangle$
1	0.731	2.113	0.137	1.110
2	1.436	1.328	0.056	1.113
3	2.100	0.608	0	1.104
4	2.707	0	0	1.081

The expected leverage humps at $\ell = 2$ and lands on $2.925/2.707 = 1.081$. The realized run of Section 2 scattered around this gentle expectation: greedy questioning and a lucky truth delivered 1.311. Figure 9 stacks the expected bits.

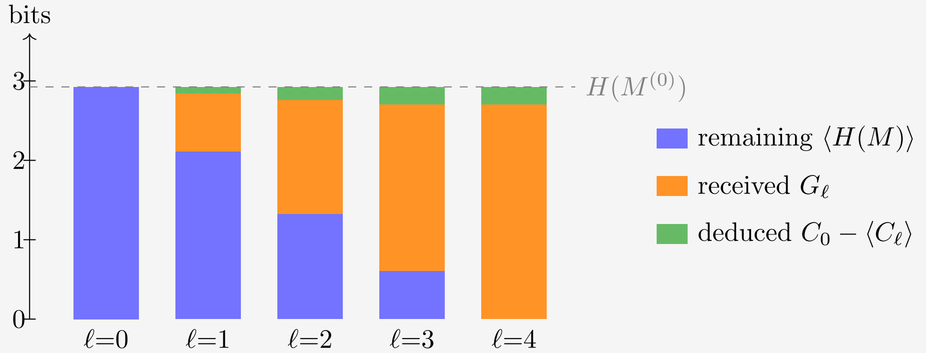


Figure 9: The expected run through the guiding example. Conservation is now exact at every ℓ: remaining plus received plus deduced reaches $H(M^{(0)})$, so the cake always tops out on the dashed line, and no overshoot is possible. The top layer is the expected withdrawal from the correlation store, $C_0 - \langle C_\ell\rangle$; it saturates at $C_0 = 0.218$ already at $\ell = 3$.

3.1 Choosing the questions

The uniform distribution on question order was a symmetry choice, not a strategy. Werner can do better along two independent axes: how far he looks ahead, and whether he commits to an order beforehand or pivots contingent on the answers so far. A *Greedy* Werner asks, at every step, the single question with the largest expected drop of $H(M)$: one step of lookahead. *ℓ-optimal* maximizes the expected knowledge after exactly ℓ questions, looking the whole horizon ahead. All four combinations are straightforward with the machinery at hand. 1-optimal coincides with Greedy at the start.

	order fixed beforehand	**contingent on answers**
greedy	Add planned questions to the docket, each maximizing the immediate expected drop of $H(M)$, averaged over the still-unseen answers. This before any answers are seen, and cannot react to luck.	Recompute the forecasts after every answer and ask the current argmax. Question order is a branching tree, each branch contingent on last answer.
ℓ-optimal	Updating commutes, so a fixed plan is just a set $\lvert\mathcal{S}\rvert = \ell - 1$: search the $\binom{Q}{\ell}$ subsets for the smallest expected remainder $\sum_{q' \notin \mathcal{S}} H(\psi(q') \mid \psi(\mathcal{S}))$. The best non-adaptive knowledge; sees correlations that the one-step view misses.	Backward induction over posteriors. The Bayes-optimal policy: recompute optimal further set for remainder at ℓ, after every answer.

Expected knowledge is non-decreasing left to right and top to bottom. The dynamic program in the last cell is exact but grows quickly: its states are the reachable posteriors, one per pair of asked set and answer pattern, $(1 + A)^Q$ in all. That is 81 for the guiding example and astronomical beyond. Its policy is also horizon-dependent, since the best first question for $\ell = k$ need not open the best plan for $\ell = k+1$. For the guiding prior of table 4, contingent greedy outperforms the uniform order, and already attains the contingent ℓ-optimum at every horizon. That is a quirk of this prior, not a law: a prior that stores its knowledge in deep shells can defeat Greedy in either column: a question worth little now may unlock a cascade later.

It turns out that both ℓ-optimal produce the same trajectory on every set we've tested. This agreement makes it unappealing to perform the full expensive calculation of the right column. Whether this is a law, or coincidence, needs further research.

3.2 The Gibbs complexity prior

Every prior so far was an arbitrary set of instructively chosen $\{p_j\}$. This section builds the prior the other way around, from a single philosophical principle: *simple explanations are likelier.* How do we define simplicity? With circuit complexity as the measure, made quantitative as a Gibbs ensemble over the hypothesis space. It is an illustration: the later chapters do not depend on the rest of this section.

3.2.1 NAND complexity

There are a number of equivalent, Turing-complete set of boolean logic gates. But it's hard to argue with a set that consists of one universal element, when ordering circuits by complexity.

The choice here for that universal element is NAND. Let the complexity X_j of a map ϕ_j be the fewest number of two-input NAND gates in any feed-forward circuit that computes it. For the bookkeeping we count in the equivalent *and-inverter* form: two-input AND gates of unit cost, with inversion free on every wire (an edge attribute, not a gate). De Morgan's laws push inverters along wires and into gates, so an and-inverter circuit converts to a NAND-only circuit of the same gate count and back; X_j is the standard AIG-size metric of logic synthesis. Reference values in this convention:

- AND, OR, NAND and NOR all cost 1;
- XOR costs 3;
- constants, projections and negations cost 0.

For $m > 1$ outputs, X_j is the size of one *joint* circuit computing all output bits together, gates shared, which is practical. It is the true minimum-hardware cost, generally smaller than the sum of per-output circuits.

The $(n, m) = (4, 1)$ and $(3, 2)$ are the largest tractable setups, at 65,536 maps a piece. The tabulation is feasible because X_j is invariant under relabeling: permuting or negating inputs, negating or (for $m \geq 2$) swapping outputs are all free operations, so the 65,536 maps collapse to 222 and 308 equivalence classes respectively. The $(4, 1)$ class values are published optimal AIG sizes[1]; the $(3, 2)$ values come from that synthesizer: circuit sizes $X = 0, 1, 2, \ldots$ are encoded as satisfiability instances, and the first satisfiable X is provably minimal, every smaller size having been refuted.

Alongside the interior cost X_j, which takes a SAT solver to see, two descriptors are visible in one pass over the truth table:

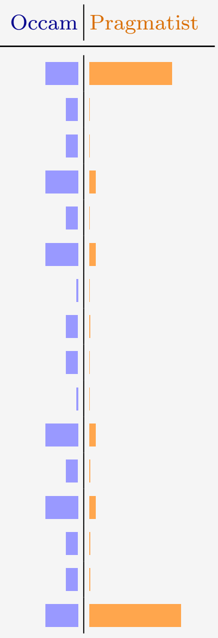
- the **footprint** $F_j = n \cdot [\text{Image}] + [\text{Support}]$, where the support counts the input bits the map actually depends on and the image size counts the distinct answers it attains (from 1 for constants up to A). The combination is bijective to the pair, as Support $\leq n$. Compare to base-n digits, quotient Image, remainder Support. The footpring can be viewed as the map's *interface*: does it react to all its inputs, does it use all its values?
- the **bias** B_j: the number of 1s minus the number of 0s among the mQ truth-table bits. This is the symmetry breaker: X_j and F_j are even under negating all outputs, B_j is odd: an energy term coupling to it acts like a magnetic field, favoring 0s or 1s off the bar. In the language of section 2.5: switching on the odd correlator shells that a pure complexity prior forbids.

[1] The `github.com/krinkin/bounds` reproducibility dataset, cross-checked against our own SAT-based synthesizer on random classes with zero mismatches.

EXAMPLE 29

Take again the $(n, m) = (2, 1)$ space. Each map's data is listed for illustration. The size of two Gibbs priors using these inputs, defined in section 3.2.2 is graphed. Occam in blue growing left, the Pragmatist in orange growing right. The Pragmatist is on a log scale, for better readability of the far-from-democratic distribution.

j	map	X_j	Support	Image	F_j	B_j	prior mass (Occam | Pragmatist)
0	FALSE	0	0	1	2	-4	
1	$\neg(b_1 \vee b_0)$ (NOR)	1	2	2	6	-2	
2	$\neg b_1 \wedge b_0$	1	2	2	6	-2	
3	$\neg b_1$	0	1	2	5	0	
4	$b_1 \wedge \neg b_0$	1	2	2	6	-2	
5	$\neg b_0$	0	1	2	5	0	
6	$b_1 \oplus b_0$ (XOR)	3	2	2	6	0	
7	$\neg(b_1 \wedge b_0)$ (NAND)	1	2	2	6	$+2$	
8	$b_1 \wedge b_0$ (AND)	1	2	2	6	-2	
9	$b_1 = b_0$ (XNOR)	3	2	2	6	0	
10	b_0 (projection)	0	1	2	5	0	
11	$b_1 \to b_0$	1	2	2	6	$+2$	
12	b_1 (projection)	0	1	2	5	0	
13	$b_0 \to b_1$	1	2	2	6	$+2$	
14	$b_1 \vee b_0$ (OR)	1	2	2	6	$+2$	
15	TRUE	0	0	1	2	$+4$	



Occam spreads its belief mostly across the six gate-free maps (0.1106 apiece) and thins gradually with cost; the Pragmatist, punishing footprint and rewarding bits with value 1, piles 0.90 of its mass onto the two constants (0.62 on TRUE against 0.28 on FALSE), the bias breaking the tie. It all but abandons the interior of the table. If the map turns out non-constant, that interior is scaled up and becomes relevant. Grouped by the full tuple (X, F, B), the sixteen maps fall into six energy classes:

(X, F, B)	members	(X, F, B)	members
$(0, 2, -4)$	1	$(1, 6, -2)$	4
$(0, 2, +4)$	1	$(1, 6, +2)$	4
$(0, 5, 0)$	4	$(3, 6, 0)$	2

Note the table skips $X = 2$ entirely: with two inputs, nothing a two-gate circuit computes is out of reach of one or zero gates.

At the larger sizes the tuple (X, F, B) breakdown runs to hundreds of rows (102 distinct tuples at $(n, m) = (4, 1)$, 196 at $(3, 2)$). It is still interesting to tabulate the complexity X shells alone, with each shell's *aggregate* prior mass under the Occam setting alongside its head count:

EXAMPLE 30

These tables illustrate the compromise between exponentially suppressed weight per map, and combinatorics putting most maps at middling complexity.

	$(4,1)$	
X	maps	Occam mass
0	10	0.041
1	48	0.072
2	256	0.141
3	940	0.190
4	2,336	0.174
5	6,464	0.177
6	10,616	0.107
7	18,984	0.070
8	17,680	0.024
9	7,882	0.004
10	320	0.000
total	65,536	1

	$(3,2)$	
X	maps	Occam mass
0	64	0.058
1	432	0.144
2	2,064	0.253
3	4,860	0.219
4	10,832	0.180
5	16,208	0.099
6	17,060	0.038
7	10,672	0.009
8	3,312	0.001
9	32	0.000
total	65,536	1

The head counts are thin at both ends: ten gate-free maps at $(4,1)$ (two constants, eight literals) against only 320 at the ceiling $X = 10$. The aggregate mass shows the thermodynamic compromise this forces on Occam: the count grows faster than e^{-X} decays until $X = 3$ at $(4,1)$ ($X = 2$ at $(3,2)$), so even under a pure simplicity prior Werner has most faith thst the map has $X_j \in \{2,3\}$. Shell degeneracy beats shell energy for the simplest maps.

3.2.2 The ensemble

Take the energy linear in the three descriptors and the prior as its Boltzmann weight,

$$E_j = \beta\, X_j + \alpha\, F_j + \mu\, B_j, \qquad p_j = \frac{e^{-E_j}}{\sum_k e^{-E_k}}, \tag{58}$$

with no overall temperature: it's absorbed in the coupling fields. Positive β suppresses complex maps, positive α suppresses large footprints, and μ couples to the signed bias like a magnet. $\mu < 0$ tilts toward 1-heavy tables and thereby breaking the output-negation symmetry. Three settings are explored below, chosen out of principle, as well as to exhibit interesting learning behavior.

	β	α	μ
Occam	1	0	0
Occamer	2	0	0
Pragmatist	1.5	1	-0.1

Occam suppresses circuit complexity and nothing else. **Occamer** doubles the same field: an even stronger simplicity bias stores more knowledge in the prior, and stores it deeper in the correlation shells, where only the later questions of a run can release it. The **Pragmatist** judges by the obvious heuristics instead: a footprint reads the visible interface rather than the interior circuit, a moderate complexity field seconds it, and the tilt $\mu = -0.1$ leans toward 1-heavy tables, breaking the output-negation symmetry that the other two settings keep. This last choice is to illustrate that Werner isn'tb bound to start with a uniform distribution M before the first question.

3.2.3 Trajectories

Figures 10–12 show the expected cumulative leverage $\langle L_\ell \rangle$ under the three settings, for the uniform-order average and the strategy grid of Section 3.1. The ℓ-optimal schedule targets the halfway horizon $\ell^* = Q/2$ and is plotted only up to it: past its own horizon the optimizer's continuation is not defined, and inside the optimal *set* the questions are asked in random order. Its contingent twin is omitted everywhere: wherever we can afford to compute it, it reproduces the fixed optimum's curve to machine precision in all three settings – a coincidence of these fields rather than a law (a stronger tilt, $\mu = -0.3$, separates the two), and one for which we have no symmetry argument yet (at $(4, 1)$ its 3^{16}-state program is in any case out of reach). Two anchors hold in every panel: all schedules agree at $\ell = 1$, because input relabeling acts transitively on the questions and the Gibbs energies are relabeling-invariant, so single questions are exchangeable; and every full-length curve ends at the same $L_Q = H(M^{(0)})/H(p)$, since any policy that asks everything receives exactly $H(p)$.

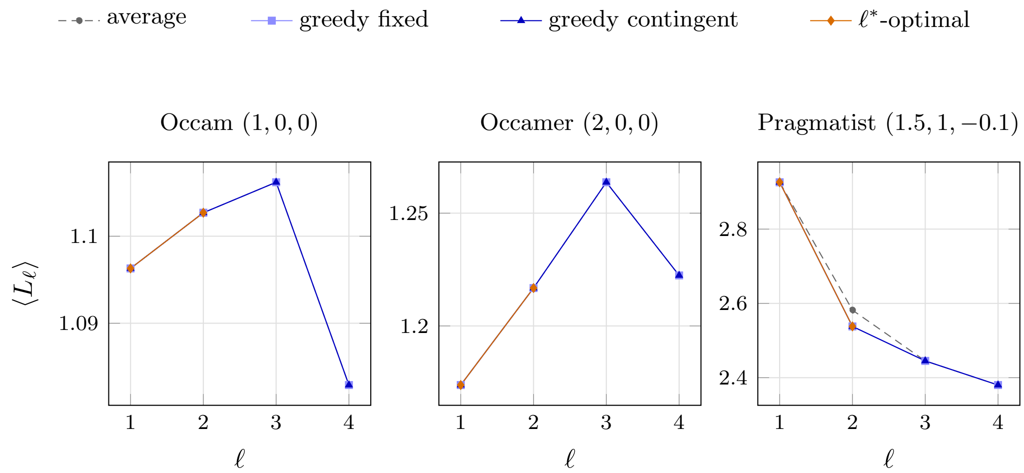


Figure 10: Expected leverage under the Gibbs priors at $(2, 1)$, $\ell^* = 2$. In both complexity-only settings every schedule coincides exactly; the Pragmatist's fields finally separate them.

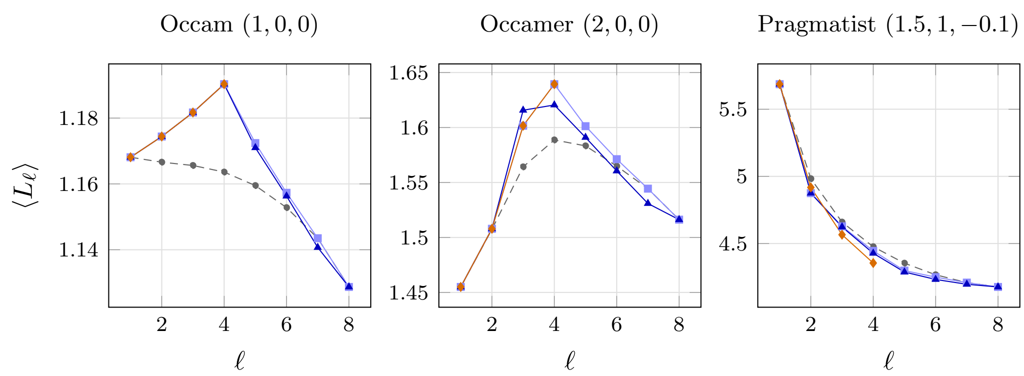


Figure 11: Expected leverage under the Gibbs priors at $(3, 2)$, $\ell^* = 4$. Greedy's fixed set is the optimal set under Occam and, up to an input negation, under Occamer; under the Pragmatist it misses – knowing less at the horizon yet out-levering the optimum.

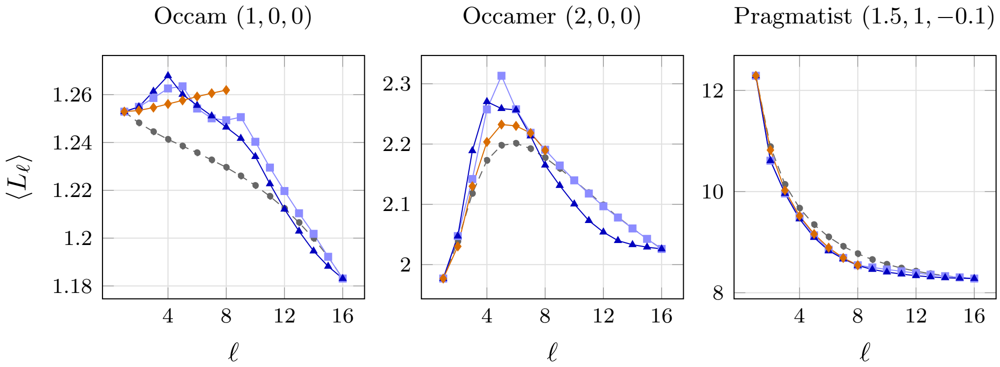


Figure 12: Expected leverage under the Gibbs priors at $(4, 1)$, $\ell^* = 8$. Occamer's deep store bends even the strategyless average curve into an interior hump; the Pragmatist is a two-sided spike on the constants, all leverage and little game.

Three observations, one per system size.

At $(2, 1)$, under *both* complexity-only settings, the strategy grid collapses exactly: all question subsets of equal size share a single joint entropy, so every schedule is the uniform order. Occamer shows what doubling the field buys within that straitjacket – the whole curve lifts (endpoint 1.2224 against Occam's 1.0829, store $C_0 = 0.728$ against 0.306 bits) without creating a single strategic choice to make. The Pragmatist's fields split the pair orbits at last, and the very first thing strategy does is post a *lower* number: greedy's pair $\{00, 11\}$ knows more at $\ell = 2$ than the random order and shows less leverage for it (2.538 against 2.582).

At $(3, 2)$, the optimal four questions are the even-weight inputs $\{000, 011, 101, 110\}$ – the maximally spread design, pairwise at Hamming distance two – under *all three* settings. Under Occam greedy builds exactly that set; under Occamer it lands on the odd-weight twin, the same design up to an input negation, and its curve coincides with the optimum's. Under the Pragmatist greedy finally misses: its set $\{000, 011, 101, 111\}$ leaves twice the ignorance of the optimal four (0.405 against 0.198 bits remaining) while receiving 0.099 bits less – and it posts the *higher* leverage, 4.445 against 4.355, with the strategyless average higher still at 4.476. The ℓ-optimal schedule maximizes knowledge, not leverage, and a ratio is perfectly happy with cheaper ignorance.

At $(4, 1)$ the settings differ far more than the schedules within any one of them, and Occamer is the showpiece: doubling β triples the store ($C_0 = 8.10$ bits against Occam's 2.48) and bends even the average curve into an interior hump, rising to 2.20 at $\ell = 6$ before decaying – the deep-shell signature promised in Section 3.2.2, knowledge that only becomes spendable once enough answers are in hand to cash the higher correlations. The optimal eight questions are a parity class of the input hypercube in every setting (even-weight under Occam and the Pragmatist, the odd twin under Occamer); greedy misses the class under both complexity-only settings (1.249 against the optimum's 1.262 at the Occam horizon) but finds it exactly under the Pragmatist – whose prior, 0.90 of its mass on the two constants, is a two-sided spike in disguise: $H(p) = 0.53$ bits, leverage running from 12.29 down to 8.28, a great deal of inference about a game that is mostly over before it starts.

4 The thermodynamic limit

The earlier chapters describe learning on a finite question set. This chapter asks what remains when the number of input bits tends to infinity. The hope is to find traces of universality: that character of learning where the microscopic details average out.

Since $Q = |\mathcal{Q}| = 2^n$, the limit $n \to \infty$ is one of an exponentially growing table, and we measure time by the fraction of that table already queried,

$$t \;=\; \frac{\ell}{Q} \;\in\; [0,1]. \tag{59}$$

The limit at fixed $t > 0$ is *macroscopic*: it records learning that occupies a nonzero fraction of the run. Learning completed in $O(1)$, in $O(\log Q)$, or more generally in $o(Q)$ questions is compressed into a boundary layer at $t = 0$. That distinction will will determine the class of solutions we find.

Throughout, $u := A^{-1} = 2^{-m}$ abbreviates the reciprocal answer alphabet, and $N = A^Q$ counts the maps $\mathcal{Q} \to \mathcal{A}$, enumerated $\phi_0, \ldots, \phi_{N-1}$ with prior weights $p_j = \Pr(\psi = \phi_j)$. One value of the truth ψ is drawn at the start of a run and supplies every answer in it; posteriors are conditional distributions of that same random map, and expected trajectories average over its prior law.

4.1 A toy example: the spike prior and its renormalization flow

The spike prior places a distinguished weight on one map and spreads the remaining mass uniformly over every other map. Take the special map to be the all-zero map, $\phi_0(q) = 0^m$ for every $q \in \mathcal{Q}$, so that its answer to any k questions is the all-zero block 0^{mk}. This is fully general by symmetry: the answer alphabet may be relabelled separately at each question. Then

$$p_0 = p, \qquad p_j = \omega := \frac{1-p}{N-1} \quad (j \neq 0). \tag{60}$$

The plain $p = p_0$ is the mass of the all-zero map alone. We will be interested in the behavior at large (but not necessarily unit) p. The interpretation is that Werner is quite sure of his hypothesis, without ruling anything out.

The example is useful because it is closed under Bayesian conditioning. A confirming answer produces a sharper spike on a smaller map space; a refuting answer removes the special map and leaves equal weights on all survivors. The posterior therefore has a two-state flow, in which the uniform family is the stable, absorbing class: once a trajectory enters it, later conditioning keeps it there. An interior spike is a source of one-way probability current into that class, while conditional on avoiding refutation the surviving spike sharpens.

4.1.1 The two-state flow and exact leverage

What happens when Werner handles the first question? He either gets a zero, or not.

The maps that produce 0^m are a proportion $u = A^{-1}$. One has probability p, and $Nu-1$ have probability ω. If we started with probability $P_0 = 1$ in this prior, Werner avoids refuting ϕ_0 with probability

$$P_1 = p + (Nu - 1)\,\omega \tag{61}$$

In this case, what exactly is the shape of the posterior? The Bayesian consistency condition tells us that unconditional probability cannot be created without measurement. At this point in our ansatz, we must have the same credence in ϕ_0. Let r_1 be the height of the peak p_0, after one question. Then we must have $r_1 \cdot P_1 = p$. The rest of the probability is spread again over the $Nu - 1$ other surviving maps. Conversely, with probability $1 - P_1$, we've refuted the zero map, and are in a uniform posterior corresponding to a nonzero answer to the question.

After repeating this process k times, asking the set of questions $\mathcal{S}$, we can infer the probability that ϕ_0 is still in the race. Let P_k be the probability that we remain in the confirming scenario: Werner has only seen zeros. A history remains on the spike branch exactly when $\psi(\mathcal{S}) = 0^{mk}$, and exactly Nu^k maps produce that confirming history, one of which is ϕ_0. Its prior probability is therefore

$$P_k \;:=\; \Pr\left(\psi(\mathcal{S}) = 0^{mk}\right) \;=\; p + (Nu^k - 1)\,\omega. \tag{62}$$

On that event the posterior is another spike on those Nu^k maps,

$$r_k \;=\; \Pr\left(\psi = \phi_0 \mid \psi(\mathcal{S}) = 0^{mk}\right) \;=\; \frac{p}{P_k}, \qquad \omega_k \;=\; \frac{\omega}{P_k}, \tag{63}$$

so r_k is the time-indexed posterior height of the spike evaluated on the realized confirming branch, with $r_0 = p$ the original weight. It's pegged inversely to P_k for all time, by the same Bayesian consistency:

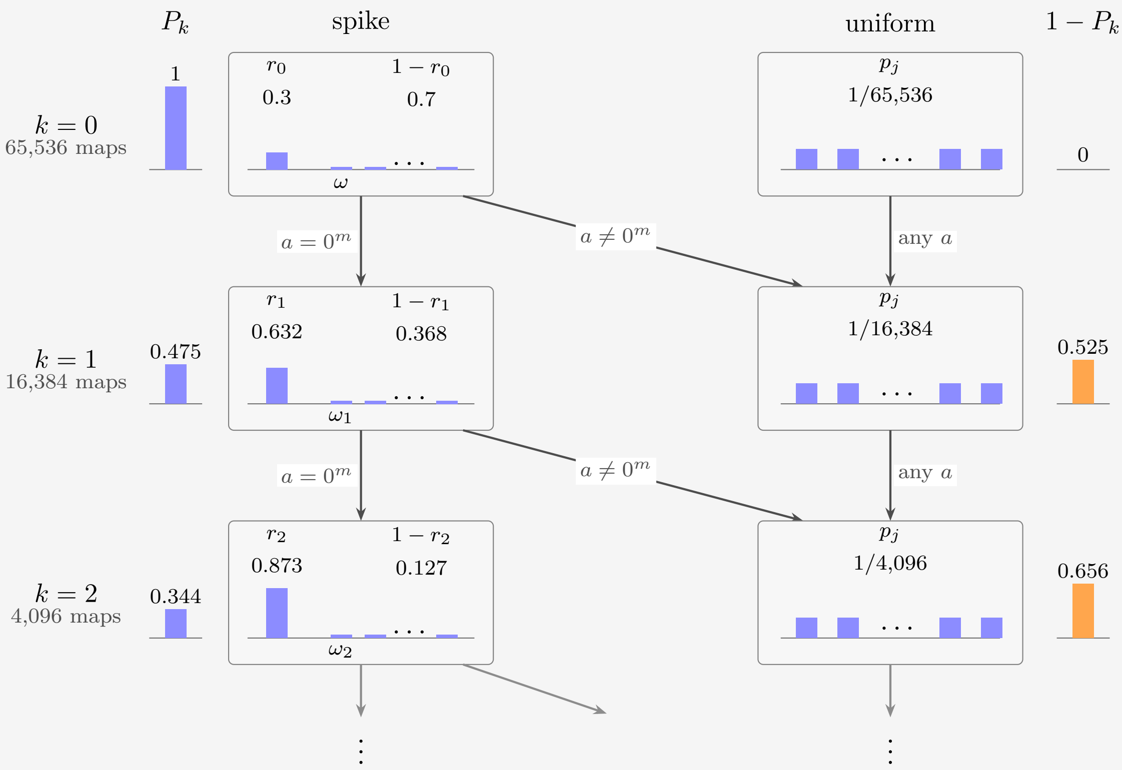
$$P_k\, r_k \;=\; p. \tag{64}$$

If any answer is nonzero then ϕ_0 is eliminated, every compatible survivor had the same mass ω_k, and the posterior is uniform; subsequent conditioning preserves uniformity. With an equivalence across system sizes, this induces a two state flow. Renormalization sharpens the spike. The two macrostates and their directions are

$$\text{spike} \longrightarrow \begin{cases} \text{sharper spike}, & \text{with probability } P_{k+1}/P_k, \\ \text{uniform}, & \text{with probability } 1 - P_{k+1}/P_k, \end{cases} \qquad \text{uniform} \longrightarrow \text{uniform}. \tag{65}$$

EXAMPLE 31

The flow at $(n, m) = (3, 2)$, so $A = 4$ and $N = 4^8 = 65{,}536$, started from $r_0 = p_0 = 0.3$. Each confirmed answer divides the surviving map space by four and renormalizes the spike upward; each refutation drops the belief into the uniform class, which nothing ever leaves.



The two columns of boxes are the macrostates of (65), each arrow is labelled by the answer that drives its transition, and each row's map space is one quarter of the row above. The outer columns track the branch probabilities P_k and $1-P_k$, and the invariant $P_k r_k = p$ is visible between the P_k bars and the $r_k = p_0$ bars: the spike climbs exactly as fast as its branch loses probability, $0.3 = 1 \times 0.3 = 0.475 \times 0.632 = 0.344 \times 0.873$.

For $p \in (0, 1)$, we can combine equations (60) and (62) to get the limiting odds on the surviving branch. Also the surviving posterior, from equation (63):

$$P_k = p + (1-p)u^k + O(N^{-1}), \qquad r_k = \frac{p}{p + (1-p)u^k} + O(N^{-1}). \tag{66}$$

Each confirmation supplies approximately $-\log_2 u = m$ bits of evidence for the spike. After Q (all) questions, the only way the spike can survive is by eliminating all alternative maps. This happens with $P_Q = p$ and hence $r_Q = 1$; the histories that confirm through the complete table are precisely those with $\psi = \phi_0$. At the uniform point $p = 1/N$ both posterior branches are uniform on their surviving map spaces.

The same P_k has a second reading: it is the probability of the single k-answer block 0^{mk}, so there is no new distribution to compute. For $1 \leq k \leq Q$ every other answer block is realized by Nu^k background maps and therefore carries the common probability

$$U_k \;:=\; \frac{1-P_k}{A^k-1} \;=\; Nu^k\,\omega, \qquad \text{so} \qquad \Pr(\psi(\mathcal{S})=y) = \begin{cases} P_k, & y = 0^{mk}, \\ U_k, & y \neq 0^{mk}, \end{cases} \tag{67}$$

for any question set $\mathcal{S}$ of size k. Note that $A = u^{-1}$. The entropy of the above block distribution is

$$G_k \;=\; -P_k \log_2 P_k - (A^k-1)\,U_k \log_2 U_k. \tag{68}$$

Every choice of k questions obeys the same law, so (68) is also the mean k-question block entropy of Chapter 3. In particular $G_0 = 0$, $G_Q = H(\psi)$ and $H(M^{(0)}) = Q\,G_1$, with the exact prior entropy

$$G_Q \;=\; h_2(p) + (1-p)\log_2(N-1), \tag{69}$$

extensive in Q at fixed $0 < p < 1$ and vanishing at $p = 1$.

4.1.2 The fixed-p thermodynamic limit

The thermodynamic limit keeps the prior mass $0 < p < 1$ fixed and sends $n \to \infty$ at a fixed asked fraction $t > 0$. The guiding principle is the exact finite law (56): the expected leverage is built from three ingredients,

1. the prior table entropy $H(M^{(0)})$
2. the received entropy G_ℓ
3. the increment $G_{\ell+1} - G_\ell$.

If we can derive each for the example, we are done. The block entropy (68) supplies every G_k through the single counting formula (62), while $H(M^{(0)}) = Q\,G_1$ holds for any prior: G_1 is by definition the mean entropy of one column. Set $\ell = \lfloor tQ \rfloor$ and take the terms of (56) in turn.

All three derivations run through one exact regrouping of (68). The $A^k - 1$ refuting blocks (67) share the mass $1 - P_k$ evenly, so substituting $U_k = (1-P_k)/(A^k-1)$ into (68) and expanding the logarithm regroups the block entropy into the binary entropy of the confirm-or-refute split plus a uniform choice among the refuting blocks:

$$G_k \;=\; h_2(P_k) \;+\; (1-P_k)\log_2\!\big(A^k-1\big). \tag{70}$$

The prior table entropy: G_1 is the one-column entropy, (70) at $k = 1$. By (62) the spike answer 0^m carries $P_1 = p + (Nu-1)\,\omega \to p + (1-p)u$: the special map's whole mass, plus the fraction u of the background maps that happen to agree with it at any one question. Substituting that limit,

$$\begin{aligned} \frac{H(M^{(0)})}{Q} \;=\; G_1 \;&=\; h_2(P_1) + (1-P_1)\log_2(A-1) \\ &\longrightarrow\; h_2\big(p + (1-p)u\big) + (1-p)(1-u)\log_2(A-1) \;=:\; h_{\text{spike}}(p,m). \end{aligned} \tag{71}$$

The received entropy: Evaluate (70) at $k = \ell = \lfloor tQ \rfloor$. By (66) $P_\ell \to p$, so the split entropy stays bounded while the block-count logarithm grows linearly, $\log_2(A^\ell - 1) = \ell m + o(1)$:

$$\frac{G_{\lfloor tQ \rfloor}}{Q} \;=\; \underbrace{\frac{h_2(P_\ell)}{Q}}_{\to\;0} \;+\; \underbrace{(1-P_\ell)}_{\to\;1-p} \; \underbrace{\frac{\log_2\!\big(A^\ell-1\big)}{Q}}_{\to\;t\,m} \;\longrightarrow\; t\,(1-p)\,m. \tag{72}$$

This is the two-state flow watched at macroscopic time: for any $t > 0$ the confirm-or-refute lottery has already resolved inside the boundary layer at $t = 0$, where $r_k \to 1$ within $o(Q)$ questions. A surviving spike answers for free; only the refuted class, probability $1 - p$, keeps paying the full m bits per answer.

The increment: Subtract consecutive copies of (70): the bounded $h_2(P_k)$ terms cancel in the limit, and the block-count logarithm grows by $\log_2(A^{\ell+1} - 1) - \log_2(A^\ell - 1) \to \log_2 A = m$, paid only by the refuted mass:

$$G_{\ell+1} - G_\ell \longrightarrow (1-p)\, m, \qquad \text{so} \qquad \frac{(Q-\ell)\big(G_{\ell+1} - G_\ell\big)}{Q} \longrightarrow (1-t)(1-p)\, m. \tag{73}$$

Filling the three limits into (56) term by term,

$$\langle L_\ell \rangle = \frac{\overbrace{H\big(M^{(0)}\big)}^{\to\, Q\, h_{\text{spike}}(p,m)} - \overbrace{(Q-\ell)\big(G_{\ell+1} - G_\ell\big)}^{\to\, Q\,(1-t)(1-p)m}}{\underbrace{G_\ell}_{\to\, Q\, t(1-p)m}},$$

and cancelling the common factor Q gives

$$L_p(t) = \frac{h_{\text{spike}}(p,m) - (1-t)(1-p)m}{t(1-p)m} = 1 + \frac{h_{\text{spike}}(p,m) - (1-p)m}{t(1-p)m}, \qquad 0 < t \le 1. \tag{74}$$

The $1/t$ term records information released in a vanishing initial fraction of the run, divided by the $O(tQ)$ information received afterwards.

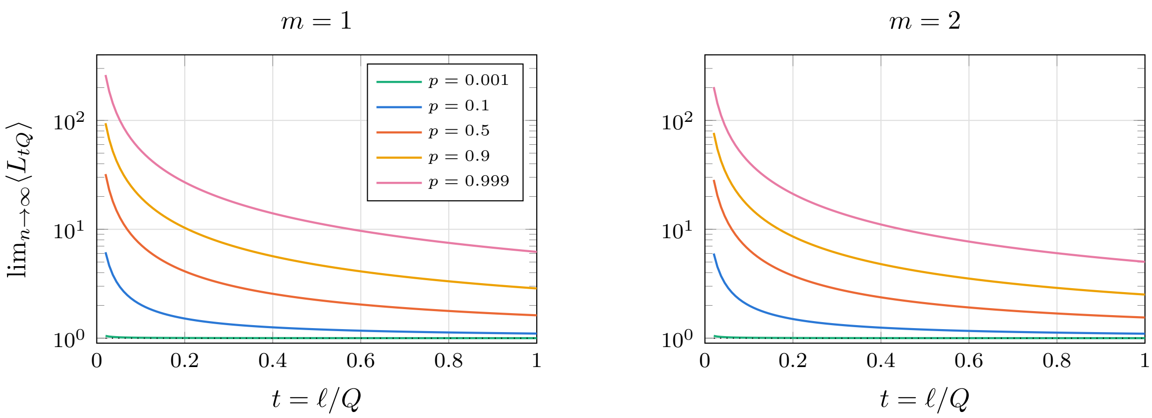


Figure 13: The fixed-p thermodynamic curve (74), on a logarithmic vertical scale. Every weight produces the same $1/t$ shape; the prior sets only the coefficient, which grows from 0.001 at $p = 0.001$ to above 5 at $p = 0.999$.

We have explored a number of other characteristics of the spike prior in Appendix A: the branch-by-branch effect of the first question, the extraction ratio, and various further limits, among them the sharp prior $p \to 1^-$ and its noncommutation with $n \to \infty$.

4.2 Thermodynamic-limit calculus

The spike is solvable because all subsets of the same size have the same entropy. The general theory replaces that symmetry by an average over subsets.

4.2.1 The finite sequence from which the limit is taken

For each fixed question set $\mathcal{S} \subseteq \mathcal{Q}$ the vector $\psi(\mathcal{S})$ collects the answers to those questions, and its entropy is taken under the prior law of ψ with $\mathcal{S}$ held fixed. The mean block entropy is that of equation (50), now carrying the system size explicitly:

$$G_{n,k} \;:=\; \binom{Q}{k}^{-1} \sum_{\substack{\mathcal{S}\subseteq\mathcal{Q}\\ |\mathcal{S}|=k}} H\big(\psi(\mathcal{S})\big), \qquad 0 \le k \le Q, \tag{75}$$

so that $G_{n,0} = 0$ and $G_{n,Q} = H(\psi) = H(p^{(n)})$, where $p^{(n)}$ is the prior law of the random map at size n, with increments $\gamma_{n,k} := G_{n,k+1} - G_{n,k}$.

By the entropy chain rule, exactly as in the derivation of equation (55), the increment is the mean conditional entropy of one fresh answer,

$$\gamma_{n,k} \;=\; \mathbb{E}_{\mathcal{S},q}\big[H\big(\psi(q) \mid \psi(\mathcal{S})\big)\big], \qquad |\mathcal{S}| = k, \quad q \in \mathcal{Q} \setminus \mathcal{S}. \tag{76}$$

There are two separate averages here: the conditional entropy averages over answer histories under the prior on the random map, while $\mathbb{E}_{\mathcal{S},q}$ averages that entropy over which questions were selected. Conditioning reduces entropy, so coupling a random k-set to a random $(k+1)$-set yields

$$m \;\ge\; \gamma_{n,0} \;\ge\; \gamma_{n,1} \;\ge\; \cdots \;\ge\; \gamma_{n,Q-1} \;\ge\; 0. \tag{77}$$

This monotonicity is deterministic, the truth and the subset having already been averaged over. It should not be called self-averaging; concentration of individual learning trajectories would be a separate result. The two laws (53) and (55) then read

$$\Big\langle \sum_{i=1}^{\ell} s_i \Big\rangle = G_{n,\ell}, \qquad \big\langle H\big(M^{(\ell)}\big)\big\rangle = (Q-\ell)\,\gamma_{n,\ell}, \tag{78}$$

for $0 \le \ell < Q$, with zero remaining entropy at $\ell = Q$, and equation (56) becomes

$$\langle L_{n,\ell}\rangle = \frac{Q\,G_{n,1} - (Q-\ell)\,\gamma_{n,\ell}}{G_{n,\ell}} \qquad (1 \le \ell < Q), \qquad \langle L_{n,Q}\rangle = \frac{Q\,G_{n,1}}{G_{n,Q}}. \tag{79}$$

4.2.2 A precise macroscopic-limit statement

It is useful to keep the initial marginal entropy separate from the macroscopic profile on the right, since that allows an $o(Q)$-question boundary layer at $t = 0$. Let $h_0 := \lim_{n\to\infty} G_{n,1}$, and suppose the staircases have an almost-everywhere limit γ on $(0,1)$, with $\gamma_{n,\lfloor tQ\rfloor} \to \gamma(t)$ at any particular t where the remaining entropy is evaluated. At such a point $\gamma(t)$ is the expected entropy of one more answer after a random t-fraction has been observed. Four hypotheses suffice:

1. **Fixed answer width.** The integer m is fixed, which supplies the uniform bound $0 \le \gamma_{n,k} \le m$.

2. **Initial-entropy convergence.** The limit h_0 exists.

3. **Profile convergence.** The staircase $t \mapsto \gamma_{n,\lfloor tQ\rfloor}$ converges almost everywhere on $(0,1)$, with direct convergence at any t where the remaining entropy is evaluated. At continuity points of a selected monotone limit this causes no ambiguity.

4. **Positive entropy density.** The limiting area is positive, $\int_0^1 \gamma(x)\,dx > 0$.

The fourth condition is the nondegenerate form of extensivity, and it implies $G_{n,Q}/Q \to \int_0^1 \gamma > 0$. Merely writing $G_{n,Q} = \Theta(Q)$ is not sufficient: the entropy density may oscillate with n, and even a convergent scalar density does not determine the profile. Conversely, profile convergence with positive area implies the extensivity required. No Gibbs representation and no full-support lower bound on individual map probabilities is needed.

Extend the finite increments to a staircase on $[0, 1]$ by $\gamma_n(t) := \gamma_{n,\min(\lfloor tQ \rfloor, Q-1)}$, the value at the single endpoint $t = 1$ being irrelevant to the integral. Telescoping gives an exact Riemann-sum identity at grid points,

$$\frac{G_{n,\ell}}{Q} = \frac{1}{Q} \sum_{k=0}^{\ell-1} \gamma_{n,k} = \int_0^{\ell/Q} \gamma_n(x)\, dx, \tag{80}$$

and since $0 \le \gamma_n \le m$, dominated convergence gives $G_{n,\ell_n}/Q \to g(t) := \int_0^t \gamma$ for $\ell_n = \lfloor tQ \rfloor$, while at a point of profile convergence $(Q - \ell_n)\gamma_{n,\ell_n}/Q \to (1-t)\gamma(t)$. Substituting both into (79) proves

$$L(t) = \frac{h_0 - (1-t)\,\gamma(t)}{g(t)}, \qquad g(t) = \int_0^t \gamma(x)\, dx, \qquad 0 < t < 1. \tag{81}$$

Since γ is nonnegative and nonincreasing, a positive integral makes $g(t) > 0$ for every $t > 0$.

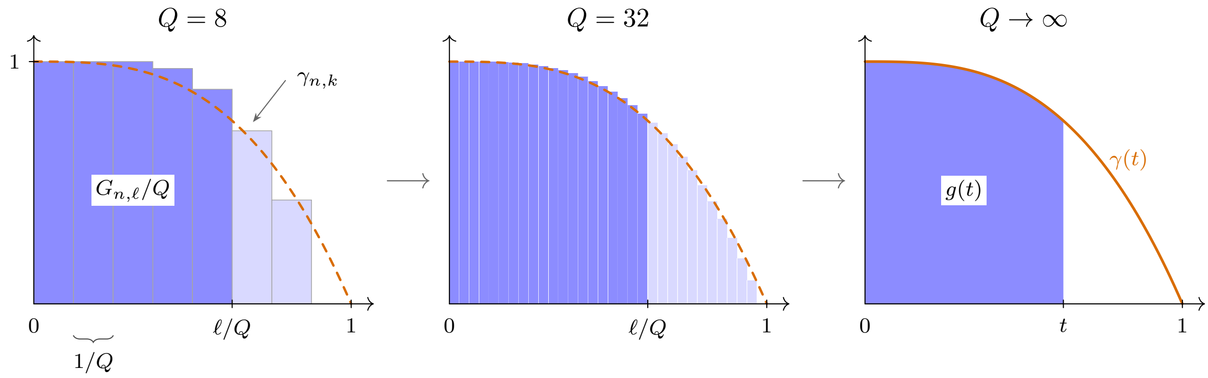


Figure 14: From the finite increments to the profile, drawn for parity blocks of size four, whose limit is $\gamma(t) = 1 - t^3$ (dashed). Each bar has width $1/Q$ and height $\gamma_{n,k}$, so the shaded bars sum – the telescoping identity (80) – to exactly $G_{n,\ell}/Q$, here at $\ell/Q = 5/8$. Refining Q shrinks the bars onto the staircase's limit, and dominated convergence turns the shaded sum into the area $g(t) = \int_0^t \gamma$: received information per question becomes the area under the profile.

Monotonicity and boundedness guarantee convergent subsequences by a Helly selection argument, but they do not guarantee that the full sequence of priors selects a unique profile: a family can alternate between two constructions on even and odd n. At the prior level one therefore needs a coherent construction, such as a fixed exchangeable directing measure or fixed local block types with convergent frequencies, or must take profile convergence itself as a hypothesis.

4.3 A constructive complexity prior

The Gibbs prior of Section 3.2 priced maps by circuit complexity: a principled Occam, but one whose complexities had to be tabulated by force, map by map, and which therefore stops at small n. This section builds an Occam prior from a complexity notion that is *constructive*

– computable from the truth table by a fast transform – and tractable at every n: the degree of the map written as a polynomial. Two ideas carry the section. First, a non-hyperbolic macroscopic curve requires a prior that is not *exchangeable*: if permuting the question labels leaves the prior invariant at every size, consistently across sizes, then de Finetti's theorem reduces it to a mixture of i.i.d. answers over a latent bias; the answers pin that bias in $o(Q)$ questions, the fresh-answer entropy settles to a constant, and the curve is a hyperbola. The polynomial structure of the question cube supplies the asymmetry without any partition of the questions into blocks. Second, weighting complexity by a single temperature fails for a counting reason made precise below: only a thin shell of maps would contribute. The repair is to *sample* the complexity shell rather than price it, which keeps every shell – and every single map – at positive weight all the way into the thermodynamic limit. Throughout this section the answer width is $m = 1$, so $u = \frac{1}{2}$ and a fresh fair answer carries one bit.

4.3.1 Polynomials over $\mathbb{F}_2$, and the butterfly

All linear algebra in this section is over the smallest field there is, the two-element field $\mathbb{F}_2$, whose two operations are

$$\mathbb{F}_2 = \{0, 1\}, \qquad b \oplus b' \;=\; b + b' \bmod 2 \quad \text{(XOR)}, \qquad b\,b' \quad \text{(AND)}. \tag{82}$$

Two consequences carry the rest. Every element is its own additive inverse, $b \oplus b = 0$, so there is no subtraction and there are no minus signs: a term contributed twice cancels. And every element is idempotent, $b^2 = b$, so no variable is ever squared, and a product of input bits is fixed by *which* bits it contains, never by how often. All matrices, vectors, ranks and inverses below are read mod 2.

A *monomial* in the input bits $b_{n-1} \cdots b_0$ of a question is therefore labelled by the set $\mathcal{B}$ of bit positions it contains: $\prod_{i \in \mathcal{B}} b_i$ is the map answering 1 exactly when every bit in $\mathcal{B}$ is 1, the AND of those bits, and the empty set gives the empty product, the constant map 1. A position is named by its bit, so we write $\mathcal{B} = \{b_1, b_0\}$ rather than $\{1, 0\}$, and the subscript on a coefficient is the monomial itself: $\rho_{b_1 b_0}$ belongs to $\mathcal{B} = \{b_1, b_0\}$, $\rho_\emptyset$ to the constant. There are $Q = 2^n$ bit sets, hence Q monomials, hence $2^Q = N$ ways to switch them on and off: exactly as many as there are maps, and the correspondence is one to one.

One more piece of vocabulary. The *support* of a question is the set of bit positions where it holds a 1,

$$\operatorname{supp}(q) \;=\; \{\, i : b_i(q) = 1 \,\}, \tag{83}$$

so a question and a set of bits are the same data in two costumes, and we use the identification freely: $\mathcal{B}$ names a monomial and equally the question whose 1s sit exactly at $\mathcal{B}$. The condition $\operatorname{supp}(q) \subseteq \mathcal{B}$ then picks out the *subcube below* $\mathcal{B}$ – the $2^{|\mathcal{B}|}$ questions that are 0 everywhere outside $\mathcal{B}$ and free inside it.

Every map now has a unique vector of single-bit coefficients $\rho_\mathcal{B} \in \{0, 1\}$, saying whether monomial $\mathcal{B}$ is present or absent, and the dictionary between the two descriptions is the *algebraic normal form*,

$$\phi(q) \;=\; \bigoplus_{\mathcal{B}} \rho_\mathcal{B} \prod_{i \in \mathcal{B}} b_i(q) \;=\; \bigoplus_{\mathcal{B} \subseteq \operatorname{supp}(q)} \rho_\mathcal{B}, \qquad \rho_\mathcal{B} \;=\; \bigoplus_{\operatorname{supp}(q) \subseteq \mathcal{B}} \phi(q). \tag{84}$$

It is worth being explicit about which symbol does what. On the left q is the free variable, the question being asked, and $\mathcal{B}$ a dummy running over all Q monomials: this is *evaluation*, coefficients in, one answer out. The middle expression is the same sum with the dead terms dropped, since $\prod_{i \in \mathcal{B}} b_i(q)$ equals 1 precisely when $\mathcal{B} \subseteq \operatorname{supp}(q)$. On the right $\mathcal{B}$ is the free variable, the coefficient being extracted, and q the dummy, running over the subcube below

$\mathcal{B}$: this is *inversion*, truth table in, one coefficient out. So the whole dictionary is one containment read in both directions, the right-hand formula being Möbius inversion. The *degree* $\deg\phi$ is the largest $|\mathcal{B}|$ carrying $\rho_{\mathcal{B}} = 1$; it is the complexity notion of this section.

Both directions are then literally one matrix. Write T for the *truth table* of a map read as a vector, $T[q] = \phi(q)$, with the questions in the descending order of Table 4, and ρ for the coefficient vector with the bit sets in the matching order. Per bit the kernel is upper triangular, and on n bits its tensor power is the subset indicator: with rows and columns descending $11, 10, 01, 00$ (per bit: $1, 0$),

$$Z = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \left(Z^{\otimes n}\right)_{\mathcal{B},q} = \mathbf{1}[\,\mathrm{supp}(q) \subseteq \mathcal{B}\,], \qquad Z^{\otimes 2} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{85}$$

where $\mathbf{1}[\cdot]$ evaluates a statement to a bit, so that row $\mathcal{B}$ is the indicator of the subcube below $\mathcal{B}$. Under the identification of bit sets with questions the two halves of (84) are that one matrix hitting the two vectors,

$$\rho \;=\; Z^{\otimes n}\, T, \qquad T \;=\; Z^{\otimes n} \rho, \tag{86}$$

at any n: to read off a map's polynomial, transform its truth table; to tabulate a polynomial, transform its coefficients. The two are the same operation because over $\mathbb{F}_2$ the kernel is its own inverse, as the one-bit case already shows: Z^2 has entries $1, 2, 0, 1$, and $2 = 0$, so $Z^2 = I$ and $(Z^{\otimes n})^2 = (Z^2)^{\otimes n} = I$ with it.

Computing ρ is then a matter of ordering the work, and here the tensor structure pays. Done literally, (84) sums each coefficient's subcube from scratch, at a cost of $\sum_{\mathcal{B}} 2^{|\mathcal{B}|} = 3^n$ XORs, re-adding the deep subcubes over and over. But $Z^{\otimes n} = Z_{n-1} \cdots Z_1 Z_0$, where Z_i applies Z along the axis of bit i and the identity on the other $n-1$ axes, and the factors commute. The transform is therefore n passes, one per bit, and a single pass is just the 2×2 kernel acting on every pair of table slots that differ in bit i alone: the slot with $b_i = 0$ is left as it stands (bottom row $(0, 1)$), the slot with $b_i = 1$ absorbs it (top row $(1, 1)$). In place, for each bit i and each slot $\mathcal{B}$ with $i \in \mathcal{B}$,

$$T[\mathcal{B}] \;\longleftarrow\; T[\mathcal{B}] \oplus T[\mathcal{B} \setminus \{i\}]. \tag{87}$$

Follow one slot through. After the pass over bit 0, slot $\mathcal{B}$ holds the XOR of the two original entries reachable by clearing bit 0; after the pass over bit 1, of the four reachable by clearing bits 0 and 1; after all n passes, of every entry reachable by clearing any subset of the bits set in $\mathcal{B}$. Those entries are exactly the q with $\mathrm{supp}(q) \subseteq \mathcal{B}$, so the slot now holds $\rho_{\mathcal{B}}$: each pass extends the subcube by one dimension, and every partial XOR is shared by all the slots above it rather than recomputed. Each pass touches half the table once, for $n\,2^{n-1}$ XORs in place of 3^n. This is the FFT butterfly with XOR in place of addition, and because $Z^2 = I$, running the very same passes again carries coefficients back to the truth table.

EXAMPLE 32

Take NAND, map $j = 7$ of Table 4, which answers 0 only when both input bits are 1, so that $T = \big(\phi(11), \phi(10), \phi(01), \phi(00)\big) = (0, 1, 1, 1)$. Its four supports are $\mathrm{supp}(11) = \{b_1, b_0\}$, $\mathrm{supp}(10) = \{b_1\}$, $\mathrm{supp}(01) = \{b_0\}$ and $\mathrm{supp}(00) = \emptyset$: the same four bit sets that label the four monomials. For each $\mathcal{B}$, gather the questions whose support fits inside it. As the set grows, so does its subcube:

$$\begin{aligned}
&\mathcal{B} = \emptyset && \{q : \mathrm{supp}(q) \subseteq \mathcal{B}\} = \{00\},\\
&\mathcal{B} = \{b_0\} && \{q : \mathrm{supp}(q) \subseteq \mathcal{B}\} = \{01, 00\},\\
&\mathcal{B} = \{b_1\} && \{q : \mathrm{supp}(q) \subseteq \mathcal{B}\} = \{10, 00\},\\
&\mathcal{B} = \{b_1, b_0\} && \{q : \mathrm{supp}(q) \subseteq \mathcal{B}\} = \{11, 10, 01, 00\}.
\end{aligned}$$

Each coefficient is the XOR of T over its own subcube, so the four lines below have identical shape and only the subcube grows:

$$\begin{aligned}
\rho_\emptyset &= \phi(00) && = 1 && = 1,\\
\rho_{b_0} &= \phi(01) \oplus \phi(00) && = 1 \oplus 1 && = 0,\\
\rho_{b_1} &= \phi(10) \oplus \phi(00) && = 1 \oplus 1 && = 0,\\
\rho_{b_1 b_0} &= \phi(11) \oplus \phi(10) \oplus \phi(01) \oplus \phi(00) && = 0 \oplus 1 \oplus 1 \oplus 1 && = 1.
\end{aligned}$$

Stacking those four lines is (86) at $n = 2$,

$$\begin{pmatrix} \rho_{b_1 b_0} \\ \rho_{b_1} \\ \rho_{b_0} \\ \rho_\emptyset \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \phi(11) \\ \phi(10) \\ \phi(01) \\ \phi(00) \end{pmatrix} = \begin{pmatrix} 0 \oplus 1 \oplus 1 \oplus 1 \\ 1 \oplus 1 \\ 1 \oplus 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

where the two middle rows show the cancellation that makes a monomial absent.

Read the other way, (86) evaluates. Ask $q = 01$: its support is $\{b_0\}$, so row 01 gives $\phi(01) = \rho_\emptyset \oplus \rho_{b_0} = 1 \oplus 0 = 1$, which is what $b_1 b_0 \oplus 1$ returns at $b_1 = 0$, $b_0 = 1$. Hence $\mathrm{NAND} = b_1 b_0 \oplus 1$, of degree 2: the top monomial is present, NAND is not affine, and its truth table knows it.

For readers of Section 2.5: Z is the triangular half of the Walsh–Hadamard kernel (37), since in the same $1, 0$ ordering

$$W = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} = Z \begin{pmatrix} -2 & 0 \\ 0 & 1 \end{pmatrix} Z^{\mathsf{T}}, \tag{88}$$

the matrix form of $\prod_q (1 - 2y_q) = \sum_T (-2)^{|T|} \prod_{q \in T} y_q$: the ledger transform is two Möbius passes with a $(-2)^{|T|}$ rescaling between them, and reducing mod 2 leaves exactly the XOR butterfly.

EXAMPLE 33

The butterfly on every map of the guiding example, truth tables $T = (\phi(11), \phi(10), \phi(01), \phi(00))$ – so that T read as a binary numeral *is* the index j – and coefficients ordered $(\rho_{b_1 b_0}, \rho_{b_1}, \rho_{b_0}, \rho_{\emptyset})$.

j	map	T	ρ	polynomial	deg
0	FALSE	$(0,0,0,0)$	$(0,0,0,0)$	0	0
1	NOR	$(0,0,0,1)$	$(1,1,1,1)$	$b_1 b_0 \oplus b_1 \oplus b_0 \oplus 1$	2
2	$\neg b_1 \wedge b_0$	$(0,0,1,0)$	$(1,0,1,0)$	$b_1 b_0 \oplus b_0$	2
3	$\neg b_1$	$(0,0,1,1)$	$(0,1,0,1)$	$b_1 \oplus 1$	1
4	$b_1 \wedge \neg b_0$	$(0,1,0,0)$	$(1,1,0,0)$	$b_1 b_0 \oplus b_1$	2
5	$\neg b_0$	$(0,1,0,1)$	$(0,0,1,1)$	$b_0 \oplus 1$	1
6	XOR	$(0,1,1,0)$	$(0,1,1,0)$	$b_1 \oplus b_0$	1
7	NAND	$(0,1,1,1)$	$(1,0,0,1)$	$b_1 b_0 \oplus 1$	2
8	AND	$(1,0,0,0)$	$(1,0,0,0)$	$b_1 b_0$	2
9	XNOR	$(1,0,0,1)$	$(0,1,1,1)$	$b_1 \oplus b_0 \oplus 1$	1
10	b_0	$(1,0,1,0)$	$(0,0,1,0)$	b_0	1
11	$b_1 \to b_0$	$(1,0,1,1)$	$(1,1,0,1)$	$b_1 b_0 \oplus b_1 \oplus 1$	2
12	b_1	$(1,1,0,0)$	$(0,1,0,0)$	b_1	1
13	$b_0 \to b_1$	$(1,1,0,1)$	$(1,0,1,1)$	$b_1 b_0 \oplus b_0 \oplus 1$	2
14	OR	$(1,1,1,0)$	$(1,1,1,0)$	$b_1 b_0 \oplus b_1 \oplus b_0$	2
15	TRUE	$(1,1,1,1)$	$(0,0,0,1)$	1	0

Table 5: The guiding example in both bases: every map of Table 4 as a truth table T and as a coefficient vector ρ, with its polynomial and its degree.

Read the table from both ends at once: j and $15-j$ are negations of each other, and their coefficient vectors differ in $\rho_{\emptyset}$ alone, negation flipping the constant and nothing else. Four maps are fixed points of the butterfly, $\rho = T$: FALSE, XOR, AND and OR. The eight with $\rho_{b_1 b_0} = 0$ are the affine ones, class $\mathrm{RM}(1,2)$, and the other eight are the curved ones that the top monomial buys.

4.3.2 Complexity classes and the sampling rule

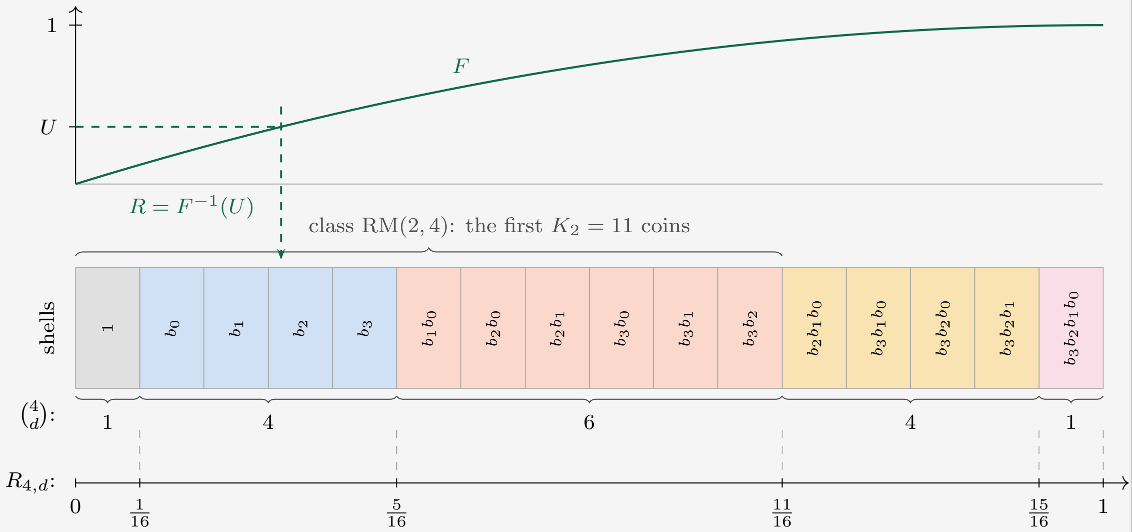
Grade the hypothesis space by degree. The class of all maps of degree at most d is a linear space of dimension $K_d = \sum_{i \le d} \binom{n}{i}$ – one coefficient bit per allowed monomial – and the classes are nested, from the two constants ($d = 0$) to the full map space ($d = n$). In coding theory this class is the Reed–Muller code $\mathrm{RM}(d, n)$. Two words are used precisely below: the *class* d collects all maps of degree at most d (nested), while its outermost layer, the maps of degree exactly d, is the *shell* d (disjoint). The class's *rate* is its dimension per question,

$$R_{n,d} = \frac{K_d}{Q} \in (0, 1], \tag{89}$$

and it will turn out to be the fraction of a full run at which the class is learned out. Consecutive rates slice $(0, 1]$ into *rate gaps* $(R_{n,d-1}, R_{n,d}]$ of lengths $\binom{n}{d}/2^n \le \sqrt{2/\pi n}$: one cell per class, a partition that refines as n grows.

EXAMPLE 34

The monomial shelf at $n = 4$: one box per monomial, grouped by degree into shells. Each box is one coefficient bit, a fair coin when drawing within a class, so the shelf prefix of length K_d *is* the class $\mathrm{RM}(d, 4)$ measured in bits, holding 2^{K_d} maps, and shell d is the $\binom{4}{d}$ boxes the class adds. The geometry is fixed once and for all – dividing the edge by $Q = 16$ puts the class boundaries at the rates $R_{4,d} = K_d/16$, with the shells as the rate gaps – and none of it depends on F. The statistic enters only through the dart: draw U uniformly on the vertical axis, bounce off the graph of F, and land at $R = F^{-1}(U)$, drawn here for $F(x) = 1 - (1 - x)^2$, the smaller of two uniform draws – the $r = 2$ member of the constant-leverage family (100) – which tilts the landing law toward the simple end. For $F(x) = x$ the bounce is trivial, and each shell is hit with probability exactly its width.



Reading the shelf against the table below it: the shell widths $1, 4, 6, 4, 1$ are the populations' logarithmic footprints, while the map *counts* 2^{K_d} explode left to right – the bulge in boxes is mild, the bulge in maps is the cliff.

Why sample a shell rather than weight the maps? The natural Occam move, $p_j \propto e^{-\beta \deg \phi_j}$, fails by a scaling argument. There are 2^{K_k} maps of degree at most k, so raising the cutoff by one buys $\binom{n}{k+1} \ln 2$ nats of entropy against a linear price β; no single temperature balances jumps of that size at more than one k. The ensemble concentrates on the shell $\binom{n}{d^*} \approx \beta/\ln 2$ – at most two adjacent degrees – and every map outside it retains only vanishing relative weight. A single temperature hears a thin shell of the hypothesis space and nothing else. The cliff in numbers: the share of maps of degree exactly d is $2^{K_d - Q} - 2^{K_{d-1} - Q}$, and

degree	share	$n = 2$	$n = 4$	$n = 6$	$n = 8$
n	$\frac{1}{2}$	50%	50%	50%	50%
$n - 1$	$\frac{1}{2} - 2^{-(n+1)}$	37.5%	46.9%	49.2%	49.8%
$n - 2$	$\approx 2^{-(n+1)}$	12.5%	3.1%	0.78%	0.20%
$n - 3$	$\approx 2^{-(n+1+\binom{n}{2})}$	—	$4.6 \cdot 10^{-4}$	$2.4 \cdot 10^{-7}$	$7.3 \cdot 10^{-12}$
$\leq n - 4$	rest	—	$3.1 \cdot 10^{-5}$	$2.3 \cdot 10^{-13}$	$1.0 \cdot 10^{-28}$

Half of all maps sit at full degree, all but a $2^{-(n+1)}$ sliver in the top two shells, and everything below is doubly exponentially rare: this is the population any per-map weight has to fight. The repair prices the *shell*, not the map:

1. draw a single random number $R \in [0, 1]$ (the *statistic*) from a fixed law μ with CDF $F(x) = \Pr(R \le x)$ and, when it exists, density $f = F'$;
2. let D be the class whose rate gap contains R;
3. draw the truth uniformly from class D: each allowed coefficient by a fair coin.

In every expectation below the random object is this one draw R; the $R_{n,d}$ are the deterministic grid it lands among. The class label D is latent – Werner never observes it – and is one value among $n+1$, worth at most $\log_2(n+1)$ bits. Integrating the lottery, and using nestedness (a map of degree e belongs to every class $d \ge e$),

$$w_d \;=\; F(R_{n,d}) - F(R_{n,d-1}), \qquad p_j \;=\; \sum_{d \ge \deg \phi_j} w_d\, 2^{-K_d}, \tag{90}$$

with $\sum_d w_d = 1$ telescoping. If F increases across every gap (any positive density), then every $w_d > 0$, and two properties follow at once: p_j is *strictly decreasing in the degree*, each step down costing a factor of order $2^{-\binom{n}{d}}$ – a genuine Occam – and $p_j \ge w_n 2^{-Q} > 0$ for every single map: full support, with no noise channel, at every n and in the limit. Every shell, and every map, gets a chance. Semantically, F budgets the *shells* while nestedness orders the *maps*: w_d is, up to a $2^{-\binom{n}{d}}$ sliver leaking in from higher classes, the total weight of the degree-d shell, split evenly within it. The per-map Occam ordering therefore holds for every F, and F decides only how much aggregate weight each shell carries – a populous shell can outweigh a simpler one even though each of its members is individually lighter.

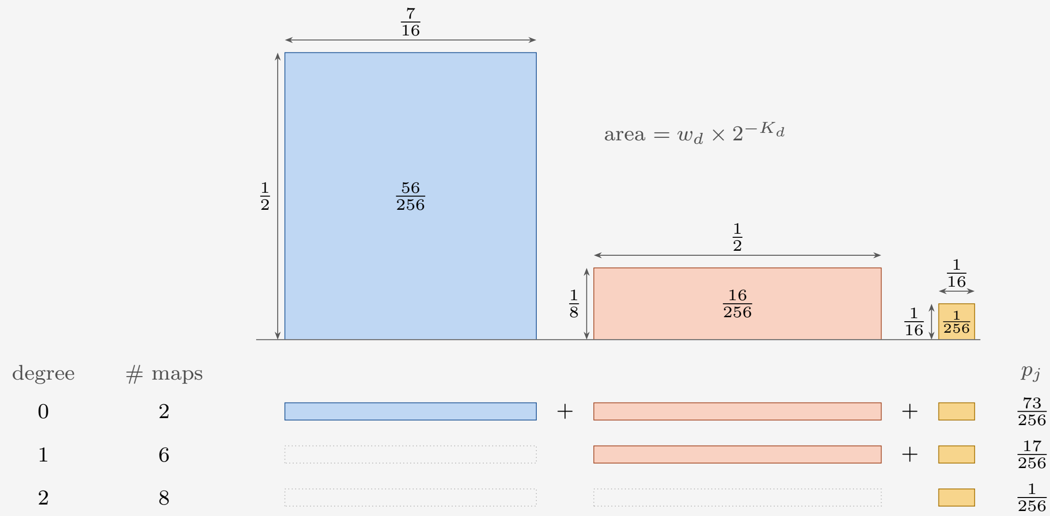
A judicious choice of F is what puts the correct portion of aggregate belief on the complexity that Werner expects from nature.

EXAMPLE 35

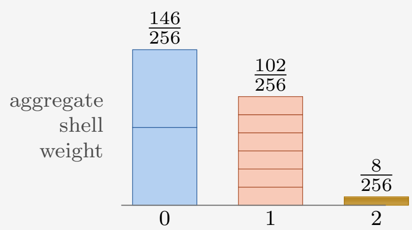
The prior (90) at $(n,m)=(2,1)$ with the guiding statistic $F(x)=1-(1-x)^2$, the $r=2$ member of (99). The three classes have

$$(K_0,K_1,K_2)=(1,3,4),$$
$$(w_0,w_1,w_2)=(\tfrac{7}{16},\tfrac{1}{2},\tfrac{1}{16}),$$

so the rates K_d/Q are $\frac{1}{4}$, $\frac{3}{4}$ and 1. Class d offers a map it contains a rectangle: width w_d, the chance the lottery calls that class, by height 2^{-K_d}, the chance the class then calls that map. A map of degree e lies in every class $d \geq e$, so it collects every block from e rightward – all the blocks a harder map collects, and one more besides.



Each step up in degree curtails the sum from the left, so p_j falls with the degree by construction rather than by fiat: the two constants of Table 4 take all three blocks, $\frac{73}{256}$, the six affine maps lose the first, $\frac{17}{256}$, and the eight curved maps keep only the last, $\frac{1}{256}$. The blocks are drawn to scale, and that is the whole Occam story in one picture. A concave F makes the constants' block the widest draw but for the middle one, and having only two members to split between makes it much the tallest, so it carries 56 of the 73 parts; the full class is a chip at both ends and contributes 1. The check $2\cdot 73+6\cdot 17+8\cdot 1=256$ closes the distribution.



4.3.3 Three terms for the leverage average

The exact finite law (56), in the chapter's notation (79), needs three ingredients: the prior table entropy $Q\,G_{n,1}$, the received entropy $G_{n,\ell}$, and the increment $\gamma_{n,\ell}$. Each reduces to a statement about F.

The table entropy. A class is a linear code; make the structure explicit once. The *generator matrix* Ω of class d is the $K_d \times Q$ matrix over $\mathbb{F}_2$ whose rows are the monomials the class allows, evaluated at every question,

$$\Omega_{\mathcal{B},q} \;=\; \prod_{i\in\mathcal{B}} b_i(q) \;=\; \mathbf{1}[\,\mathcal{B} \subseteq \operatorname{supp}(q)\,], \qquad |\mathcal{B}| \le d, \tag{91}$$

so that a uniform class member is $\psi = \xi^{\mathsf{T}}\Omega$ with ξ uniform on $\{0,1\}^{K_d}$. Every column of Ω is nonzero (its constant-monomial entry is 1), so each single answer is the image of a uniform ξ under a surjective linear map onto $\mathbb{F}_2$: a fair bit, in every class, and a mixture of fair bits is a fair bit. At every finite n, with no limit taken and no reference to μ,

$$G_{n,1} = 1, \qquad H\big(M^{(0)}\big) = Q. \tag{92}$$

The prior predicts nothing about any single answer; its entire content is correlational.

EXAMPLE 36

The generator matrix at $n = 2$: the questions across the top in descending order, the monomials down the side in degree order, and the entry where they meet the value (91) gives. The three boxes are the three classes, each keeping the first K_d rows, which is what makes them nest:

$$\Omega \;=\; \begin{array}{c|cccc|l}
\mathcal{B}\backslash\mathbf{q} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} & \\
\hline
1 & 1 & 1 & 1 & 1 & d=0 \\
b_0 & 1 & 0 & 1 & 0 & \\
b_1 & 1 & 1 & 0 & 0 & d=1 \\
b_1b_0 & 1 & 0 & 0 & 0 & d=2
\end{array}$$

A class member is $\psi = \xi^{\mathsf{T}}\Omega^{(d)}$, the XOR of the rows that the coin string ξ switches on. The K_d rows are independent, so $\xi \mapsto \psi$ is a bijection onto the class: flipping K_d fair coins, one per allowed monomial, lands on each of the 2^{K_d} maps exactly once, which is step 3 of the sampling rule and why no map inside a class is favoured over another. Three draws, each extending the last:

$$\begin{array}{llll}
\xi = (1) & \Rightarrow & \psi = (1,1,1,1) & = \text{TRUE}, \\
\xi = (1,0,1) & \Rightarrow & \psi = (1,1,1,1) \oplus (1,1,0,0) = (0,0,1,1) & = \neg b_1, \\
\xi = (1,0,1,1) & \Rightarrow & \psi = (0,0,1,1) \oplus (1,0,0,0) = (1,0,1,1) & = b_1 \to b_0.
\end{array}$$

Every column carries the constant row's 1, so flipping $\xi_\emptyset$ flips the answer to every question at once: the 2^{K_d} maps pair off into complementary halves and $M_{a,q} = \frac{1}{2}$ at every question, in every class, for every F. The table starts knowing nothing, $H(M^{(0)}) = Q$. Read against (85), Ω is $Z^{\otimes n}$ transposed, its rows sorted by degree so that every class is a prefix.

The received entropy. Take the easy case first: suppose Werner were told the class. Inside class d the truth is $\psi = \xi^{\mathsf{T}}\Omega$ with ξ a string of K_d fair coins, so his answers are uniform on the image of the columns he has asked and their entropy is that submatrix's rank. Every fresh column independent of its predecessors buys one whole bit; once the rank reaches K_d the coins are pinned, ξ is known, and every answer after that is forced and worth nothing. Average that behaviour over the classes with weight $w_d = \Pr(D = d)$ and the received entropy would be settled.

Two things spoil it at finite n, and both dissolve in the limit. Asking K_d questions is not the same as spending the rank, because the columns are not in general position: the smallest dependent set has 2^{d+1} members, the minimum distance of the dual code $\mathrm{RM}(n-d-1, n)$, and for the middle classes that falls well short of K_d – four questions against $K_1 = 5$ at $n = 4$, eight against $K_2 = 37$ at $n = 8$. Four questions whose bit patterns XOR to zero already determine each other for an affine map, whatever else is unsettled. So a question can be redundant early and the rank often is not full until late: the switch from one bit to none is a smeared step, not a sharp one, though it encloses area K_d regardless. Only the two ends escape, the constants and the full space being the codes for which every K_d columns really are independent. And Werner is not told the class; he has to read it off the answers, and that costs. Computing $H(\psi(\mathcal{S}), D)$ by the chain rule in both orders prices the second of these,

$$H\big(\psi(\mathcal{S})\big) = \sum_d w_d\, H_d\big(\psi(\mathcal{S})\big) + I\big(D; \psi(\mathcal{S})\big), \qquad 0 \le I \le \log_2(n+1): \tag{93}$$

mixture entropy is mean class entropy plus what the answers reveal about the label, and the label is cheap – $o(1)$ bits per question. Within class d the answers on $\mathcal{S}$ are $\xi^{\mathsf{T}}\Omega_{\cdot,\mathcal{S}}$, uniform on the image of the column submatrix, so $H_d(\psi(\mathcal{S})) = \mathrm{rank}(\Omega_{\cdot,\mathcal{S}})$ exactly: the number of $\mathbb{F}_2$-linearly independent columns, the number of bits needed to describe those answers.

EXAMPLE 37

Where the classes part is what a set of columns spans. Ask 11, then 10, then 01 of class 1: the boxed submatrix $\Omega^{(1)}_{\cdot,\mathcal{S}}$ is square and of full rank 3, so each of the three answers is a fresh bit.

$$\Omega^{(1)} = \begin{array}{c|cccc} \mathcal{B}\backslash\mathbf{q} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} \\ \hline 1 & 1 & 1 & 1 & 1 \\ b_0 & 1 & 0 & 1 & 0 \\ b_1 & 1 & 1 & 0 & 0 \end{array}$$

But $\mathrm{rank}\,\Omega^{(1)} = K_1 = 3$ across all four columns, and the box has already used the rank up. The fourth column is the XOR of the boxed three, so the answer to 00 is spoken for before it is asked: the rows span the even-weight code, whose parity check reads $\psi(11) \oplus \psi(10) \oplus \psi(01) \oplus \psi(00) = 0$. An affine truth pays one bit a question and runs out at its own rate $R_{2,1} = \frac{3}{4}$. Class 2 has full rank 4 and never runs out; class 0 is spent after a single answer, at $R_{2,0} = \frac{1}{4}$.

Three facts pin the limit of the class average $G^{(d)}_{n,\ell} = \mathbb{E}\,\mathrm{rank}$. The mean increment $\gamma^{(d)}_{n,\ell} = \Pr[\text{a fresh column is independent of } \ell \text{ random ones}]$ is nonincreasing in ℓ, by rank submodularity (what is dependent on few columns is dependent on more). The increments telescope to the dimension, $\sum_{\ell<Q} \gamma^{(d)}_{n,\ell} = K_d$: the finite-$n$ area theorem, received equals learnable. And the one imported result – Reed–Muller codes achieve capacity on the binary erasure channel (Kudekar, Kumar, Mondelli, Pfister, Şaşoğlu and Urbanke, 2017) – says that revealing any fraction beyond the rate determines a fresh answer with probability tending to one:

$$R_{n,d_n} \to R \in (0,1) \quad \Longrightarrow \quad \lim_{n\to\infty} \gamma^{(d_n)}_{n,\lfloor tQ \rfloor} = 0 \quad \text{for every fixed } t > R. \tag{94}$$

A bounded nonincreasing sequence with area R_d and no mass above R_d has nowhere to sit but at height 1 below, so the class increments converge to a pure step, written with the

Heaviside function $\theta(x) = \mathbf{1}[x > 0]$:

$$\lim_{n\to\infty} \gamma^{(d)}_{n,\lfloor tQ\rfloor} = \theta(R_d - t), \qquad \frac{G^{(d)}_{n,\lfloor tQ\rfloor}}{Q} \longrightarrow \int_0^t \theta(R_d - x)\,dx = \min(t, R_d): \tag{95}$$

each class pays one full bit per question until its clock stops at its own rate, and nothing after. Finally, round the draw R up to the top of its rate gap: since $\min(t,\cdot)$ is 1-Lipschitz and the rounding moves R by at most $\sqrt{2/\pi n}$, the class average converges to an average over the statistic,

$$\frac{G_{n,\lfloor tQ\rfloor}}{Q} \longrightarrow \mathbb{E}_\mu\big[\min(t,R)\big] = \int_0^t \big(1 - F(x)\big)\,dx =: g(t), \tag{96}$$

the last step by layer cake: pointwise $\min(t,R) = \int_0^t \theta(R-x)\,dx$, and $\mathbb{E}\,\theta(R-x) = 1 - F(x)$. Received information per question is an average of clipped clocks, the area under the survival function.

The increment. The sequence $\ell \mapsto G_{n,\ell}$ is concave (increments nonincreasing), and slopes of concave functions converge wherever the limit is differentiable, so at every continuity point of F

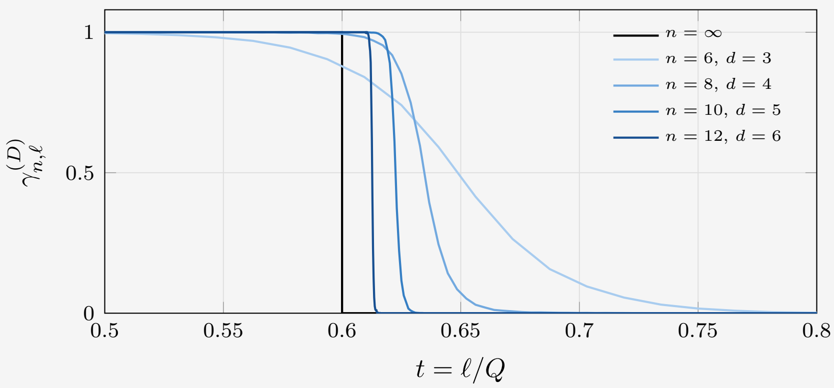
$$\gamma_{n,\lfloor tQ\rfloor} \longrightarrow g'(t) = 1 - F(t) = \Pr(R > t). \tag{97}$$

This is the heart of the construction: the mean conditional entropy of a fresh answer is the probability that the true class is not yet exhausted. The survival function of the statistic is not a postulate; it is the increment term of the finite law.

EXAMPLE 38

Where γ comes from, at $F(x) = 1 - (1-x)^2$. Conditional on the latent class D, the answers on $\mathcal{S}$ are $\xi^{\mathsf{T}}\Omega_{\cdot,\mathcal{S}}$, so their entropy is the rank of that column submatrix, and $\gamma^{(D)}_{n,\ell}$ is the chance a fresh column is still independent of the ℓ already asked. Nothing in that depends on F; the statistic enters only by choosing which class to watch.

Hold the draw fixed at $R = 0.6$ and let n grow. The sampling rule hands back whichever class's rate gap contains it, so the class index climbs with n, taking $d = 3, 4, 5, 6$ below, while R itself stays put; by (94) the increment tends to the black step at R. Outside the window γ is flat: one to the left, nothing to the right.

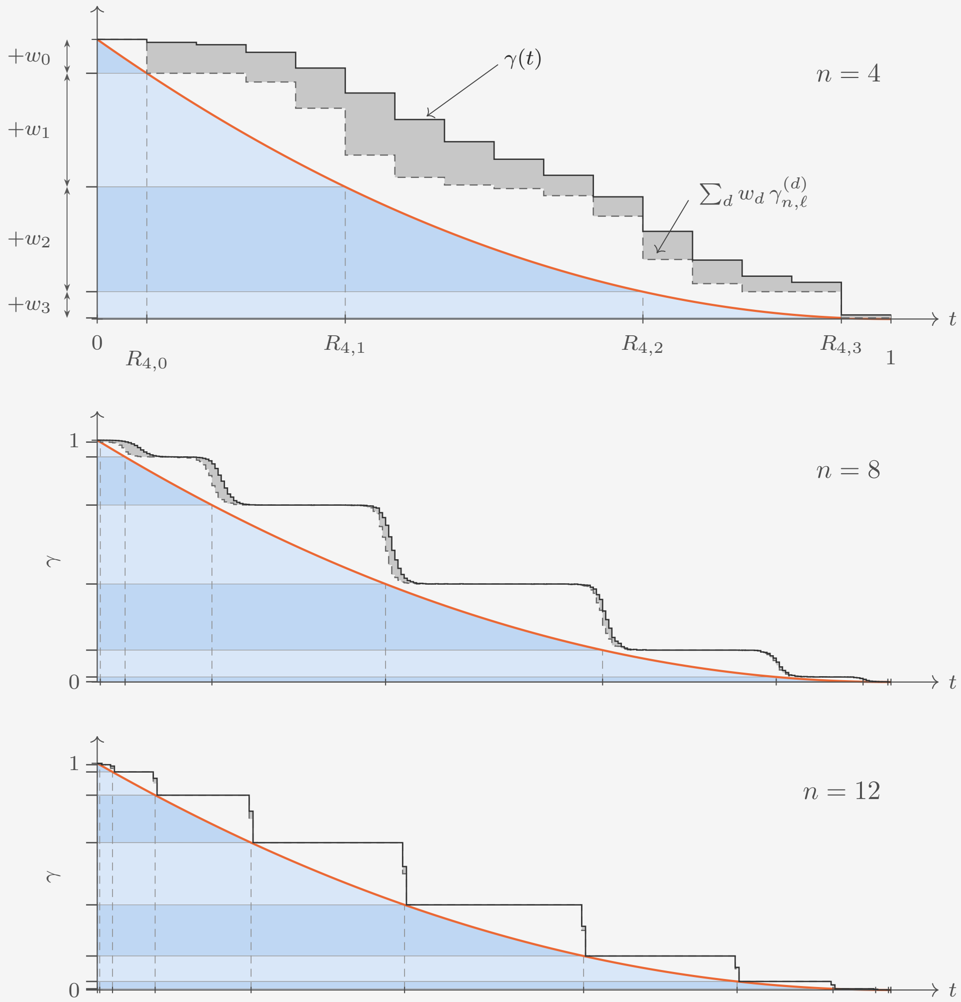


Two things converge, at very different speeds. The *shape* goes fast: the smear from $\gamma = 0.9$ down to 0.1 measures six questions at every n here, so as a fraction of the run it closes like $1/Q$, from 0.094 at $n = 6$ to 0.0015 at $n = 12$. The *position* goes slowly: each curve steps at its own rate, which is R rounded up to the grid and so overshoots by at most the gap width $\binom{n}{d}/2^n \le \sqrt{2/\pi n}$ – here $0.056, 0.037, 0.023, 0.013$, closing on R but only at that $1/\sqrt{n}$, the same rate the rounding step before (96) has to absorb. Whatever the overshoot, each curve encloses exactly its own rate: the increments telescope to the dimension, $\sum_{\ell<Q} \gamma^{(d)}_{n,\ell} = K_d$, at every n and with no approximation.

EXAMPLE 39

Averaging over D. Slice the belief by class: a horizontal slab of thickness w_d each, stacked longest-lived first, the arrows on the axis reading off the weights. The right edge is the survival curve itself, which enters slab d at the rate $R_{n,d-1}$ and leaves it at $R_{n,d}$ – the dashed verticals – so the shaded region has height $\gamma(t) = 1 - F(t)$ everywhere and area $g(t)$ to the left of t: the layer cake, drawn. Raising n does not move the region, it only cuts it into more slabs, the gaps closing like $\binom{n}{d}/2^n \le \sqrt{2/\pi n}$.

Over it sit the two exact increments, one question answered under two states of knowledge. Dashed: what a fresh answer is worth to someone already *told* the class, $\sum_d w_d \gamma^{(d)}_{n,\ell}$. Solid: what it is worth to Werner, who is not. By (93) the mixture is more uncertain than its components by exactly what the answers say about the label, so the shaded wedge between the two is the label, and the white between the dashed curve and the region is the coarseness of the rate grid.



The relative value of the label vanishes with n as $\log_2(n+1)/Q$. The steps emerge because at any time a class has either been settled, and contributes nothing, or is still running, and contributes its full bit at every question, until the next class boundary is reached. The steps hug $1 - F$ as $n \to \infty$.

4.3.4 The macroscopic curve

Substitute (92), (96) and (97) into (56) and cancel the common factor Q; the numerator tidies to $1-(1-t)(1-F(t)) = t+(1-t)F(t)$, and

$$L_F(t) \;=\; \frac{t+(1-t)\,F(t)}{\displaystyle\int_0^t \big(1-F(x)\big)\,dx}. \tag{98}$$

This is the master law (81) realized with $h_0 = 1$, $\gamma(t) = 1-F(t)$ and $g(t) = \int_0^t (1-F)$; the hypotheses hold for free, γ being bounded and nonincreasing by construction. The numerator has its own reading: destroyed density is the asked fraction t plus $F(t)$ per unasked column, since an unasked answer is deducible exactly when the class died before t. Asked plus deduced, over received. Expanding at the ends, $L_F(0^+) = 1 + f(0^+)$ and $L_F(1) = 1/\mathbb{E}_\mu[R]$: the curve starts at baseline plus the density of the statistic at zero, and completes at the reciprocal mean rate. Conversely, any nonincreasing profile with values in $[0,1]$ is the survival function of exactly one law μ (read the layer cake backwards), so (98) is not one family among many: the sampling rule realizes every admissible curve of (81) at $m = 1$, and the statistic is the profile. A parallel construction from blocks of questions, mathematically complete in the same sense but less aligned with modeling a learner, is collected in Appendix B.

4.3.5 Toy statistics: min and max draws

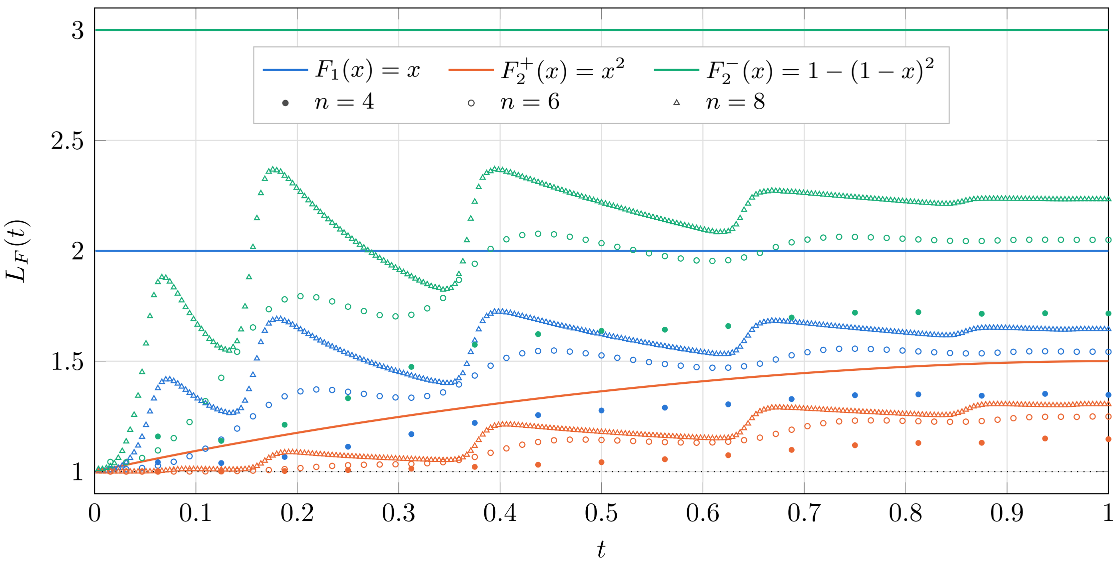


Figure 15: The two families of (100) and (102) at $r = 2$, meeting at $r = 1$; their profiles are in Figure 16. The larger of two draws ($F_2^+(x) = x^2$, orange) delays its deductions and climbs from the baseline to $L_2^+(1) = \frac{3}{2}$; the smaller of two draws ($F_2^-(x) = 1 - (1-x)^2$, green) holds the constant $L_2^- \equiv 3$; between them lies the single uniform draw ($F_1(x) = x$, blue), the one statistic the families share, at $L \equiv 2$. Dots are finite-n leverage: filled $n = 4$ (all 2^{16} question sets enumerated, exact), circles $n = 6$ and triangles $n = 8$ (Monte Carlo over question orders, subset entropies exact via the nested rank profile, latent-label information included). Convergence is slow from below: the label costs $O(\log n)$ of the Q available bits, and the mean rate approaches its limit only as $1/\sqrt{n}$. The left edge is a boundary layer: $L_F(0^+) = 1 + f(0^+)$ presumes $1 \ll \ell \ll Q$, while at $\ell = 1$ the only exhaustible class is the constants and $\langle L_1 \rangle = 1 + (Q-1)\, w_0^2/(2 \ln 2) + O(w_0^3)$, the early deductions being spent on identifying the latent label. The finite curves also scallop, most visibly at $n = 8$: each surge is one class being learned out near its rate K_d/Q, a discreteness the limit smooths away as the rate grid refines.

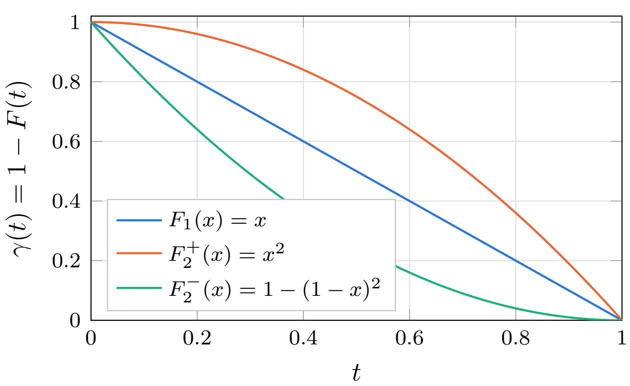


Figure 16: The profiles behind Figure 15: the survival function $\gamma(t) = 1 - F(t)$ of each statistic, which by (97) is the mean entropy of a fresh answer at macroscopic time t. Convex, straight and concave F give concave, straight and convex profiles; the smaller of two draws spends its predictive power earliest, the larger of two hoards it.

Two one-parameter families are enough to see the law work. Draw r independent uniforms on $[0,1]$ and keep one of them: keeping the smaller drags the rate toward zero, keeping the larger pushes it toward one, and r sets how hard. Every curve in Figure 15 comes from one of the two.

The smaller of r draws. All r land above x with probability $(1-x)^r$, so

$$F_r^-(x) = 1-(1-x)^r, \qquad \gamma_r^-(t) = (1-t)^r, \qquad \mathbb{E}[R] = \frac{1}{r+1}. \tag{99}$$

The density $r(1-x)^{r-1}$ decreases, so budget is pushed onto the low shells. Substituting into (98), the numerator folds up and the denominator comes back carrying the same bracket,

$$L_r^-(t) \;=\; \frac{t+(1-t)\big(1-(1-t)^r\big)}{\displaystyle\int_0^t (1-x)^r\,dx} \;=\; \frac{1-(1-t)^{r+1}}{\big(1-(1-t)^{r+1}\big)/(r+1)} \;=\; r+1 \tag{100}$$

at every macroscopic time: a whole ladder of constant curves, one rung per r, r deduced bits riding on each received one. Both endpoint identities below (98) return that same number, $1+f(0^+)=1+r$ at the start and $1/\mathbb{E}[R]=r+1$ at the finish, and with a constant curve they have no choice but to agree.

The larger of r draws. All r land below x with probability x^r, so

$$F_r^+(x) = x^r, \qquad \gamma_r^+(t) = 1-t^r, \qquad \mathbb{E}[R] = \frac{r}{r+1}, \tag{101}$$

an increasing density, budget on the high shells. Nothing telescopes now and the leverage moves,

$$L_r^+(t) \;=\; \frac{t+(1-t)\,t^r}{t-t^{r+1}/(r+1)} \;=\; \frac{(r+1)\big(t+t^r(1-t)\big)}{(r+1)\,t-t^{r+1}}, \tag{102}$$

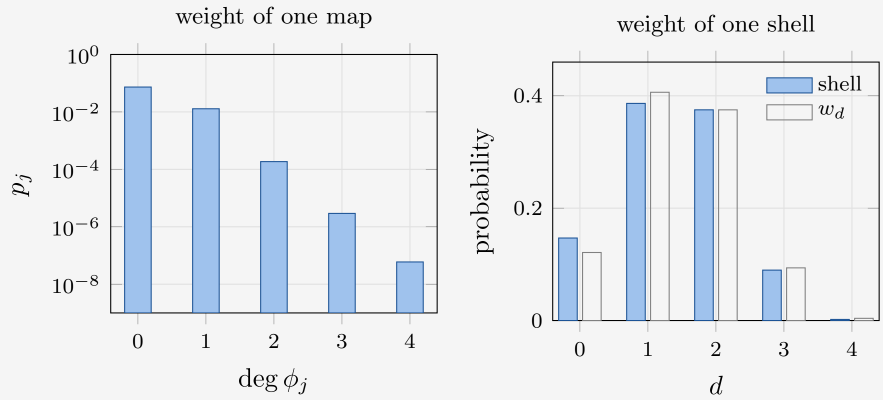
rising monotonically across the run. At the start the density vanishes and $L_r^+(0^+) = 1+f(0^+) = 1$ for every $r>1$: no leverage at all, each early answer paid in full. At the finish $L_r^+(1) = 1/\mathbb{E}[R] = (r+1)/r$, so the deductions are late-loaded and even the last of them buys only $1/r$ extra. The two finishes are complements, $1/L_r^-(1)+1/L_r^+(1)=1$, because the mean of the smaller and the mean of the larger of the same r draws sum to one. Pushing r below 1 gives up the min-max reading and flips the tilt of F_r^+ back to concave, but with an infinite density at the origin ($F=\sqrt{x}$ at $r=\frac{1}{2}$), and $L_r^+(0^+)$ diverges: front-loaded deduction, the shape of a boundary layer, no longer a constant.

Where they meet. At $r=1$ there is nothing to choose between and both families collapse onto the single uniform draw $F(x)=x$, whose numerator $t(2-t)$ cancels its denominator $t(2-t)/2$ to leave $L\equiv 2$ exactly. That is the hinge of Figure 15: the blue line is the shared $r=1$ point, and the curves either side of it are the $r=2$ members of the two families, L_2^+ climbing from 1 to $\frac{3}{2}$ and $L_2^-\equiv 3$. The uniform draw is also the flattest Occam on offer, each shell receiving exactly the share of coefficient space its degree opens, $w_d=\binom{n}{d}/2^n$. It does not identify its microscopics: pairing questions into equality cliques (Appendix B) produces the same constant 2 by a completely different mechanism.

A tilt, not a switch. The concave direction may look like the natural one, a simplicity prior ought to favour simple worlds, but F never decides *whether* Werner is an Occamist. The classes are nested, so the weight p_j of a single map falls with its degree for every F whatsoever, and even the convex F_r^+, which loads the high shells, still ranks each complicated map below each simple one. What F moves is the aggregate: which shell carries the belief, and with it the leverage, from r deduced bits per received one under F_r^- down to at most $1/r$ under F_r^+.

EXAMPLE 40

The whole construction, priced out at $n = 4$, $m = 1$, with the statistic of the shelf above: $F(x) = 1 - (1 - x)^2$, the smaller of two uniform draws. The five classes have dimensions $K_d = 1, 5, 11, 15, 16$ and rates $R_{4,d} = K_d/16$, so (90) hands out shell budgets $w_d = \frac{31}{256}, \frac{104}{256}, \frac{96}{256}, \frac{24}{256}, \frac{1}{256}$ and, to a map of degree d, the weight $p_j = 7.34 \cdot 10^{-2}$, $1.29 \cdot 10^{-2}$, $1.86 \cdot 10^{-4}$, $2.92 \cdot 10^{-6}$, 2^{-24}.



The two panels answer different questions and their axes say so. On the left, one map: the weight drops by 5.7, then 69, 64, 49 – the factor $2^{\binom{4}{d}}$ that the next class costs in dimension, tilted by the budget ratio w_{d-1}/w_d – six decades across five degrees, which is why the axis is logarithmic. Occam is priced per map, and it is brutal. On the right, one shell: the $2, 30, 2016, 30720, 32768$ maps of each degree carry $14.7\%, 38.6\%, 37.5\%, 9.0\%, 0.2\%$ of the belief, on a linear axis, the peak one step away from the simplest shell and the next shell all but level with it. Degree 1 outweighs the constants better than five to two although each of its thirty members is nearly six times lighter: population beats simplicity in the aggregate, while the per-map ordering never wavers. The open bars are the budgets w_d, and the daylight between the two is the $2^{-\binom{4}{d}}$ shrinkage: half of w_4 at full degree, invisible at $d = 2$ where it is repaid almost exactly by what leaks down from above, and the surplus piling onto the two constants, 0.147 against a budget of 0.121.

Appendix A continues the spike-prior example of Section 4.1. Appendix B collects the block-prior construction and its Bernstein profile language, a block-based route to the same family of macroscopic curves.

A More characteristics of the spike prior

This appendix continues the toy example of Section 4.1: the anatomy of a single question, the fraction of the dependency stock it extracts, and the sharp-prior limit with its order-of-limits structure.

A.1 One question, branch by branch

For a single question abbreviate $P := P_1$ and $U := U_1$, so that $P + (A - 1)U = 1$: the zero answer has probability P, while each particular nonzero answer has probability U. Here *correct* means that the observed answer agrees with the spike map at this question; it does not assert that the entire spike map is the truth. The surprisals are

$$s_{\text{correct}} = -\log_2 P, \qquad s_{\text{wrong}} = -\log_2 U, \qquad \langle s\rangle = P\, s_{\text{correct}} + (A-1)U\, s_{\text{wrong}} = G_1, \tag{103}$$

a particular wrong answer having probability U, not $1-P$. Before asking, every one of the Q columns carries the same answer distribution $(P, U, \ldots, U)$ and hence entropy G_1, so $H_{\text{before}} = H(M^{(0)}) = Q\,G_1$. Conditional on confirmation the next unasked answer has probabilities P_2/P for zero and U_2/P for each nonzero symbol, with entropy

$$H_{\text{c}} := -\frac{P_2}{P}\log_2\frac{P_2}{P} - (A-1)\frac{U_2}{P}\log_2\frac{U_2}{P}, \qquad H_{\text{after,correct}} = (Q-1)H_{\text{c}}, \tag{104}$$

the asked column being fixed while all $Q-1$ unasked columns share that conditional marginal. Refutation instead leaves equal posterior weights on all compatible maps, so every unasked answer is uniform on A symbols and $H_{\text{after,wrong}} = (Q-1)m$. Weighting the two branches,

$$\big\langle H\big(M^{(1)}\big)\big\rangle = P\,H_{\text{after,correct}} + (A-1)U\,H_{\text{after,wrong}} = (Q-1)(G_2 - G_1), \tag{105}$$

the first equality ordinary branch averaging, the second the $\ell = 1$ case of (55).

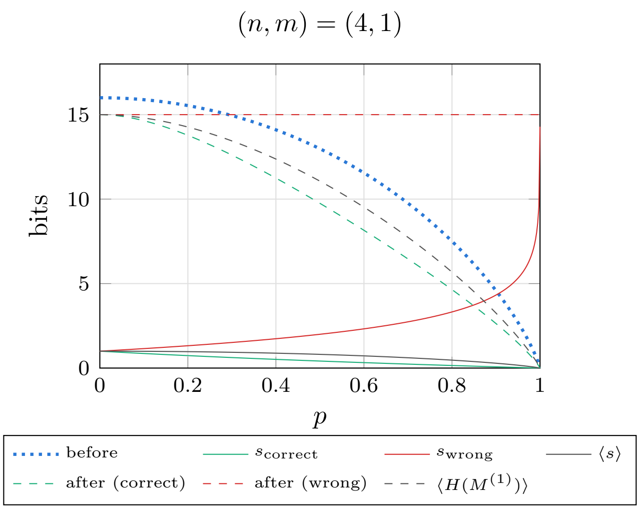


Figure 17: One-question anatomy of the spike prior against the special map's mass p: the branch surprisals (103) (solid), the answer-matrix entropy before the question (dotted) and on each branch after it (104) (dashed). Both panels carry $Qm = 16$ bits of table entropy at the uniform point.

For $p < 1$ the branch leverages divide the vertical distance between the pre-question curve and each branch curve by that branch's own surprisal,

$$L_{\text{correct}} = \frac{QG_1 - (Q-1)H_{\text{c}}}{-\log_2 P}, \qquad L_{\text{wrong}} = \frac{QG_1 - (Q-1)m}{-\log_2 U}, \tag{106}$$

whereas the expected leverage is the ratio of the two expected totals,

$$\langle L_1\rangle = \frac{P\big[QG_1 - (Q-1)H_{\text{c}}\big] + (A-1)U\big[QG_1 - (Q-1)m\big]}{G_1} = \frac{QG_1 - (Q-1)(G_2 - G_1)}{G_1}. \tag{107}$$

The branch formulas use realized surprisals; the expected formula divides the expected entropy drop by the expected surprisal G_1 and is therefore *not* the probability-weighted average of the two branch ratios.

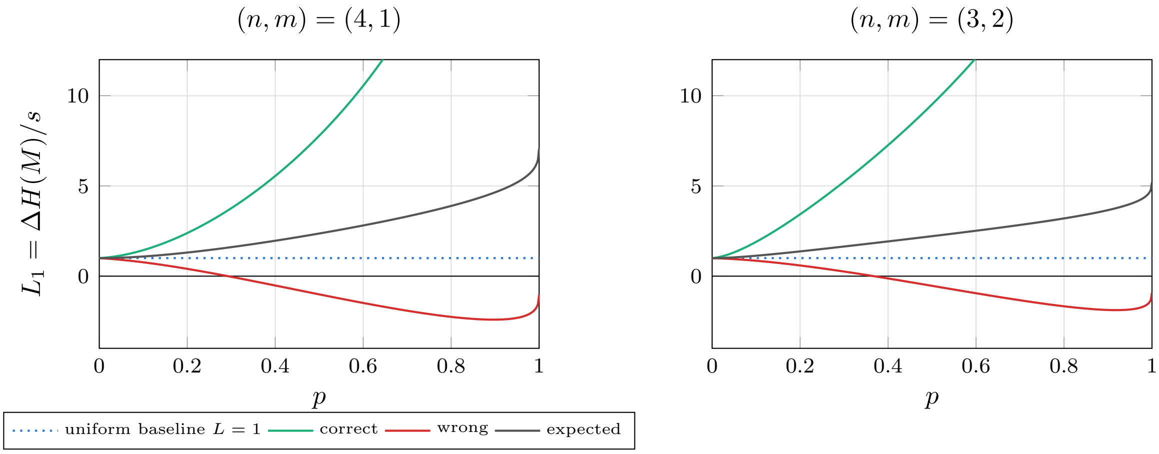


Figure 18: Branchwise (106) and expected (107) leverage of the first question. A refuting answer carries so much surprisal that its own ratio falls below the uniform baseline and eventually below zero, yet the expectation stays well above 1 because the confirming branch is both likely and cheap.

A.2 The extraction ratio

The total correlation $C := H(M^{(0)}) - G_Q = QG_1 - G_Q$ is the information stored in dependencies among answers rather than visible in their separate marginals. Only the part of the expected entropy drop beyond the received entropy G_1 comes out of that store, so when $C > 0$ the share of it freed by a single question is

$$R_1 := \frac{\langle \Delta H(M) \rangle - G_1}{C} = \frac{(Q-1)(2G_1 - G_2)}{QG_1 - G_Q}. \tag{108}$$

The two ratios differ only in their denominator: R_1 measures against the whole dependency stock, leverage against the received surprisal, and

$$R_1 = \frac{G_1}{C}\big(\langle L_1 \rangle - 1\big), \qquad \langle L_1 \rangle = 1 + \frac{C}{G_1} R_1. \tag{109}$$

Thus R_1 is a stock-extraction fraction while $\langle L_1 \rangle - 1$ is released dependency information per received bit. At $p = 1$ both the numerator of R_1 and C vanish; expanding (68) with $\varepsilon = 1 - p \to 0^+$ gives the exact finite-(n, m) ratio limit

$$\lim_{p \to 1^-} R_1(p) = \frac{(Q-1)N(1-u)^2}{QN(1-u) - (N-1)}, \qquad \frac{(Q-1)(1-u)^2}{Q(1-u) - 1} \tag{110}$$

the second expression being the additional large-N approximation. At $(n, m) = (2, 1)$ the exact value is $12/17 \simeq 0.706$ against the approximation's 0.75. Both share the thermodynamic limit

$$\lim_{n \to \infty} \lim_{p \to 1^-} R_1(p) = 1 - u = 1 - 2^{-m}. \tag{111}$$

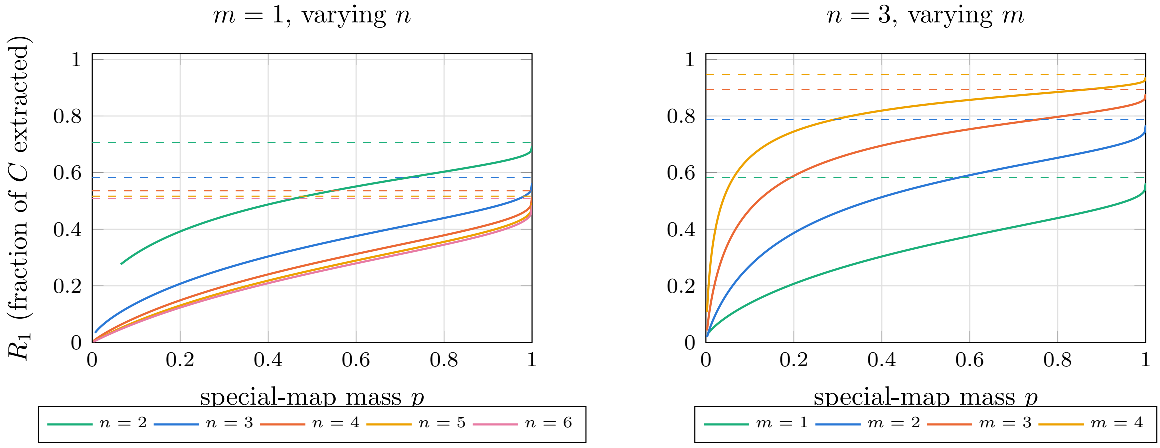


Figure 19: The exact one-question extraction ratio (108) of the spike prior, plotted only where it is defined ($1/N < p < 1$). Dashed lines are the exact finite-N limits (110) as $p \to 1^-$. Growing n drives them down to $1 - u = 1/2$; growing m lifts them toward one.

For the spike-plus-uniform family (111) is a theorem. Extending it to a broader class of uninformative hedges would require a precise definition and a separate extremal argument: full support alone is not sufficient.

A.3 The sharp-prior limit and the order of limits

The sharp-prior calculation takes $\varepsilon = 1 - p \to 0^+$ with n and m fixed. It is a limit toward the delta prior, not evaluation at $p = 1$, where leverage is $0/0$. With the entropy coefficient $\iota_k := \frac{N}{N-1}(1 - u^k)$ we have $P_k = 1 - \varepsilon\iota_k$, the other $u^{-k} - 1$ outcomes share mass $\varepsilon\iota_k$, and

$$G_k = h_2(\varepsilon\iota_k) + \varepsilon\iota_k \log_2(u^{-k} - 1) \;\sim\; \varepsilon\,\iota_k \log_2 \frac{1}{\varepsilon} \qquad (\varepsilon \downarrow 0) \tag{112}$$

at fixed finite Q and k. Substituting into (56) cancels the common factor $\varepsilon \log_2(1/\varepsilon)$ and leaves, for $1 \le k \le Q$,

$$\lim_{p\to 1^-} \langle L_k \rangle = (1-u)\,\frac{Q - (Q-k)u^k}{1 - u^k} = \underbrace{Q(1-u)}_{\text{extensive plateau}} + \underbrace{(1-u)\frac{k\,u^k}{1-u^k}}_{O(1)\text{ boundary layer}} . \tag{113}$$

The second term decays geometrically in k and does not depend on Q at all. In particular $\lim_{p\to 1^-} \langle L_1 \rangle = Q - (Q-1)2^{-m}$, so

$$\lim_{n\to\infty} \lim_{p\to 1^-} \frac{\langle L_1 \rangle}{Q} \;=\; 1 - 2^{-m}. \tag{114}$$

The coefficient $1 - 2^{-m}$ is the fraction of the hypothesis count eliminated by one answer. Its equality with the normalized leverage follows from the entropy asymptotics and should not be read as literally settling that fraction of every individual column. The limiting operation is necessary: at $p = 1$ the learner already assigns probability one to the truth, so received and destroyed information both vanish and leverage is $0/0$. The finite value is the ratio at which the two vanish as $p \to 1^-$. Taking $k = \lfloor tQ \rfloor$ in (113) gives the same constant at every $t > 0$, with the uniform bound

$$\sup_{1\le k\le Q} \left| \frac{1}{Q} \lim_{p\to 1^-} \langle L_k \rangle - (1-u) \right| = O(Q^{-1}), \tag{115}$$

so the microscopic boundary term survives only as an $O(1)$ correction to the unnormalized leverage.

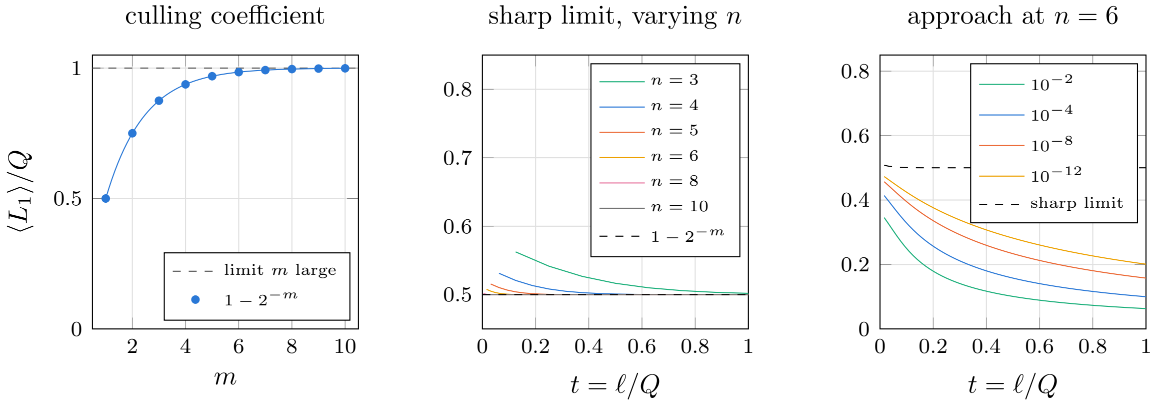


Figure 20: Left: the culling coefficient (114) against the answer width. Middle: the sharp-prior curve (113) normalized by Q, whose $O(1)$ boundary layer is squeezed into the origin as n grows, leaving the plateau at $1-u$. Right: the approach at fixed $n = 6$, $m = 1$, labelled by $1-p$. Tightening the prior lifts the curve only logarithmically, so the plateau is approached pointwise in t but never uniformly.

As $p \to 1^-$ the fixed-p coefficient of (74) diverges, $\big[h_{\text{spike}}(p,m) - (1-p)m\big] / \big[(1-p)m\big] \sim \frac{1-2^{-m}}{m} \log_2 \frac{1}{1-p}$, which is the noncommutation in explicit form. Writing $L_{p,n}(t) := \langle L_{\lfloor tQ \rfloor} \rangle$ and comparing under the same normalization,

$$\lim_{n\to\infty} \lim_{p\to 1^-} \frac{L_{p,n}(t)}{Q} = 1 - 2^{-m}, \qquad \lim_{p\to 1^-} \lim_{n\to\infty} \frac{L_{p,n}(t)}{Q} = 0. \tag{116}$$

Before dividing by Q the first ordering is extensive while the second diverges only logarithmically in $1/(1-p)$. The two orders are the endpoints of a family of joint scalings. Choose $p_n \to 1$ with prescribed exponential sharpness

$$\frac{\log_2 \big(1/(1-p_n)\big)}{Q} \longrightarrow \lambda \in (0, \infty), \tag{117}$$

so that $\lambda = 0$ recovers the scaling behind the second limit and the formal endpoint $\lambda = \infty$ the first. Putting $\varepsilon_n = 1 - p_n$, the leading entropies for $0 < t < 1$ and $k = \lfloor tQ \rfloor$ are $G_1 \sim (1-u)\varepsilon_n \lambda Q$, $G_k \sim \varepsilon_n Q(\lambda + mx)$ and $G_{k+1} - G_k \sim \varepsilon_n m$, and substitution into (56) gives the crossover

$$\frac{L_{p_n,n}(t)}{Q} \longrightarrow (1-u)\, \frac{\lambda}{\lambda + mx}. \tag{118}$$

At $t = 1$ the increment $G_{k+1} - G_k$ is undefined and the same value follows instead from the completion formula. The two familiar orders are thus endpoints of a larger family.

B Block priors and the Bernstein profile language

A stable, projectively consistent exchangeable prior cannot produce a smooth non-hyperbolic macroscopic curve: de Finetti reduces it to an i.i.d. mixture, whose interior profile is constant and whose macroscopic curve is a hyperbola. To obtain a profile that changes throughout $0 < t < 1$, the prior must retain structure tied to particular groups or addresses of questions, and a fixed partition into local blocks is the simplest solvable construction. The statement needs its consistency qualification: a separate finite exchangeable law can be chosen at every n without forming one infinite exchangeable family, and such a sequence need be neither an i.i.d. mixture nor possessed of any thermodynamic limit.

B.1 Independent blocks of questions

Partition $\mathcal{Q}$ into independent blocks drawn from a fixed finite set of types, writing $\mathcal{P}_n$ for the partition and indexing types by κ. At finite n let $K_{\kappa,n}$ count the type-κ blocks and set $w_{\kappa,n} := r_\kappa K_{\kappa,n}/Q$, the fraction of all questions belonging to them, assuming $w_{\kappa,n} \to w_\kappa$ with $\sum_\kappa w_\kappa = 1$. A type κ carries a size r_κ, a fixed joint distribution on its r_κ answers shared by every block of that type up to relabelling, and a mean subset-entropy profile defined as follows. Let $\mathcal{V} \subseteq \mathcal{Q}$ be one particular block of type κ, fix an ordering of it once and for all, and give every subset $\mathcal{J} \subseteq \mathcal{V}$ the inherited order, so that $\psi(\mathcal{J}) \in \mathcal{A}^{|\mathcal{J}|}$ is its random answer vector. For $j \in \{0, \dots, r_\kappa\}$ define

$$H_\kappa(j) := \binom{r_\kappa}{j}^{-1} \sum_{\substack{\mathcal{J} \subseteq \mathcal{V} \\ |\mathcal{J}|=j}} H\big(\psi(\mathcal{J})\big). \tag{119}$$

Here j is the *number of questions selected from the block*, not a question label, and $H(\psi(\mathcal{J}))$ measures the learner's uncertainty in their *joint answers*, not uncertainty about which questions were selected: the set $\mathcal{J}$ is known. No symmetry is required for this averaged definition, although every zoo member below has the stronger property of **entropy homogeneity**, meaning that $H(\psi(\mathcal{J}))$ itself depends only on $|\mathcal{J}|$. Full permutation invariance is sufficient for that but not necessary: MDS code blocks are entropy homogeneous even though their coordinate distribution is not invariant under every permutation.

The profile records how much answer uncertainty remains visible after looking at j locations inside a block, and its purpose is that successive differences have an operational meaning. Averaging the chain rule over a uniform i-subset $\mathcal{J}$ and then a fresh question drawn uniformly from $\mathcal{V} \setminus \mathcal{J}$ gives the within-block increments

$$\eta_\kappa(i) := H_\kappa(i+1) - H_\kappa(i) = \mathbb{E}_{\mathcal{J},q}\big[H\big(\psi(q) \mid \psi(\mathcal{J})\big)\big], \qquad 0 \le i \le r_\kappa - 1, \tag{120}$$

the mean uncertainty in one fresh answer after i of its block partners have been answered. This is why the block-entropy profile is useful: it converts a joint law on an entire block into exactly the conditional entropies the learning curve needs. The endpoints make the meaning concrete, $H_\kappa(0) = 0$ while $H_\kappa(r_\kappa) = H(\psi(\mathcal{V}))$ is the entropy of the whole block taken jointly. Two independent fair binary answers have profile $(0,1,2)$ and increments $(1,1)$; the same fair bit copied into both questions has profile $(0,1,1)$ and increments $(1,0)$, since after either answer is known the other carries no remaining uncertainty.

For a fresh question in a block of size r_κ, the number of its $r_\kappa - 1$ partners already present in a uniformly random ℓ-set is hypergeometric, so exactly at finite Q, for $0 \le \ell < Q$,

$$\gamma_{n,\ell} = \sum_\kappa w_{\kappa,n} \sum_{i=0}^{r_\kappa - 1} \frac{\binom{r_\kappa - 1}{i}\binom{Q - r_\kappa}{\ell - i}}{\binom{Q-1}{\ell}} \eta_\kappa(i). \tag{121}$$

Holding every r_κ fixed, taking $\ell = \lfloor tQ \rfloor$ and assuming $w_{\kappa,n} \to w_\kappa$, sampling without replacement approaches independent sampling,

$$\frac{\binom{r-1}{i}\binom{Q-r}{\ell-i}}{\binom{Q-1}{\ell}} \longrightarrow \underbrace{\binom{r-1}{i} t^i (1-t)^{r-1-i}}_{\beta_{r-1,i}(t)}, \tag{122}$$

which simultaneously defines the Bernstein basis polynomial $\beta_{r-1,i}$ and, probabilistically, is the limiting probability that exactly i of the fresh question's $r-1$ partners have been asked by macroscopic time t. The limiting profile is therefore

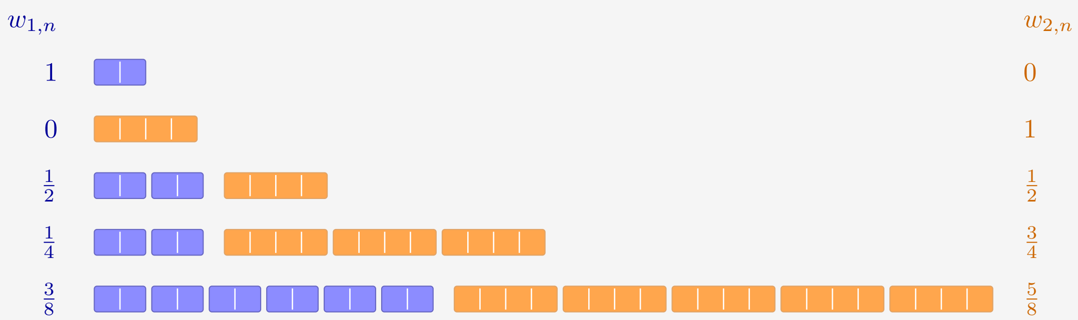
$$\gamma(t) = \sum_{\kappa} w_\kappa \sum_{i=0}^{r_\kappa-1} \beta_{r_\kappa-1,i}(t)\, \eta_\kappa(i). \tag{123}$$

At the start of the run no partner has been asked, so only the $i = 0$ term survives and $h_0 = \gamma(0) = \sum_\kappa w_\kappa \eta_\kappa(0) = \sum_\kappa w_\kappa H_\kappa(1)$: the same block-entropy profile supplies both the initial marginal entropy and the entire macroscopic conditional-entropy curve.

To construct a sequence with prescribed fractions, choose integer block counts with $r_\kappa K_{\kappa,n}/Q \to w_\kappa$ and $\sum_\kappa r_\kappa K_{\kappa,n} \le Q$, assigning only $o(Q)$ leftover sites, whose treatment does not affect the limit; at finite n any leftovers can be recorded as an additional block type, so (121) still includes every question and remains exact. If block sizes grow with n the proof changes, a useful sufficient condition for the hypergeometric-to-binomial approximation being $\left(\max_{\mathcal{V}\in\mathcal{P}_n} |\mathcal{V}|\right)^2/Q \to 0$. For countably many types, pointwise frequency convergence must be replaced by an ℓ^1 or tightness condition.

EXAMPLE 41

Two types tile the growing table: pairs ($r_1 = 2$, blue) and quads ($r_2 = 4$, orange), aiming at $w_1 = 3/8$ and $w_2 = 5/8$. Integer counts force detours. At $n = 1$ only a pair fits and at $n = 2$ only a quad, while at $Q = 8$ and $Q = 16$ the best leftover-free tilings still miss the target shares by $1/8$.



Each cell is one question and each rounded group is one block, so the counts $K_{\kappa,n}$ can be read off: $(K_1, K_2) = (1,0),\ (0,1),\ (2,1),\ (2,3),\ (6,5)$. The pair share $w_{1,n} = 1,\ 0,\ \frac{1}{2},\ \frac{1}{4},\ \frac{3}{8}$ closes in on its target by halving steps, and at $n = 5$ the tiling is exact: $6 \times 2 + 5 \times 4 = 32 = Q$.

B.2 Block laws and their map priors

The profile machinery never displayed the prior itself. It is constructive: each block law prices the restriction of a map to its block, and independence across blocks multiplies the prices. A map ϕ_j restricts to the answer tuple $\phi_j(\mathcal{V})$ on block $\mathcal{V}$, and

$$p_j = \prod_{\mathcal{V}\in\mathcal{P}_n} \Pr\left(\psi(\mathcal{V}) = \phi_j(\mathcal{V})\right). \tag{124}$$

The classical block laws, with the factor each contributes to (124) and the curve it produces:

block law	a size-r block is sampled by	factor $\Pr(\psi(\mathcal{V}) = y)$	leverage
clique	one value $a_\mathcal{V} \sim \nu_\mathcal{V}$ copied to every question	$\nu_\mathcal{V}(a)$ if $y = (a, \ldots, a)$, else 0	$L \equiv r$, exact at every n
parity (σ)	uniform on the coset $\bigoplus_q y_q = \sigma$	$2^{-(r-1)}$ on the coset, else 0	rises $1 \to r/(r-1)$, late
tilted parity	i.i.d. Bernoulli(θ) bits conditioned on the coset	$\theta^{\lvert y\rvert}(1-\theta)^{r-\lvert y\rvert}/\zeta_{r,\sigma}$ on the coset, else 0	same shape, bent by $\varepsilon = 1 - 2\theta$
$[r,k]_A$ MDS	uniform on an MDS code ($A \geq r$)	A^{-k} on codewords, else 0	$L(1) = r/k$; interior peak when mixed with fresh sites
lapse (δ)	any law above, replaced by the uniform block with probability δ	$(1-\delta)P_\mathcal{V}(y) + \delta A^{-r}$	full support; $\gamma(1) > 0$

The supports multiply as well: cliques carry the $A^{Q/r}$ maps constant on every block, parity carries the $2^{(r-1)Q/r}$ maps landing in every coset, a code carries $A^{kQ/r}$ maps, and the lapse restores $p_j > 0$ everywhere, playing the role the top class played in Section 4.3.

EXAMPLE 42

Cliques at $(n, m) = (2, 1)$: partition the four questions by their high bit, $\mathcal{V}_1 = \{00, 01\}$ and $\mathcal{V}_2 = \{10, 11\}$, with $\nu_1 = (0.8, 0.2)$ and $\nu_2 = (0.55, 0.45)$ on the answers $(0, 1)$. A supported map must be constant on both blocks, so it can depend only on b_1: of the sixteen maps of Table 4, only FALSE ($j = 0$), $\neg b_1$ ($j = 3$), b_1 ($j = 12$) and TRUE ($j = 15$) survive, with product weights $p_0 = 0.8 \times 0.55 = 0.44$, $p_3 = 0.11$, $p_{12} = 0.36$, $p_{15} = 0.09$, and $p_j = 0$ for the other twelve. One answer from either block settles that block's remaining question for free: $L \equiv 2$.

From any row of the table the route to leverage is the one already walked in Section 4.3. The block law fixes the subset entropies (119), their differences give the increments η (120), and the hypergeometric average (121) is exact at every finite n; its binomial limit is the Bernstein profile (123). No capacity theorem is needed: blocks are bounded, so elementary hypergeometric-to-binomial convergence replaces it. The finite law (79) then gives $\langle L_\ell \rangle$ exactly at every n, and the master law (81) gives the curve, with $h_0 = \sum_\kappa w_\kappa \eta_\kappa(0)$.

B.3 Bernstein polynomials as a profile language

For a single block type, suppressing the type label, equation (123) reads

$$\gamma(t) = \eta(0)(1-t)^2 + 2\eta(1)t(1-t) + \eta(2)t^2 \tag{125}$$

at $r = 3$, and $\eta(0)(1-t)^3 + 3\eta(1)t(1-t)^2 + 3\eta(2)t^2(1-t) + \eta(3)t^3$ at $r = 4$. Some useful coefficient vectors, with h_{fresh} the entropy of one independent fresh answer and h_{clique} the entropy of the value copied through a clique:

block	$(\eta(0), \dots, \eta(r-1))$	profile
independent fresh answers	$(h_{\text{fresh}}, \dots, h_{\text{fresh}})$	h_{fresh}
clique	$(h_{\text{clique}}, 0, \dots, 0)$	$h_{\text{clique}}(1-t)^{r-1}$
parity	$(m, \dots, m, 0)$	$m(1-t^{r-1})$
(r,k) MDS	k entries m, then zeros	$m \Pr[\text{Bin}(r-1,t) \le k-1]$

Entropy submodularity gives $m \ge \eta_\kappa(0) \ge \cdots \ge \eta_\kappa(r_\kappa - 1) \ge 0$, and the derivative identity for Bernstein polynomials makes the macroscopic monotonicity explicit,

$$\gamma'(t) = \sum_\kappa w_\kappa (r_\kappa - 1) \sum_{i=0}^{r_\kappa - 2} \big(\eta_\kappa(i+1) - \eta_\kappa(i)\big) \beta_{r_\kappa - 2, i}(t) \;\le\; 0, \tag{126}$$

so γ is nonincreasing as the general entropy calculus requires. Leverage itself need not be monotone.

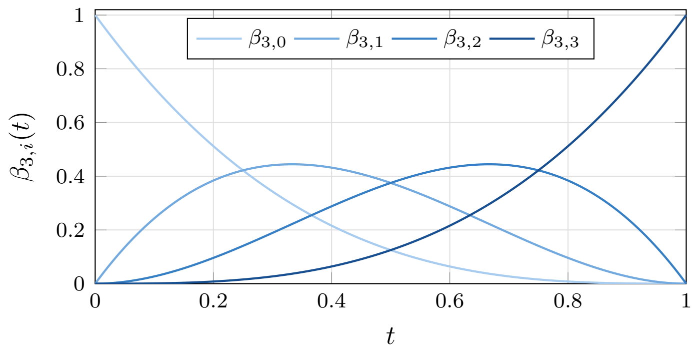


Figure 21: The degree-three Bernstein basis: bump functions scanning left to right, the i-th peaked at $i/3$. A block profile is a positive combination of such bumps weighted by the microscopic conditional entropies $\eta(i)$, so the shape of L becomes a question about *where the coefficients drop.*

Equation (123) describes every prior in the stated class of independent blocks drawn from a fixed finite list of bounded types, and mixing types mixes their profiles linearly. There is also a useful approximation result when the answer alphabet is allowed to grow: mixtures of common-length MDS blocks realize any nonincreasing Bernstein coefficient sequence, and Bernstein polynomials approximate every continuous nonincreasing profile. A Reed–Solomon block of length r needs $A \ge r$, however, so sending $r \to \infty$ in that approximation also requires $m \to \infty$, or a different family of realizable entropy vectors. It is not a completeness theorem at fixed answer width.

B.4 Leverage curves at finite n

Because (121) is exact, the finite-n curves need no simulation: the hypergeometric increments feed the finite law (79) directly, and the limits below are the master law (81) applied to the Bernstein profiles of the table above. Three representative members, at $Q = 16$ and $Q = 64$ against their limits:

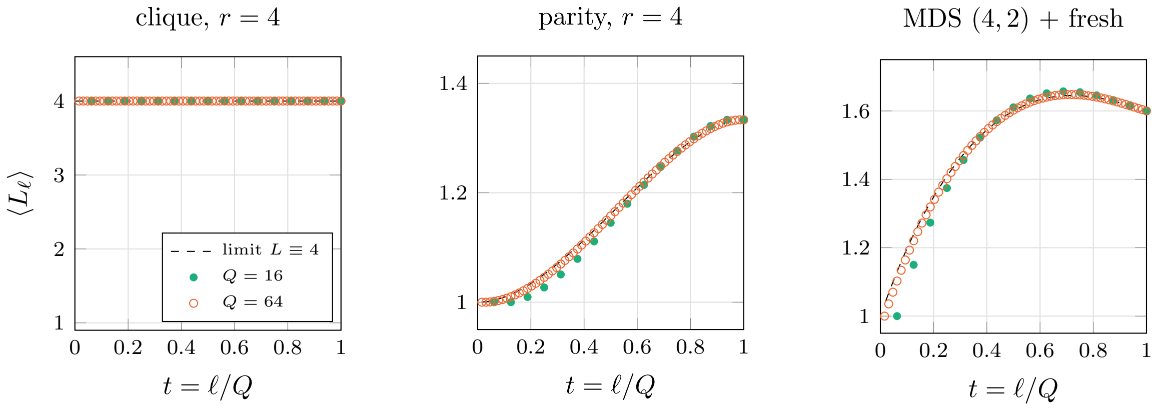


Figure 22: Exact finite-n leverage (dots, $Q = 16$ and 64) against the macroscopic limits (dashed), from (121) and (79) with no simulation. Left: cliques of four with a fair shared bit; the dots sit exactly on the constant $L \equiv 4$ at every finite n. Middle: parity blocks of four; deductions concentrate late and the finite curves approach the rise to $r/(r-1) = \frac{4}{3}$ from below. Right: a 3:1 mixture of $[4, 2]$ MDS code blocks ($m = 2$) with fresh questions; the code releases its deductions around the threshold and the fresh sites keep receiving afterwards, an interior maximum of 1.645 at $t \simeq 0.72$ against the completion value 1.6.